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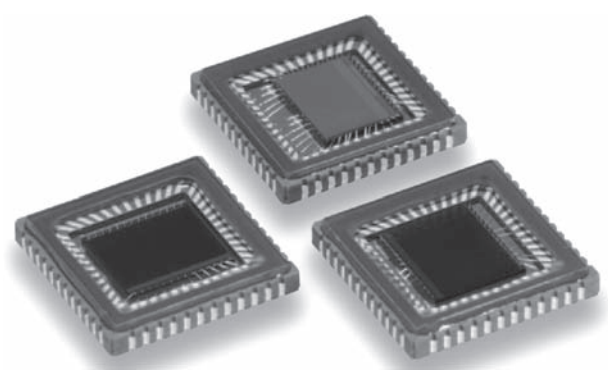
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CMOS image sensors with wide dynamic range. (*Courtesy of New Imaging Technologies (NIT), France*)

Physicists use the wave theory on Mondays, Wednesdays and Fridays and the particle theory on Tuesdays, Thursdays and Saturdays.

—Sir William Henry Bragg[1]



Augustin Jean Fresnel (1788–1827) was a French physicist and a civil engineer for the French government who was one of the principal proponents of the wave theory of light. He made a number of distinct contributions to optics including the well-known Fresnel lens that was used in lighthouses in the nineteenth century. He fell out with Napoleon in 1815 and was subsequently put under house arrest until the end of Napoleon's reign. During his enforced leisure time he formulated his wave ideas of light into a mathematical theory. *(© INTERFOTO/Alamy.)*

If you cannot saw with a file or file with a saw, then you will be no good as an experimentalist.

—Attributed to Augustin Fresnel

[1] As quoted in Alan Mackay, *Dictionary of Scientific Quotes*, 2nd Edition (Institute of Physics Publishing, Bristol, 1991), p. 37.

1 WAVE NATURE OF LIGHT

1.1 LIGHT WAVES IN A HOMOGENEOUS MEDIUM

A. Plane Electromagnetic Wave

The wave nature of light, quite aside from its photonic behavior, is well recognized by such phenomena as interference and diffraction. We can treat light as an electromagnetic (EM) wave with time-varying electric and magnetic fields, E_x and B_y, respectively, which are propagating through space in such a way that they are always perpendicular to each other and the direction of propagation z as illustrated in Figure 1.1. The simplest traveling wave is a sinusoidal wave that, for propagation along z, has the general mathematical form

Traveling wave along z

$$E_x = E_o \cos(\omega t - kz + \phi_o) \tag{1.1.1}$$

in which E_x is the electric field at position z at time t, k is the **propagation constant**[2] given by $2\pi/\lambda$, where λ is the wavelength, ω is the angular frequency, E_o is the amplitude of the wave, and ϕ_o is a phase constant, which accounts for the fact that at $t = 0$ and $z = 0$; E_x may or may not necessarily be zero depending on the choice of origin. The argument $(\omega t - kz + \phi_o)$ is called the **phase** of the wave and denoted by ϕ. Equation (1.1.1) describes a **monochromatic plane wave** of infinite extent traveling in the positive z direction as depicted in Figure 1.2. In any plane perpendicular to the direction of propagation (along z), the phase of the wave, according to Eq. (1.1.1), is constant, which means that the field in this plane is also constant. A surface over which the phase of a wave is constant at a given instant is referred to as a **wavefront**. A wavefront of a plane wave is obviously an infinite plane perpendicular to the direction of propagation as shown in Figure 1.2.

We know from electromagnetism that time-varying magnetic fields result in time-varying electric fields (Faraday's law) and vice versa. A time-varying electric field would set up a time-varying magnetic field with the same

[2] Some authors also call k the *wave number*. However, in spectroscopy, the wave number implies $1/\lambda$, reciprocal wavelength. To avoid any confusion, propagation constant would be preferred for k.

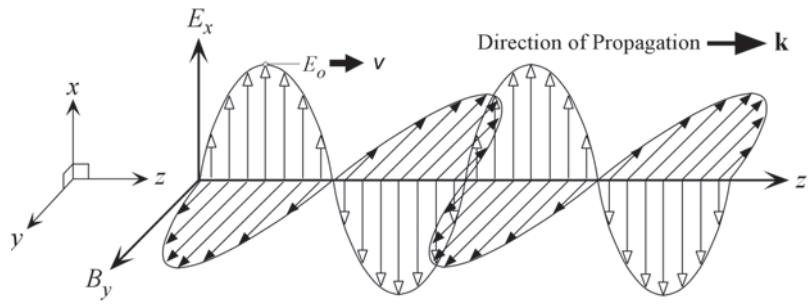


FIGURE 1.1 An electromagnetic wave in a homogenous and isotropic medium is a traveling wave that has time-varying electric and magnetic fields which are perpendicular to each other and the direction of propagation z. This is a snapshot at a given time of a particular harmonic or a sinusoidal EM wave. At a time δt later, a point on the wave, such as the maximum field, would have moved a distance $v\delta t$ in the z-direction.

frequency. According to electromagnetic principles,[3] a traveling electric field E_x as represented by Eq. (1.1.1) would always be accompanied by a traveling magnetic field B_y with the same wave frequency and propagation constant (ω and k) but the directions of the two fields would be orthogonal as in Figure 1.1. Thus, there is a similar traveling wave equation for the magnetic field component B_y. We generally describe the interaction of a light wave with a non-conducting matter (conductivity, $\sigma = 0$) through the electric field component E_x rather than B_y because it is the electric field that displaces the electrons in molecules or ions in the crystal and thereby gives rise to the polarization of matter. However, the two fields are linked, as in Figure 1.1, and there is an intimate relationship between them. The **optical field** refers to the electric field E_x.

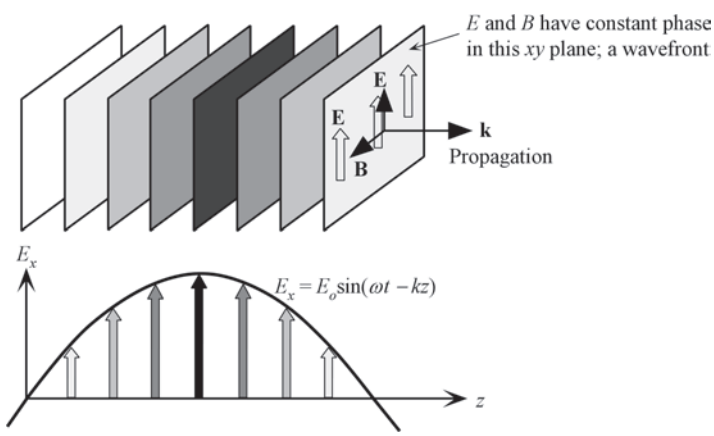


FIGURE 1.2 A plane EM wave traveling along z has the same E_x (or B_y) at any point in a given xy plane. All electric field vectors in a given xy plane are therefore in phase. The xy planes are of infinite extent in the x and y directions.

[3] Maxwell's equations formulate electromagnetic phenomena and provide relationships between the electric and magnetic fields and their space and time derivatives. We need only to use a few selected results from Maxwell's equation without delving into their derivations. The magnetic field B is also called the magnetic induction or magnetic flux density. The magnetic field *intensity* H and magnetic field B in a non-magnetic material are related by $B = \mu_o H$ in which μ_o is the absolute permeability of the medium.

We can also represent a traveling wave using the exponential notation since $\cos\phi = \mathsf{Re}\left[\exp(j\phi)\right]$ in which Re refers to the real part. We then need to take the real part of any complex result at the end of calculations. Thus, we can write Eq. (1.1.1) as

$$E_x(z, t) = \mathsf{Re}\left[E_o \exp(j\phi_o) \exp j(\omega t - kz)\right]$$

or

$$E_x(z, t) = \mathsf{Re}\left[E_c \exp j(\omega t - kz)\right] \tag{1.1.2}$$

Traveling wave along z

in which $E_c = E_o \exp(j\phi_o)$ is a complex number that represents the amplitude of the wave and includes the constant phase information ϕ_o. Note that in Eq. (1.1.2), $\exp j(\omega t - kz)$ represents $e^{j(\omega t - kz)}$.

We indicate the direction of propagation with a vector **k**, called the **wave vector** (or **propagation vector**), whose magnitude is the **propagation constant**, $k = 2\pi/\lambda$. It is clear that **k** is perpendicular to constant phase planes as indicated in Figure 1.2. Consider an electromagnetic wave that is propagating along some arbitrary direction **k**, as indicated in Figure 1.3. The electric field $E(\mathbf{r}, t)$ at an arbitrary point **r** is given by

$$E(\mathbf{r}, t) = E_o \cos(\omega t - \mathbf{k}\cdot\mathbf{r} + \phi_o) \tag{1.1.3}$$

Traveling wave in 3D with wave vector **k**

because the dot product $\mathbf{k}\cdot\mathbf{r}$ is along the direction of propagation similar to kz as indicated in Figure 1.3. The latter can be shown by drawing a plane that has the point **r** and is perpendicular to **k** as illustrated in Figure 1.3. The dot product is the product of **k** and the projection of **r** onto **k**, which is $\mathbf{r}'$ in Figure 1.3, so that $\mathbf{k}\cdot\mathbf{r} = kr'$. Indeed, if propagation is along z, $\mathbf{k}\cdot\mathbf{r}$ becomes kz. In general, if **k** has components k_x, k_y, and k_z along x, y, and z, then from the definition of the dot product, $\mathbf{k}\cdot\mathbf{r} = k_x x + k_y y + k_z z$.

The relationship between time and space for a given phase, ϕ for example, that corresponds to a maximum field, according to Eq. (1.1.1), is described by

$$\phi = \omega t - kz + \phi_o = \text{constant}$$

During a time interval δt, this constant phase (and hence the maximum field) moves a distance δz. The phase velocity of this wave is therefore $\delta z/\delta t$. Thus the **phase velocity** v is

$$v = \frac{dz}{dt} = \frac{\omega}{k} = \upsilon\lambda \tag{1.1.4}$$

Phase velocity

in which υ is the frequency ($\omega = 2\pi\upsilon$) of the EM wave. For an EM wave propagating in free space v is the speed of light in vacuum or c.

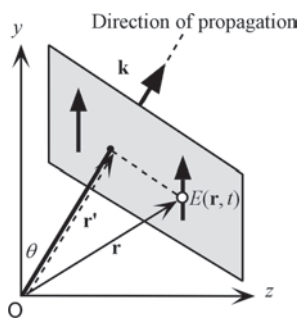


FIGURE 1.3 A traveling plane EM wave along a direction **k**.

We are often interested in the phase difference $\Delta\phi$ at a given time between two points on a wave (Figure 1.1) that are separated by a certain distance. If the wave is traveling along z with a wave vector k, as in Eq. (1.1.1), then the phase difference between two points separated by Δz is simply $k\Delta z$ since ωt is the same for each point. If this phase difference is 0 or multiples of 2π then the two points are in phase. Thus phase difference $\Delta\phi$ can be expressed as $k\Delta z$ or $2\pi\Delta z/\lambda$.

B. Maxwell's Wave Equation and Diverging Waves

Consider the plane EM wave in Figure 1.2. All constant phase surfaces are xy planes that are perpendicular to the z-direction. A cut of a plane wave parallel to the z-axis is shown in Figure 1.4 (a) in which the parallel dashed lines at right angles to the z-direction are wavefronts. We normally show wavefronts that are separated by a phase of 2π or a whole wavelength λ as in the figure. The vector that is normal to a wavefront surface at a point such as P represents the direction of wave propagation (**k**) at that point P. Clearly, the propagation vectors everywhere are all parallel and the plane wave propagates without the wave diverging; *the plane wave has no divergence.* The amplitude of the planar wave E_o does not depend on the distance from a reference point, and it is the same at all points on a given plane perpendicular to **k** (*i.e.*, independent of x and y). Moreover, as these planes extend to infinity there is infinite energy in the plane wave. A plane wave such as the one in Figure 1.4 (a) is an idealization that is useful in analyzing many wave phenomena. In reality, however, the electric field in a plane at right angles to **k** does not extend to infinity since the light beam would have a finite cross-sectional area and finite power. We would need an infinitely large EM source with infinite power to generate a perfect plane wave!

In practice there are many types of possible EM waves. These waves must obey a special wave equation that describes the time and space dependence of the electric field. In an isotropic and linear dielectric medium, the relative permittivity (ε_r) is the same in all directions and is independent of the electric field. The field E in such a medium obeys **Maxwell's EM wave equation**

Maxwell's wave equation

$$\frac{\partial^2 E}{\partial x^2} + \frac{\partial^2 E}{\partial y^2} + \frac{\partial^2 E}{\partial z^2} - \varepsilon_o \varepsilon_r \mu_o \frac{\partial^2 E}{\partial t^2} = 0 \qquad \textbf{(1.1.5)}$$

in which μ_o is the absolute permeability, ε_o is the absolute permittivity, and ε_r is the relative permittivity of the medium. Equation (1.1.5) assumes an isotropic medium (as discussed later) and

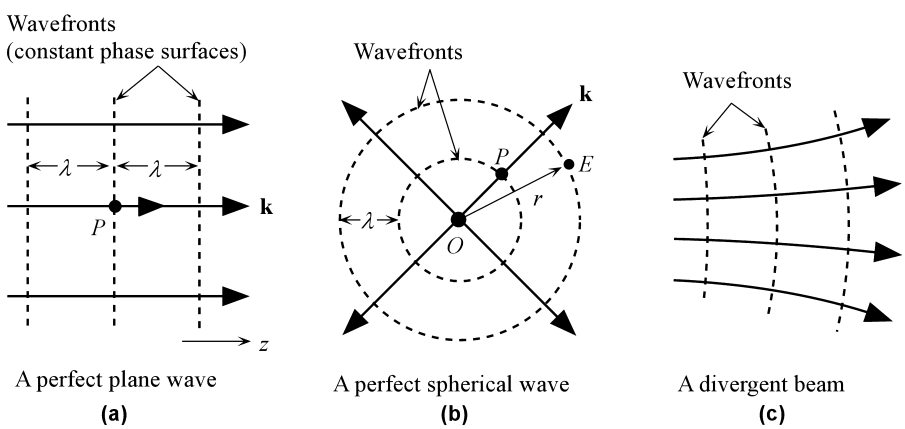


FIGURE 1.4 Examples of possible EM waves. (a) A perfect plane wave. (b) A perfect spherical wave. (c) A divergent beam.

that the conductivity of the medium is zero. To find the time and space dependence of the field, we must solve Eq. (1.1.5) in conjunction with the initial and boundary conditions. We can easily show that the plane wave in Eq. (1.1.1) satisfies Eq. (1.1.5). There are many possible waves that satisfy Eq. (1.1.5) that can therefore exist in nature.

A **spherical wave** is described by a traveling field that emerges from a point EM source and whose amplitude decays with distance r from the source. At any point r from the source, the field is given by

$$E = \frac{A}{r}\cos(\omega t - kr) \qquad \textbf{(1.1.6)}$$

Spherical wave

in which A is a constant. We can substitute Eq. (1.1.6) into Eq. (1.1.5) to show that Eq. (1.1.6) is indeed a solution of Maxwell's equation (transformation from Cartesian to spherical coordinates would help). A cut of a spherical wave is illustrated in Figure 1.4 (b) where it can be seen that *wavefronts are spheres* centered at the point source O. The direction of propagation $\mathbf{k}$ at any point such as P is determined by the normal to the wavefront at that point. Clearly $\mathbf{k}$ vectors diverge out and, as the wave propagates, the constant phase surfaces become larger. **Optical divergence** refers to the angular separation of wave vectors on a given wavefront. The spherical wave has 360° of divergence in all planes through the point source. It is apparent that plane and spherical waves represent two extremes of wave propagation behavior from perfectly parallel to fully diverging wave vectors. They are produced by two extreme sizes of EM wave source: an infinitely large source for the plane wave and a point source for the spherical wave. In reality, an EM source is neither of infinite extent nor in point form, but would have a finite size and finite power. Figure 1.4 (c) shows a more practical example in which a light beam exhibits some inevitable divergence while propagating; the wavefronts are slowly bent away thereby spreading the wave. Light rays of geometric optics are drawn to be normal to constant phase surfaces (wavefronts). Light rays therefore follow the wave vector directions. Rays in Figure 1.4 (c) slowly diverge away from each other. The reason for favoring plane waves in many optical explanations is that, at a distance far away from a source, over a *small spatial region*, the wavefronts will appear to be plane even if they are actually spherical. Figure 1.4 (a) may be a small part of a huge spherical wave.

Many light beams, such as the output from a laser, can be described by assuming that they are **Gaussian beams**. Figure 1.5 illustrates a Gaussian beam traveling along the z-axis. The beam still has an $\exp j(\omega t - kz)$ dependence to describe propagation characteristics but the amplitude varies spatially away from the beam axis and also *along* the beam axis. Such a beam has similarities to that in Figure 1.4 (c); it slowly diverges[4] and is the result of radiation from a source of finite extent. The light intensity (*i.e.*, the radiation energy flow per unit area per unit time) distribution across the beam cross-section anywhere along z is Gaussian as shown in Figures 1.5 (b) and (c). The **beam diameter** $2w$ at any point z is defined in such a way that the cross-sectional area πw^2 at that point contains 86% of the beam power. Thus, the beam diameter $2w$ increases as the beam travels along z. The Gaussian beam shown in Figure 1.5 (a) starts from O with a finite width $2w_o$ where the wavefronts are parallel and then the beam slowly diverges as the wavefronts curve out during propagation along z. The finite width $2w_o$ where the wavefronts are parallel is called the **waist** of the beam; w_o is the **waist radius**

[4]The divergence is due to the self-diffraction of the beam—the beam is diffraction limited. Diffraction is covered later in this chapter. Further, the intensity of light will be defined quantitatively in Section 1.4. For the present discussion it represents the radiation energy flow per unit time per unit area.

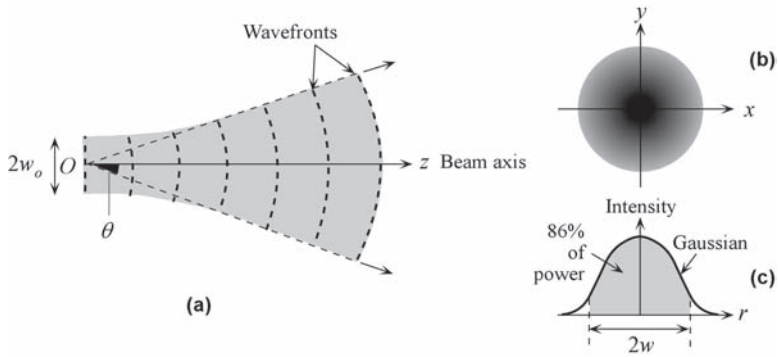


FIGURE 1.5 (a) Wavefronts of a Gaussian light beam. (b) Light intensity across a beam cross-section. (c) Light intensity vs. radial distance r from beam axis (z).

and $2w_o$ is the **spot size**. Far away from the source, the beam diameter $2w$ increases linearly with distance z. The increase in beam diameter $2w$ with z makes an angle 2θ at O, as shown in Figure 1.5, which is called the **beam divergence**. The greater the waist, the narrower the divergence. The two are related by

Far field Gaussian beam divergence

$$2\theta = \frac{4\lambda}{\pi(2w_o)} \tag{1.1.7}$$

It can be shown that the Gaussian beam is a solution of Maxwell's equations when the beam divergence is small,[5] and the intensity decays slowly with distance z. Suppose that we reflect the Gaussian beam back on itself (by using a spherical mirror that has the right curvature to match the incident wavefront) so that the beam is traveling in the $-z$ direction and converging toward O; simply reverse the direction of travel in Figure 1.5 (a). The wavefronts would be "straightening out," and at O they would be parallel again. The beam would still have the same finite diameter $2w_o$ (waist) at O. From then on, the beam again diverges out just as it did traveling in $+z$ direction as illustrated in Figure 1.6 (a). When we try to focus a Gaussian beam using a lens or a spherical mirror, as in Figure 1.6 (a), the beam cannot be brought a point but to a finite spot size $2w_o$. The relationship in Eq. (1.1.7) for the beam divergence can be used to find this minimum spot size $2w_o$ to which a Gaussian beam can be focused.

At a certain distance z_o from O, the beam diameter becomes $2^{1/2}(2w_o)$ as illustrated in Figure 1.6 (a). The distance z_o is called the **Rayleigh range**, and is given by

Gaussian beam Rayleigh range

$$z_o = \frac{\pi w_o^2}{\lambda} \tag{1.1.8}$$

The Rayleigh range is also known as the **depth of focus**. $2z_o$ is called the **confocal parameter**. The region along z far away from the Rayleigh region, $z \gg z_o$, is called the **far-field region**.

The width $2w$ of a Gaussian beam at a position along z increases with z and is given by

Gaussian beam width at distance

$$2w = 2w_o\left[1 + \left(\frac{z}{z_o}\right)^2\right]^{1/2} = 2w_o\left[1 + \left(\frac{z\lambda}{\pi w_o^2}\right)^2\right]^{1/2} \tag{1.1.9a}$$

[5] This is called the *paraxial approximation* in which the normals to the wavefronts, the wave vectors, make small angles with the z-axis.

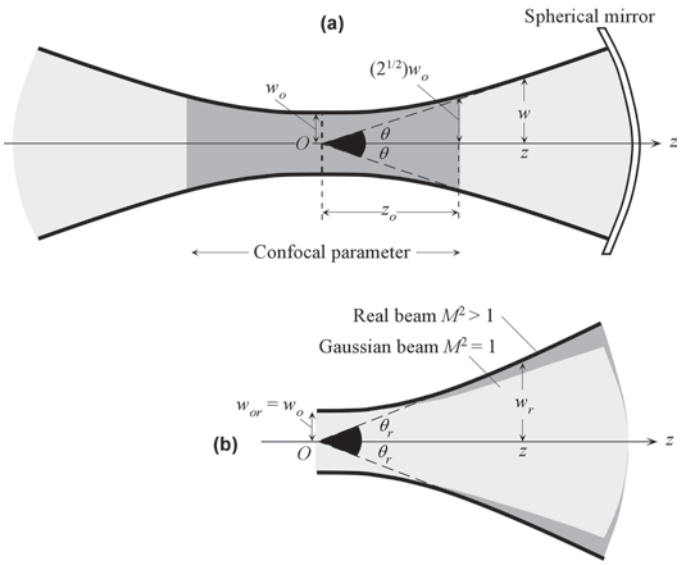


FIGURE 1.6 (a) Gaussian beam definitions. The region z_o is called the Rayleigh range and also the depth of focus. (b) Comparison of a real beam with $M^2 > 1$ with a Gaussian beam with $M^2 = 1$ and the same waist $2w_o$.

Far away from the Rayleigh range, for $z \gg z_o$, in the far-field region, the beam width increases *linearly* with z, that is,

$$2w \approx (2w_o)\frac{z}{z_o} \qquad \textbf{(1.1.9b)}$$

Gaussian beam width at distance z

Notice that the product of the beam radius w_o (half the beam waist) and half the divergence angle θ from Eq. (1.1.7) is given by $w_o\theta = \lambda/\pi$, that is, it depends only on the wavelength and is a well-defined constant for a given wavelength. The product $w_o\theta$ is called the **beam parameter product**.

The Gaussian beam concept is so useful in photonics that a special quantity, called the M^2-factor, has been introduced to compare a given laser beam to an ideal Gaussian beam. The **M^2 factor** measures the deviation of the real laser beam from the Gaussian characteristics, in which $M^2 = 1$ for an ideal (theoretical) Gaussian beam shape. Suppose that $2\theta_r$ and $2w_{or}$ are the divergence and waist, respectively, of the *real* laser beam, and 2θ and $2w_o$ are those for the ideal Gaussian. The M^2 *factor* is defined by[6]

$$M^2 = \frac{w_{or}\theta_r}{w_o\theta} = \frac{w_{or}\theta_r}{(\lambda/\pi)} \qquad \textbf{(1.1.10)}$$

M^2 factor definition

where we have used $w_o\theta = \lambda/\pi$ for an ideal Gaussian beam.

According to Eq. (1.1.10), M^2 is the ratio of the beam parameter product of the real beam to that of a Gaussian beam, and hence M^2 gauges the **beam quality** of the laser beam. For many lasers, M^2 is greater than unity, and can be as high as 10–30 in multimode lasers.

[6]Some authors define M, instead of M^2, as $M = \theta_r/\theta = w_{or}/w_o$. In addition, the reader should not be too concerned with the terms "multimode" and "single mode" at this point, except that they represent the types of radiation that is emitted from lasers.

For stable He-Ne single mode lasers, it is very close to unity. We can apply the above Gaussian equations to a real laser beam by replacing w_o with w_{or}/M^2. For example, the width at a distance z becomes

Beam width at distance z

$$2w = 2w_{or}\left[1 + \left(\frac{z\lambda M^2}{\pi w_{or}}\right)^2\right]^{1/2} \tag{1.1.11}$$

Far away from the Rayleigh range, $2w_r = M^2(2w)$, where $2w$ is the ideal Gaussian beam width at the same location. Suppose that we put a Gaussian beam with a waist w_o onto the real beam and adjust w_o to be the same as w_{or}, *i.e.*, $w_{or} = w_o$. Then, from Eq. (1.1.10) the divergence of the real beam is greater inasmuch as $\theta_r = M^2\theta$, which is shown in Figure 1.6 (b).

EXAMPLE 1.1.1 A diverging laser beam

Consider a He-Ne laser beam at 633 nm with a spot size of 1 mm. Assuming a Gaussian beam, what is the divergence of the beam? What are the Rayleigh range and the beam width at 25 m?

Solution

Using Eq. (1.1.7), we find

$$2\theta = \frac{4\lambda}{\pi(2w_o)} = \frac{4(633 \times 10^{-9}\,\text{m})}{\pi(1 \times 10^{-3}\,\text{m})} = 8.06 \times 10^{-4}\,\text{rad} = 0.046^\circ$$

The Rayleigh range is

$$z_o = \frac{\pi w_o^2}{\lambda} = \frac{\pi\left[(1 \times 10^{-3}\,\text{m})/2\right]^2}{(633 \times 10^{-9}\,\text{m})} = 1.24\,\text{m}$$

The beam width at a distance of 25 m is

$$2w = 2w_o\left[1 + (z/z_o)^2\right]^{1/2} = (1 \times 10^{-3}\,\text{m})\left\{1 + \left[(25\,\text{m})/(1.24\,\text{m})\right]^2\right\}^{1/2}$$
$$= 0.0202\,\text{m} \quad \text{or} \quad 20\,\text{mm}.$$

1.2 REFRACTIVE INDEX AND DISPERSION

When an EM wave is traveling in a dielectric medium, the oscillating electric field polarizes the molecules of the medium at the frequency of the wave. Indeed, the EM wave propagation can be considered to be the propagation of this polarization in the medium. The field and the induced molecular dipoles become coupled. The relative permittivity ε_r measures the ease with which the medium becomes polarized and hence it indicates the extent of interaction between the field and the induced dipoles. To find the nature of propagation of an EM wave in a dielectric medium, and hence the phase velocity, we have to solve Maxwell's equations in a dielectric medium. If we assume the medium is insulating, nonmagnetic, and also **isotropic**, that is the relative permittivity is independent of the direction of propagation of the EM wave and the optical field, then the solution becomes quite simple and leads to Maxwell's wave equation stated in Eq. (1.1.5). We can continue to represent the EM wave in a similar fashion to its propagation in vacuum but

we need to assign to it a new phase velocity v, and a new wavelength, both of which depend on ε_r. In a dielectric medium of relative permittivity ε_r, the phase velocity v is given by

$$v = \frac{1}{\sqrt{\varepsilon_r \varepsilon_o \mu_o}} \tag{1.2.1}$$

Phase velocity in a medium

It is important to use the relative permittivity at the frequency of operation in Eq. (1.2.1) since ε_r depends on the frequency. Typical frequencies that are involved in optoelectronic devices are in the infrared (including far infrared), visible, and UV, and we generically refer to these frequencies as *optical frequencies*; they cover a somewhat arbitrary range from roughly 10^{12} Hz to 10^{16} Hz.

For an EM wave traveling in free space, $\varepsilon_r = 1$ and $v_{\text{vacuum}} = 1/(\varepsilon_o \mu_o)^{1/2} = c = 3 \times 10^8\,\text{m s}^{-1}$, the velocity of light in vacuum. The ratio of the speed of light in free space to its speed in a medium is called the **refractive index** n of the medium, that is,

$$n = \frac{c}{v} = \sqrt{\varepsilon_r} \tag{1.2.2}$$

Definition of refractive index

If k is the propagation constant ($k = 2\pi/\lambda$) and λ is the wavelength, both in free space, then in the medium[7] $k_{\text{medium}} = nk$ and $\lambda_{\text{medium}} = \lambda/n$. Equation (1.2.2) is in agreement with our intuition that light propagates more slowly in a denser medium that has a higher refractive index. We should note that the frequency υ (or ω) remains the same.[8]

The refractive index of a medium is not necessarily the same in all directions. In noncrystalline materials such as glasses and liquids, the material structure is the same in all directions and n does not depend on the direction. The refractive index is then **isotropic**. In crystals, however, the atomic arrangements and interatomic bonding are different along different directions. Crystals, in general, have nonisotropic, or **anisotropic**, properties. Depending on the crystal structure, the relative permittivity ε_r is different along different crystal directions. This means that, in general, the refractive index n seen by a propagating electromagnetic wave in a crystal will depend on the value of ε_r along the direction of the oscillating electric field (*i.e.*, along the direction of polarization). For example, suppose that the wave in Figure 1.1 is traveling along the z-direction in a particular crystal with its electric field oscillating along the x-direction. If the relative permittivity along this x-direction is ε_{rx}, then $n_x = (\varepsilon_{rx})^{1/2}$. The wave therefore propagates with a phase velocity that is c/n_x. The variation of n with direction of propagation and the direction of the electric field depends on the particular crystal structure. With the exception of cubic crystals (such as diamond), all crystals exhibit a degree of optical anisotropy that leads to a number of important applications as discussed in Chapter 6. Typically, noncrystalline solids, such as glasses and liquids, and cubic crystals are **optically isotropic**; they possess only one refractive index for all directions.

Relative permittivity ε_r or the dielectric constant of materials, in general, depends on the frequency of the electromagnetic wave. The relationship $n = (\varepsilon_r)^{1/2}$ between the refractive index n and ε_r must be applied at the same frequency for both n and ε_r. The relative permittivity for many materials can be vastly different at high and low frequencies because different polarization

[7] On occasions, we will need to use k_o and λ_o for the free-space propagation constant and wavelength (as in the next section) and use k and λ for those values inside the medium. In each case, these quantities will be clearly defined to avoid confusion.

[8] We are accustomed to describing light in terms of its wavelength and often quote wavelengths in nm (or Å) in the visible and IR regions. However, there would be certain advantages to using the frequency instead of the wavelength, for example, terahertz (THz) instead of nm, one of which is that the frequency v does not change in the medium. (See Roger A. Lewis, *Am. J. Phys*, *79*, 341, 2011.)

TABLE 1.1 Low-frequency (LF) relative permittivity ε_r(LF) and refractive index n

Material	ε_r(LF)	$[\varepsilon_r(\text{LF})]^{1/2}$	n (at λ)	Comment
Si	11.9	3.44	3.45 (at 2.15 μm)	Electronic bond polarization up to optical frequencies
Diamond	5.7	2.39	2.41 (at 590 nm)	Electronic bond polarization up to UV light
GaAs	13.1	3.62	3.30 (at 5 μm)	Ionic polarization contributes to ε_r(LF)
SiO_2	3.84	2.00	1.46 (at 600 nm)	Ionic polarization contributes to ε_r(LF)
Water	80	8.9	1.33 (at 600 nm)	Dipolar polarization contributes to ε_r(LF), which is large

mechanisms operate at these frequencies.[9] At low frequencies all polarization mechanisms present can contribute to ε_r, whereas at optical frequencies only the electronic polarization can respond to the oscillating field. Table 1.1 lists the relative permittivity ε_r(LF) at low frequencies (*e.g.*, 60 Hz or 1 kHz as would be measured, for example, using a capacitance bridge in the laboratory) for various materials. It then compares $\left[\varepsilon_r(\text{LF})\right]^{1/2}$ with n.

For silicon and diamond there is an excellent agreement between $\left[\varepsilon_r(\text{LF})\right]^{1/2}$ and n. Both are covalent solids in which electronic polarization (electronic bond polarization) is the only polarization mechanism at low and high frequencies. Electronic polarization involves the displacement of light electrons with respect to positive ions of the crystal. This process can readily respond to the field oscillations up to optical or even ultraviolet frequencies.

For GaAs and SiO_2 $\left[\varepsilon_r(\text{LF})\right]^{1/2}$ is larger than n because at low frequencies both of these solids possess a degree of ionic polarization. The bonding is not totally covalent and there is a degree of ionic bonding that contributes to polarization at frequencies below far-infrared wavelengths.

In the case of water, the ε_r(LF) is dominated by orientational or dipolar polarization, which is far too sluggish to respond to high-frequency oscillations of the field at optical frequencies.

It is instructive to consider what factors affect n. The relative permittivity depends on the polarizability α per molecule (or atom) in the solid. (α is defined as the induced electric dipole moment per unit applied field.) The simplest and approximate expression for the relative permittivity is

$$\varepsilon_r \approx 1 + \frac{N\alpha}{\varepsilon_o}$$

in which N is the number of molecules per unit volume. Both the atomic concentration, or density, and polarizability therefore increase n. For example, glasses of given type but with greater density tend to have higher n.

The frequency or wavelength dependence of ε_r and hence n is called the **dispersion relation**, or simply dispersion. There are various theoretical and empirical models that describe the n vs. λ behavior. The Cauchy dispersion equation in its simplest form is given by[10]

Cauchy short form dispersion equation

$$n = A + \frac{B}{\lambda^2} + \frac{C}{\lambda^4} \qquad \textbf{(1.2.3)}$$

[9] Chapters 7 and 9 in *Principles of Electronic Materials and Devices*, 3rd Edition, S. O. Kasap (McGraw-Hill, 2006) provides a semiquantitative description of the frequency dependence of ε_r and hence the wavelength dependence of n.

[10] Dispersion relations like the one in Eq. (1.2.3) are always in terms of the free-space wavelength λ. (It does not make sense to give them in terms of the actual wavelength in the medium.)

TABLE 1.2 Sellmeier and Cauchy coefficients

Sellmeier	A_1	A_2	A_3	λ_1 (μm)	λ_2 (μm)	λ_3 (μm)
SiO_2 (fused silica)	0.696749	0.408218	0.890815	0.0690660	0.115662	9.900559
86.5%SiO_2-13.5%GeO_2	0.711040	0.451885	0.704048	0.0642700	0.129408	9.425478
GeO_2	0.80686642	0.71815848	0.85416831	0.068972606	0.15396605	11.841931
Sapphire	1.023798	1.058264	5.280792	0.0614482	0.110700	17.92656
Diamond	0.3306	4.3356	–	0.1750	0.1060	–

Cauchy	Range of $h\upsilon$ (eV)	n_{-2} (eV^2)	n_0	n_2 (eV^{-2})	n_4 (eV^{-4})
Diamond	0.05–5.47	-1.07×10^{-5}	2.378	8.01×10^{-3}	1.04×10^{-4}
Silicon	0.002–1.08	-2.04×10^{-8}	3.4189	8.15×10^{-2}	1.25×10^{-2}
Germanium	0.002–0.75	-1.0×10^{-8}	4.003	2.2×10^{-1}	1.4×10^{-1}

Source: Sellmeier coefficients combined from various sources. Cauchy coefficients from D. Y. Smith *et al.*, *J. Phys. CM*, *13*, 3883, 2001.

where A, B, and C are material-specific constants. A more general Cauchy dispersion relation is of the form

Cauchy dispersion equation in photon energy

$$n = n_{-2}(h\upsilon)^{-2} + n_0 + n_2(h\upsilon)^2 + n_4(h\upsilon)^4 \tag{1.2.4}$$

where $h\upsilon$ is the photon energy, and n_0, n_{-2}, n_2, and n_4 are constants; values for diamond, Si, and Ge are listed in Table 1.2. The general Cauchy equation is usually applicable over a wide photon energy range.

Another useful dispersion relation that has been widely used, especially in optical fibers, is the Sellmeier equation given by

Sellmeier equation

$$n^2 = 1 + \frac{A_1\lambda^2}{\lambda^2 - \lambda_1^2} + \frac{A_2\lambda^2}{\lambda^2 - \lambda_2^2} + \frac{A_3\lambda^2}{\lambda^2 - \lambda_3^2} \tag{1.2.5}$$

where A_1, A_2, A_3 and λ_1, λ_2, λ_3 are constants, called **Sellmeier coefficients.**[11] Equation (1.2.5) turns out to be quite a useful semi-empirical expression for calculating n at various wavelengths if the Sellmeier coefficients are known. Higher terms involving A_4 and higher A coefficients can generally be neglected in representing n vs. λ behavior over typical wavelengths of interest. For example, for diamond, we only need the A_1 and A_2 terms. The Sellmeier coefficients are listed in various optical data handbooks.

EXAMPLE 1.2.1 Sellmeier equation and diamond

Using the Sellmeier coefficients for diamond in Table 1.2, calculate its refractive index at 610 nm (red light) and compare with the experimental quoted value of 2.415 to three decimal places.

[11] This is also known as the Sellmeier-Herzberger formula.

Solution

The Sellmeier dispersion relation for diamond is

$$n^2 = 1 + \frac{0.3306\lambda^2}{\lambda^2 - 175\,\text{nm}^2} + \frac{4.3356\lambda^2}{\lambda^2 - 106\,\text{nm}^2}$$

$$n^2 = 1 + \frac{0.3306(610\,\text{nm})^2}{(610\,\text{nm})^2 - (175\,\text{nm})^2} + \frac{4.3356(610\,\text{nm})^2}{(610\,\text{nm})^2 - (106\,\text{nm})^2} = 5.8308$$

So that

$$n = 2.4147$$

which is 2.415 to three decimal places and matches the experimental value.

EXAMPLE 1.2.2 Cauchy equation and diamond

Using the Cauchy coefficients for diamond in Table 1.2, calculate the refractive index at 610 nm.

Solution

At $\lambda = 610$ nm, the photon energy is

$$h\upsilon = \frac{hc}{\lambda} = \frac{(6.626 \times 10^{-34}\,\text{J s})(2.998 \times 10^{8}\,\text{m s}^{-1})}{(610 \times 10^{-9}\,\text{m})} \times \frac{1}{1.602 \times 10^{-19}\,\text{J eV}^{-1}} = 2.0325\,\text{eV}$$

Using the Cauchy dispersion relation for diamond with coefficients from Table 1.2,

$$\begin{aligned} n &= n_{-2}(h\upsilon)^{-2} + n_0 + n_2(h\upsilon)^2 + n_4(h\upsilon)^4 \\ &= (-1.07 \times 10^{-5})(2.0325)^{-2} + 2.378 + (8.01 \times 10^{-3})(2.0325)^2 \\ &\quad + (1.04 \times 10^{-4})(2.0325)^4 \\ &= 2.4140 \end{aligned}$$

which is slightly different than the value calculated in Example 1.2.1; one reason for the discrepancy is due to the Cauchy coefficients quoted in Table 1.2 being applicable over a wider wavelength range at the expense of some accuracy. Although both dispersion relations have four parameters, A_1, A_2, λ_1, λ_2 for Sellmeier and n_{-2}, n_0, n_2, n_4 for Cauchy, the functional forms are different.

1.3 GROUP VELOCITY AND GROUP INDEX

Since there are no perfect monochromatic waves in practice, we have to consider the way in which a group of waves differing slightly in wavelength will travel along the z-direction. Figure 1.7 shows how two perfectly harmonic waves of slight different frequencies $\omega - \delta\omega$ and $\omega + \delta\omega$ interfere to generate a periodic **wave packet** that contains an oscillating field at the mean frequency ω that is amplitude modulated by a slowly varying field of frequency $\delta\omega$. We are interested in the velocity of this wave packet. The two sinusoidal waves of frequencies $\omega - \delta\omega$ and $\omega + \delta\omega$ will propagate with propagation constants $k - \delta k$ and $k + \delta k$ respectively inside the material so that their sum will be

$$E_x(z, t) = E_o \cos\left[(\omega - \delta\omega)t - (k - \delta k)z\right] + E_o \cos\left[(\omega + \delta\omega)t - (k + \delta k)z\right]$$

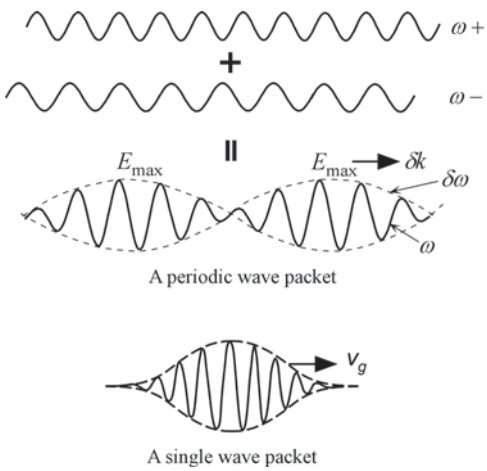


FIGURE 1.7 Two waves of slightly different wavelengths traveling in the same direction result in a wave packet that has an amplitude variation which travels at the group velocity.

By using the trigonometric identity $\cos A + \cos B = 2\cos\left[\frac{1}{2}(A - B)\right]\cos\left[\frac{1}{2}(A + B)\right]$ we arrive at

$$E_x(z, t) = 2E_o \cos\left[(\delta\omega)t - (\delta k)z\right] \cos(\omega t - kz)$$

As illustrated in Figure 1.7, this represents a sinusoidal wave of frequency ω, which is amplitude modulated by a very slowly varying sinusoid of frequency $\delta\omega$. The system of waves, that is, the modulation, travels along z at a speed determined by the modulating term, $\cos\left[(\delta\omega)t - (\delta k)z\right]$. The maximum in the field occurs when $\left[(\delta\omega)t - (\delta k)z\right] = 2m\pi = \text{constant}$ (m is an integer), which travels with a velocity

$$\frac{\delta z}{\delta t} = \frac{\delta\omega}{\delta k}$$

or[12]

$$v_g = \frac{d\omega}{dk} \qquad \textbf{(1.3.1)}$$

Group velocity

where the velocity v_g is the **group velocity** of the waves, since it determines the speed of propagation of the maximum electric field along z. The group velocity represents the speed with which energy or information is propagated since it defines the speed of the envelope of the amplitude variation. The maximum electric field in Figure 1.7 advances with a velocity v_g whereas the phase variations in the electric field propagate at the phase velocity v.

The wave packet we generated by adding two slightly different harmonic waves is periodic with a period $2\pi/\delta\omega$. By adding many such harmonic waves with slightly different frequencies but with the right amplitudes, we can generate a *single wave packet* that is nonperiodic as illustrated in Figure 1.7. This wave packet travels with a group velocity v_g.

[12] It is quite common to replace small changes in expressions like $\delta z/\delta t$ by differentials, and write it as dz/dt, which we will often do in the future.

We know that in vacuum, $\omega = ck$ and the group velocity is

Group velocity in vacuum

$$v_g(\text{vacuum}) = \frac{d\omega}{dk} = c = \text{phase velocity} \tag{1.3.2}$$

In vacuum or air, the group velocity is the same as the phase velocity.

For an EM wave in a medium, k in Eq. (1.3.2) is the propagation constant inside the medium, which can be written $k = 2\pi n/\lambda_o$ where λ_o is the free pace wavelength. The group velocity then is not necessarily the same as the phase velocity v, which depends on ω/k and is given by c/n. The group velocity v_g, on the other hand, is $d\omega/dk$, which depends on how the propagation changes in the medium, δk, with the change in frequency $\delta\omega$, and $d\omega/dk$, is not necessarily the same as ω/k when the refractive index has a wavelength dependence. Suppose that the refractive index $n = n(\lambda_o)$ is a function of (free space) wavelength λ_o, perhaps being described by one of the expressions in Section 1.2. Its gradient would be $dn/d\lambda_o$. We can easily find the group velocity, as shown in Example 1.3.1, by first finding $d\omega$ and dk in terms of dn and $d\lambda_o$, and then using Eq. (1.3.2),

Group velocity in a medium

$$v_g(\text{medium}) = \frac{d\omega}{dk} = \frac{c}{n - \lambda_o\left(\dfrac{dn}{d\lambda_o}\right)} \tag{1.3.3}$$

This can be written as

Group velocity in a medium

$$v_g(\text{medium}) = \frac{c}{N_g} \tag{1.3.4}$$

in which

Group index

$$N_g = n - \lambda_o \frac{dn}{d\lambda_o} \tag{1.3.5}$$

is defined as the **group index of the medium**. Equation (1.3.5) defines the group refractive index N_g of a medium and determines the effect of the medium on the group velocity via Eq. (1.3.4). What is important in Eqs. (1.3.4) and (1.3.5) is the gradient of the refractive index, $dn/d\lambda_o$. If the refractive index is constant and independent of the wavelength, at least over the wavelength range of interest, then $N_g = n$; and the group and phase velocities are the same.

In general, for many materials the refractive index n and hence the group index N_g depend on the wavelength of light by virtue of ε_r being frequency dependent. Then, both the phase velocity v and the group velocity v_g depend on the wavelength and the medium is called a **dispersive medium**. The refractive index n and the group index N_g of pure SiO_2 (silica) glass are important parameters in optical fiber design in optical communications. Both of these parameters depend on the wavelength of light as shown in Figure 1.8. Around 1300 nm, N_g is minimum, which means that for wavelengths close to 1300 nm, N_g is wavelength independent. Thus, light waves with wavelengths around 1300 nm travel with the same group velocity and do not experience dispersion. This phenomenon is significant in the propagation of light in optical fibers as discussed in Chapter 2.

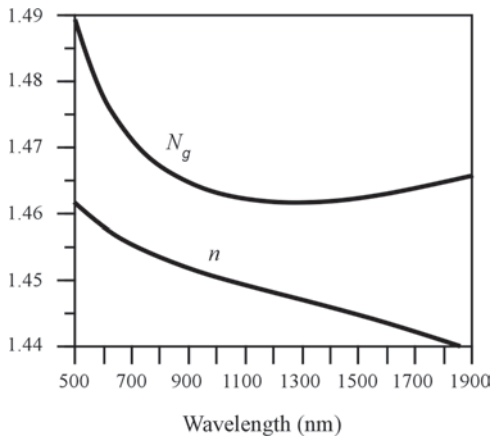


FIGURE 1.8 Refractive index n and the group index N_g of pure SiO_2 (silica) glass as a function of wavelength.

EXAMPLE 1.3.1 Group velocity

Consider two sinusoidal waves that are close in frequency, that is, waves of frequencies $\omega - \delta\omega$ and $\omega + \delta\omega$ as in Figure 1.7. Their propagation constant will be $k - \delta k$ and $k + \delta k$. The resultant wave will be

$$E_x(z, t) = E_o\cos\left[(\omega - \delta\omega)t - (k - \delta k)z\right] + E_o\cos\left[(\omega + \delta\omega)t - (k + \delta k)z\right]$$

By using the trigonometric identity $\cos A + \cos B = 2\cos\left[\frac{1}{2}(A - B)\right]\cos\left[\frac{1}{2}(A + B)\right]$ we arrive at

$$E_x(z, t) = 2E_o\cos\left[(\delta\omega)t - (\delta k)z\right]\cos(\omega t - kz)$$

As illustrated in Figure 1.7, this represents a sinusoidal wave of frequency ω, which is amplitude modulated by a very slowly varying sinusoid of frequency $\delta\omega$. The system of waves, that is, the modulation, travels along z at a speed determined by the modulating term, $\cos\left[(\delta\omega)t - (\delta k)z\right]$. The maximum in the field occurs when $\left[(\delta\omega)t - (\delta k)z\right] = 2m\pi =$ constant (m is an integer), which travels with a velocity

$$\frac{\delta z}{\delta t} = \frac{\delta\omega}{\delta k} \quad \text{or} \quad v_g = \frac{d\omega}{dk}$$

This is the group velocity of the waves, as stated in Eq. (1.3.1), since it determines the speed of propagation of the maximum electric field along z.[13]

EXAMPLE 1.3.2 Group velocity and index

Consider $\omega = 2\pi c/\lambda_o$ and $k = 2\pi n/\lambda_o$, where λ_o is the free-space wavelength. By finding expressions for $d\omega$ and dk in terms of dn and $d\lambda_o$ derive Eq. (1.3.4) for the group velocity v_g.

Solution

Differentiate $\omega = 2\pi c/\lambda_o$ to get $d\omega = -(2\pi c/\lambda_o^2)d\lambda_o$, and then differentiate $k = 2\pi n/\lambda_o$ to find

$$dk = 2\pi n(-1/\lambda_o^2)d\lambda_o + (2\pi/\lambda_o)\left(\frac{dn}{d\lambda_o}\right)d\lambda_o = -(2\pi/\lambda_o^2)\left(n - \lambda_o\frac{dn}{d\lambda_o}\right)d\lambda_o$$

[13] It is left as an exercise to show that the same result can be obtained by using a sine instead of cosine function for the waves.

We can now substitute for $d\omega$ and dk in Eq. (1.3.1),

$$v_g = \frac{d\omega}{dk} = \frac{-(2\pi c/\lambda_o^2)d\lambda_o}{-(2\pi/\lambda_o^2)\left(n - \lambda_o \dfrac{dn}{d\lambda_o}\right)d\lambda_o} = \frac{c}{n - \lambda_o \dfrac{dn}{d\lambda_o}}$$

EXAMPLE 1.3.3 Group and phase velocities

Consider a light wave traveling in a pure SiO_2 (silica) glass medium. If the wavelength of light is 1 μm and the refractive index at this wavelength is 1.450, what is the phase velocity, group index (N_g), and group velocity (v_g)?

Solution

The phase velocity is given by

$$v = c/n = (3 \times 10^8 \,\mathrm{m\,s^{-1}})/(1.450) = 2.069 \times 10^8 \,\mathrm{m\,s^{-1}}$$

From Figure 1.8, at $\lambda = 1\,\mu\mathrm{m}, N_g = 1.463$, so that

$$v_g = c/N_g = (3 \times 10^8 \,\mathrm{m\,s^{-1}})/(1.463) = 2.051 \times 10^8 \,\mathrm{m\,s^{-1}}$$

The group velocity is about ~0.9% smaller than the phase velocity.

1.4 MAGNETIC FIELD, IRRADIANCE, AND POYNTING VECTOR

Although we have considered the electric field component E_x of the electromagnetic wave, we should recall that the magnetic field (magnetic induction) component B_y always accompanies E_x in an EM wave propagation. In fact, if v is the phase velocity of an EM wave in an isotropic dielectric medium and n is the refractive index, then according to electromagnetism, at all times and anywhere in an EM wave,[14]

Fields in an EM wave

$$E_x = vB_y = \frac{c}{n} B_y \tag{1.4.1}$$

in which $v = (\varepsilon_o \varepsilon_r \mu_o)^{-1/2}$ and $n = \varepsilon_r{}^{1/2}$. Thus, the two fields are simply and intimately related for an EM wave propagating in an isotropic medium. Any process that alters E_x also intimately changes B_y in accordance with Eq. (1.4.1).

As the EM wave propagates in the direction of the wave vector **k** as shown in Figure 1.9, there is an energy flow in this direction. The wave brings with it electromagnetic energy. A small region of space in which the electric field is E_x has an energy density, that is, energy per unit volume, given by $(1/2)\varepsilon_o \varepsilon_r E_x^2$. Similarly, a region of space where the magnetic field is B_y has an energy density $(1/2\mu_o)B_y^2$. Since the two fields are related by Eq. (1.4.1), the energy densities in the E_x and B_y fields are the same.

Energy densities in an EM wave

$$\frac{1}{2} \varepsilon_o \varepsilon_r E_x^2 = \frac{1}{2\mu_o} B_y^2 \tag{1.4.2}$$

The total energy density in the wave is therefore $\varepsilon_o \varepsilon_r E_x^2$. Suppose that an ideal "energy meter" is placed in the path of the EM wave so that the receiving area A of this meter is

[14] This is actually a statement of Faraday's law for EM waves. In vector notation it is often expressed as $\omega \mathbf{B} = \mathbf{k} \times \mathbf{E}$.

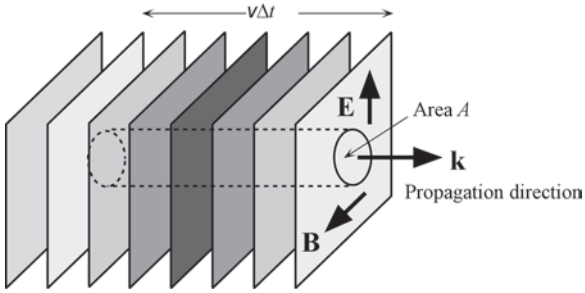


FIGURE 1.9 A plane EM wave traveling along **k** crosses an area A at right angles to the direction of propagation. In time Δt, the energy in the cylindrical volume $Av\Delta t$ (shown dashed) flows through A.

perpendicular to the direction of propagation. In a time interval Δt, a portion of the wave of spatial length $v\Delta t$ crosses A. Thus, a volume $Av\Delta t$ of the EM wave crosses A in time Δt. The energy in this volume consequently becomes received. If S is the EM power flow per unit area, then

$$S = \text{Energy flow per unit time per unit area}$$

giving

$$S = \frac{(Av\Delta t)(\varepsilon_o \varepsilon_r E_x^2)}{A\Delta t} = v\varepsilon_o \varepsilon_r E_x^2 = v^2 \varepsilon_o \varepsilon_r E_x B_y \tag{1.4.3}$$

In an isotropic medium, the energy flow is in the direction of wave propagation. If we use the vectors **E** and **B** to represent the electric and magnetic fields in the EM wave, then the wave propagates in a direction $\mathbf{E} \times \mathbf{B}$ because this direction is perpendicular to both **E** and **B**. The EM power flow per unit area in Eq. (1.4.3) can be written as,

$$\mathbf{S} = v^2 \varepsilon_o \varepsilon_r \mathbf{E} \times \mathbf{B} \tag{1.4.4}$$

Poynting vector

in which **S**, called the **Poynting vector**, represents the energy flow per unit time per unit area in a direction determined by $\mathbf{E} \times \mathbf{B}$ (direction of propagation). Its magnitude, power flow per unit area, is called the **irradiance**.[15]

The field E_x at the receiver location (say, $z = z_1$) varies sinusoidally, which means that the energy flow also varies sinusoidally. The irradiance in Eq. (1.4.3) is the **instantaneous irradiance**. If we write the field as $E_x = E_o \sin(\omega t)$ and then calculate the average irradiance by averaging S over one period we would find the **average irradiance**,

$$I = S_{\text{average}} = \frac{1}{2} v\varepsilon_o \varepsilon_r E_o^2 \tag{1.4.5}$$

Average irradiance (intensity)

Since $v = c/n$ and $\varepsilon_r = n^2$ we can write Eq. (1.4.5) as

$$I = S_{\text{average}} = \frac{1}{2} c\varepsilon_o n E_o^2 = (1.33 \times 10^{-3}) n E_o^2 \tag{1.4.6}$$

Average irradiance (intensity)

The instantaneous irradiance can be measured only if the power meter can respond more quickly than the oscillations of the electric field, and since this is in the optical frequencies range,

[15]The term *intensity* is widely used and interpreted by many engineers as power flow per unit area even though the strictly correct term is *irradiance.* Many optoelectronic data books simply use intensity to mean irradiance. By the way, the Poynting vector was named after John H. Poynting (1851–1914), an English physicist, who was a physics professor at the University of Birmingham (known as Mason Science College at the time).

all practical measurements invariably yield the average irradiance. This is because all detectors have a response rate much slower than the frequency of the wave.

The irradiance of an EM wave depends on the distance from the EM source. For an *ideal* plane wave, the irradiance at a distance r from the "infinite extended plane EM source" is independent of r. On the other hand, for a spherical wave, the irradiance drops as $1/r^2$ because the electric field drops as $1/r$. Suppose that the total power emitted by a point source placed at some point O is P_o as shown in Figure 1.10. Then the irradiance I at a distance r from O is

Irradiance of a spherical wave

$$I = \frac{P_o}{4\pi r^2} \tag{1.4.7}$$

For a Gaussian beam, the irradiance has a Gaussian distribution across the beam cross-section, as illustrated in Figure 1.5 (b) and (c), and also decreases as the beam propagates along z; the power in the beam becomes spread over larger and larger wavefront surfaces as the wave propagates along z. The irradiance I at a point z from O, and at a radial distance r from the beam axis (Figures 1.5 and 1.6) is given by

Irradiance of a Gaussian beam

$$I(z, r) = I_o\left(\frac{w_o}{w}\right)^2 \exp\left(-\frac{2r^2}{w^2}\right) \tag{1.4.8}$$

where w_o is the beam waist, w is the beam width at a distance z from O, and I_o is the maximum beam irradiance, which occurs at $z = 0$ when $w = w_o$. Since w depends on z through $2w = 2w_o\left[1 + (z/z_o)^2\right]^{1/2}$, the irradiance also depends on z and I decreases with z. At far away from the Rayleigh range, $z \gg z_o$, the irradiance on the beam axis is

Irradiance of a Gaussian beam on axis

$$I_{\text{axis}}(z) = I_o\frac{z_o^2}{z^2} \tag{1.4.9}$$

which shows that the decay of light intensity with distance is similar to that for a spherical wave. The radial dependence, of course, remains Gaussian.

The total optical power P_o is the EM power carried by a wave, and can be found by integrating the irradiance. For a Gaussian beam P_o and I_o are related by

Total power and irradiance of a Gaussian beam

$$P_o = \frac{1}{2}\left[I_o(\pi w_o^2)\right] \tag{1.4.10}$$

where it can be seen that the apparent "cross-sectional area" of the beam, πw_o^2, is used to multiply the maximum irradiance with a factor of half to yield the total optical power.

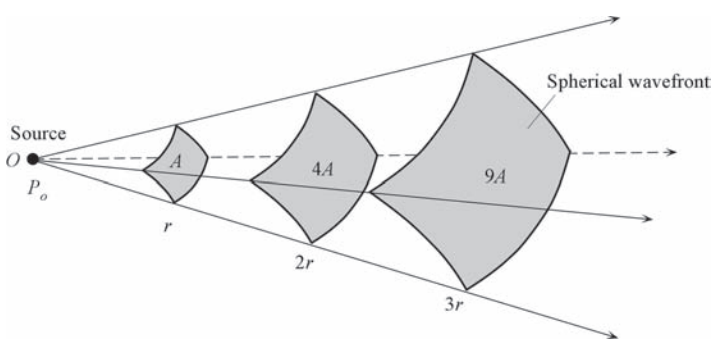


FIGURE 1.10 The irradiance of a spherical wave decreases with distance from the source because the area through which the power flows increases as r^2.

EXAMPLE 1.4.1 Electric and magnetic fields in light

The intensity (irradiance) of the red laser beam from a He-Ne laser at a certain location has been measured to be $1\,\mathrm{mW\,cm^{-2}}$. What are the magnitudes of the electric and magnetic fields? What are the magnitudes if this beam were in a glass medium with a refractive index $n = 1.45$ and the irradiance was still $1\,\mathrm{mW\,cm^{-2}}$?

Solution

Using Eq. (1.4.6) for the average irradiance, the field in air is

$$E_o = \sqrt{\frac{2I}{c\varepsilon_o n}} = \sqrt{\frac{2(1 \times 10^{-3} \times 10^4\,\mathrm{W\,m^{-2}})}{(3 \times 10^8\,\mathrm{m\,s^{-1}})(8.85 \times 10^{-12}\,\mathrm{F\,m^{-1}})(1)}}$$

so that

$$E_o = 87\,\mathrm{V\,m^{-1}} \quad \text{or} \quad 0.87\,\mathrm{V\,cm^{-1}}$$

The corresponding magnetic field is

$$B_o = E_o/c = (87\,\mathrm{V\,m^{-1}})/(3 \times 10^8\,\mathrm{m\,s^{-1}}) = 0.29\ \mu\mathrm{T}$$

If the same intensity were in a glass medium of $n = 1.45$, then

$$E_o(\text{medium}) = \sqrt{\frac{2I}{c\varepsilon_o n}} = \sqrt{\frac{2(1 \times 10^{-3} \times 10^4\,\mathrm{W\,m^{-2}})}{(3 \times 10^8\,\mathrm{m\,s^{-1}})(8.85 \times 10^{-12}\,\mathrm{F\,m^{-1}})(1.45)}}$$

or

$$E_o(\text{medium}) = 72\ \mathrm{V\,m^{-1}}$$

and

$$B_o(\text{medium}) = nE_o(\text{medium})/c = (1.45)(72\ \mathrm{V\,m^{-1}})/(3 \times 10^8\,\mathrm{m\,s^{-1}}) = 0.35\ \mu\mathrm{T}$$

EXAMPLE 1.4.2 Power and irradiance of a Gaussian beam

Consider a 5 mW He-Ne laser that is operating at 633 nm, and has a spot size of 1 mm. Find the maximum irradiance of the beam and the axial (maximum) irradiance at 25 m from the laser.

Solution

The 5 mW rating refers to the total optical power P_o available, and 633 nm is the free-space output wavelength λ. Apply Eq. (1.4.10), $P_o = (1/2)\left[I_o(\pi w_o^2)\right]$,

$$5 \times 10^{-3}\,\mathrm{W} = \frac{1}{2} I_o(\pi)\left(\frac{1}{2} \times 1 \times 10^{-3}\,\mathrm{m}\right)^2$$

which gives

$$I_o = 1.273 \times 10^4\,\mathrm{W\,m^{-2}} = 1.273\,\mathrm{W\,cm^{-2}}$$

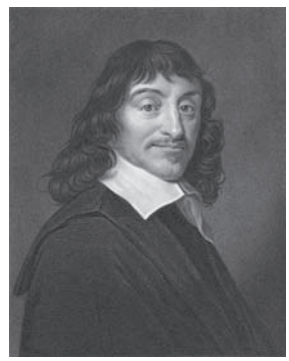
The Rayleigh range z_o was calculated previously as $z_o = \pi w_o^2/\lambda = 1.24\,\text{m}$ in Example 1.1.1. At $z = 25\,\text{m}$, the axial irradiance is

$$I_{\text{axis}} = (1.273 \times 10^4\,\text{W m}^{-2})\frac{(1.24\,\text{m})^2}{(25\,\text{m})^2} = 31.3\,\text{W m}^{-2} = 3.13\,\text{mW cm}^2$$

1.5 SNELL'S LAW AND TOTAL INTERNAL REFLECTION (TIR)



Willebrord Snellius (Willebrord Snel van Royen, 1580–1626) was a Dutch astronomer and a mathematician, who was a professor at the University of Leiden. He discovered his law of refraction in 1621 which was published by Réne Descartes in France 1637; it is not known whether Descartes knew of Snell's law or formulated it independently. *(Courtesy of AIP Emilio Segre Visual Archives, Brittle Books Collection.)*



René Descartes (1596–1650) was a French philosopher who was also involved with mathematics and sciences. He has been called the "Father of Modern Philosophy." Descartes was responsible for the development of Cartesian coordinates and analytical geometry. He also made significant contributions to optics, including reflection and refraction. *(Courtesy of Georgios Kollidas/Shutterstock.com.)*

We consider a traveling plane EM wave in a medium (1) of refractive index n_1 propagating toward a medium (2) with a refractive index n_2. Constant phase fronts are joined with broken lines and the wave vector $\mathbf{k}_i$ is perpendicular to the wavefronts as shown in Figure 1.11. When the wave reaches the plane boundary between the two media, a transmitted wave in medium 2 and a reflected wave in medium 1 appear. The transmitted wave is called the **refracted light**. The angles θ_i, θ_t, θ_r define the directions of the incident, transmitted, and reflected waves, respectively, with respect to the normal to the boundary plane as shown in Figure 1.11. The wave vectors of the reflected and transmitted waves are denoted as $\mathbf{k}_r$ and $\mathbf{k}_t$. Since both the incident and reflected waves are in the same medium, the magnitudes of $\mathbf{k}_r$ and $\mathbf{k}_i$ are the same, $k_r = k_i$.

Simple arguments based on constructive interference can be used to show that there can be only one reflected wave that occurs at an angle equal to the incidence angle. The two waves along A_i and B_i are in phase. When these waves are reflected to become waves A_r and B_r then they must still be in phase, otherwise they will interfere destructively and destroy each other. The only way the two waves can stay in phase is if $\theta_r = \theta_i$. All other angles lead to the waves A_r and B_r being out of phase and interfering destructively.

The refracted waves A_t and B_t are propagating in a medium of refracted index $n_2(<n_1)$ that is different than n_1. Hence the waves A_t and B_t have different velocities than A_i and B_i. We consider what happens to a wavefront such as AB, corresponding perhaps to the maximum field, as it propagates from medium 1 to 2. We recall that the points A and B on this front are always in phase. During the time it takes for the phase B on wave B_i to reach B', phase A on wave A_t has

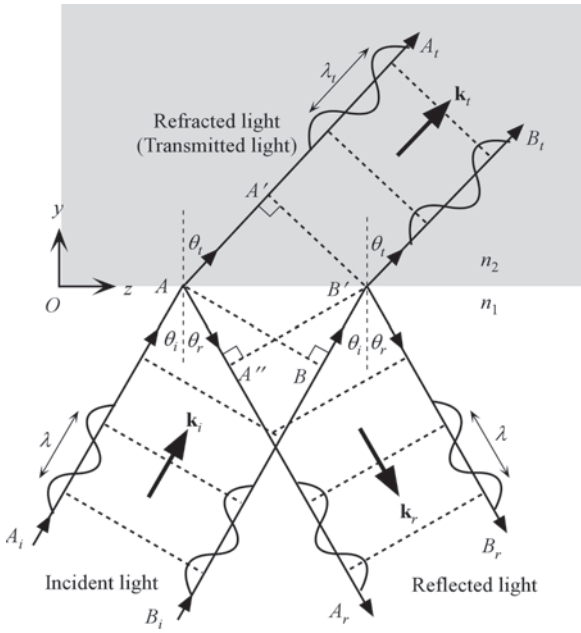


FIGURE 1.11 A light wave traveling in a medium with a greater refractive index ($n_1 > n_2$) suffers reflection and refraction at the boundary. (Notice that λ_t is slightly longer than λ.)

progressed to A'. The wavefront AB thus becomes the front $A'B'$ in medium 2. Unless the two waves at A' and B' still have the same phase, there will be no transmitted wave. A' and B' points on the front are in phase only for one particular transmitted angle, θ_t.

If it takes time t for the phase at B on wave B_i to reach B', then $BB' = v_1 t = ct/n_1$. During this time t, the phase A has progressed to A' where $AA' = v_2 t = ct/n_2$. A' and B' belong to the same front just like A and B so that AB is perpendicular to $\mathbf{k}_i$ in medium 1 and $A'B'$ is perpendicular to $\mathbf{k}_t$ in medium 2. From geometrical considerations, $AB' = BB'/\sin\theta_i$ and $AB' = AA'/\sin\theta_t$ so that

$$AB' = \frac{v_1 t}{\sin\theta_i} = \frac{v_2 t}{\sin\theta_t}$$

or

$$\frac{\sin\theta_i}{\sin\theta_t} = \frac{v_1}{v_2} = \frac{n_2}{n_1} \qquad \textbf{(1.5.1)}$$

Snell's Law

This is **Snell's law**,[16] which relates the angles of incidence and refraction to the refractive indices of the media.

If we consider the reflected wave, the wavefront AB becomes $A''B'$ in the reflected wave. In time t, phase B moves to B' and A moves to A''. Since they must still be in phase to constitute the reflected wave, BB' must be equal to AA''. Suppose it takes time t for the wavefront B to move to B' (or A to A''). Then, since $BB' = AA'' = v_1 t$, from geometrical considerations,

$$AB' = \frac{v_1 t}{\sin\theta_i} = \frac{v_1 t}{\sin\theta_r}$$

so that $\theta_i = \theta_r$. Angles of incidence and reflection are the same.

[16] Snell's law is known as Descartes's law in France as he was the first to publish it in his "Discourse on Method" in 1637.

When $n_1 > n_2$ then obviously the transmitted angle is greater than the incidence angle as apparent in Figure 1.11. When the refraction angle θ_t reaches 90°, the incidence angle is called the **critical angle** θ_c, which is given by

Total internal reflection (TIR)

$$\sin\theta_c = \frac{n_2}{n_1} \tag{1.5.2}$$

When the incidence angle θ_i exceeds θ_c then there is no transmitted wave but only a reflected wave. The latter phenomenon is called **total internal reflection (TIR)**. The effect of increasing the incidence angle is shown in Figure 1.12. It is the TIR phenomenon that leads to the propagation of waves in a dielectric medium surrounded by a medium of smaller refractive index as shown in Chapter 2. Although Snell's law for $\theta_i > \theta_c$ shows that $\sin\theta_t > 1$ and hence θ_t is an "imaginary" angle of refraction, there is however a wave called the evanescent wave, whose amplitude decays exponentially with distance into the second medium as discussed below. The wave exists only in the interface region from which the reflected wave emerges (not outside).

Snell's law can also be viewed as the **k**-vector of light parallel to the interface being continuous through the interface, that is, having the same value on both sides of the interface. In medium n_1, $\mathbf{k}_i$ parallel to the interface is $k_i \sin\theta_i$ or $kn_1 \sin\theta_i$, where $k_i = kn_1$, and k is the magnitude of the wave vector in free space. In medium n_2, $\mathbf{k}_t$ parallel to the interface is $k_t \sin\theta_t$ or $kn_2 \sin\theta_t$. If **k**'s component tangential to the interface remains constant, $kn_1 \sin\theta_i = kn_2 \sin\theta_t$, then we obtain Snell's law in Eq. (1.5.1). Put differently, Snell's law is equivalent to

Snell's Law

$$n\sin\theta = \text{constant through an interface between different media} \tag{1.5.3}$$

Snell's law of refraction and TIR play a very important role in many optoelectronic and photonic devices. A prism is a transparent optical component that can deflect a light beam as illustrated in Figure 1.13. There are two basic types of prism. In a **refracting prism**, the light deflection is caused by refractions whereas in a **reflecting prism** it is caused by one or more TIRs. (Some prisms such as composite prisms need both refraction and TIR to achieve their desired deflection.) The deflection δ depends not only on the incidence angle of the light beam on the prism, the prism material (n), and geometry, but also on the wavelength and the polarization state of the incident light. The reason is that the refractive index n of the prism material normally depends on the wavelength, and further, for certain materials (*e.g.*, quartz, calcite), it depends on the polarization state (direction of the electric field) of light as well.

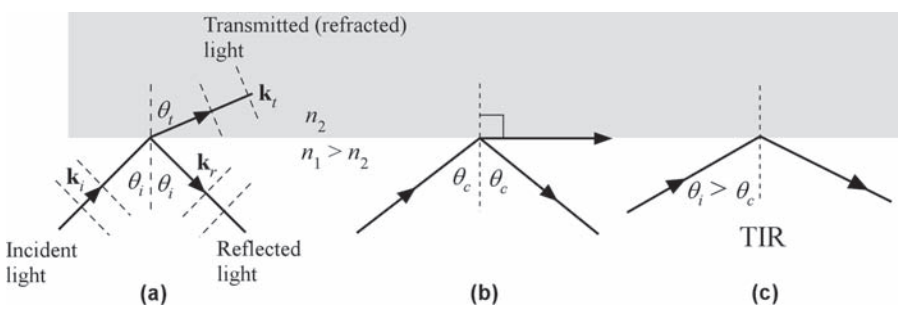


FIGURE 1.12 Light wave travelling in a more dense medium strikes a less dense medium. Depending on the incidence angle with respect to θ_c, which is determined by the ratio of the refractive indices, the wave may be transmitted (refracted) or reflected. (a) $\theta_i < \theta_c$ (b) $\theta_i = \theta_c$ (c) $\theta_i > \theta_c$ and total internal reflection. (Wavefronts are only indicated in (a).)

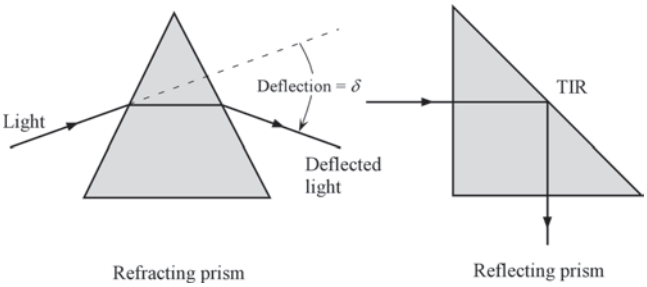


FIGURE 1.13 Basic types of prism: refracting and reflecting prisms.

EXAMPLE 1.5.1 Beam displacement

Lateral displacement of light, or **beam displacement**, occurs when a beam of light passes obliquely through a plate of transparent material, such as a glass plate. When a light beam is incident on a plate of transparent material of refractive index n, it emerges from the other side traveling parallel to the incident light but displaced from it by a distance d, called *lateral displacement*, as illustrated in Figure 1.14. Find the displacement d in terms of the incidence angle the plate thickness. What is d for a glass of $n = 1.600$, $d = 10$ mm if the incidence angle is 45°?

Solution

The displacement $d = BC = AB\sin(\theta_i - \theta_t)$. Further, $L/AB = \cos\theta_t$ so that combining these two equation we find

$$d = L\left[\frac{\sin(\theta_i - \theta_t)}{\cos\theta_t}\right]$$

We can expand $\sin(\theta_i - \theta_t) = \sin\theta_i\cos\theta_t - \cos\theta_i\sin\theta_t$, use $\cos\theta_t = \sqrt{1 - \sin^2\theta_t}$ and then apply Snell's law $n\sin\theta_t = n_o\sin\theta_i$ at the top surface to find

$$\frac{d}{L} = \sin\theta_i\left[1 - \frac{\cos\theta_i}{\sqrt{(n/n_o)^2 - \sin^2\theta_i}}\right]$$

which is maximum with $d = L$ when $\theta_i \simeq 90°$, glazing incidence. Substituting $n = 1.600, n_o = 1, \theta_i = 45°$, and $L = 10$ mm, we find, $d = 3.587$ mm. If the refractive index increases by 1%, $n = 1.616$, then $d = 3.630$ and the change in d is 0.043 mm or 43 μm, which can be measured electronically by using, for example, CCD or CMOS photodiode arrays.

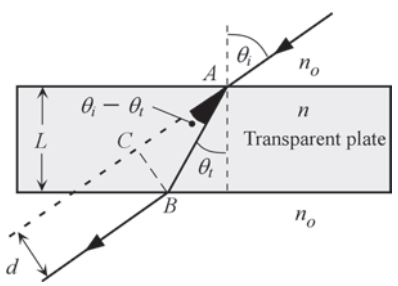


FIGURE 1.14 Lateral displacement of light passing obliquely through a transparent plate.

1.6 FRESNEL'S EQUATIONS

A. Amplitude Reflection and Transmission Coefficients (*r* and *t*)

Although the ray picture with constant phase wavefronts is useful in understanding refraction and reflection, to obtain the magnitude of the reflected and refracted waves and their relative phases, we need to consider the electric field in the light wave. The electric field in the wave must be perpendicular to the direction of propagation as shown in Figure 1.15. We can resolve the field E_i of the incident wave into two components, one in the plane of incidence, $E_{i,//}$, and the other perpendicular to the plane of incidence, $E_{i,\perp}$. The plane of incidence is defined as the plane containing the incident and the reflected rays, which in Figure 1.15 corresponds to the plane of the paper.[17] Similarly for both the reflected and transmitted waves, we will have field components parallel and perpendicular to the plane of incidence, that is, $E_{r,//}$, $E_{r,\perp}$, and $E_{t,//}$, $E_{t,\perp}$.

As apparent from Figure 1.15, the incident, transmitted, and reflected wave all have a wave vector component along the z-direction; that is, they have an effective velocity along z. The fields $E_{i,\perp}$, $E_{r,\perp}$, and $E_{t,\perp}$ are all perpendicular to the z-direction. These waves are called **transverse electric field** (TE) waves. On the other hand, waves with $E_{i,//}$, $E_{r,//}$, and $E_{t,//}$ have only their magnetic field components perpendicular to the z-direction, and these are called **transverse magnetic field** (TM) waves.

We will describe the incident, reflected, and refracted waves each by the exponential representation of a traveling wave:

Incident wave

$$E_i = E_{io}\exp j(\omega t - \mathbf{k_i}\cdot\mathbf{r}) \tag{1.6.1}$$

Reflected wave

$$E_r = E_{ro}\exp j(\omega t - \mathbf{k_r}\cdot\mathbf{r}) \tag{1.6.2}$$

Transmitted wave

$$E_t = E_{to}\exp j(\omega t - \mathbf{k_t}\cdot\mathbf{r}) \tag{1.6.3}$$

in which $\mathbf{r}$ is the position vector, the wave vectors $\mathbf{k_i}$, $\mathbf{k_r}$, and $\mathbf{k_t}$ describe the directions of the incident, reflected, and transmitted waves, and E_{io}, E_{ro}, and E_{to} are the respective amplitudes. Any phase changes such as ϕ_r and ϕ_t in the reflected and transmitted waves with respect to the phase of the incident wave are incorporated into the complex amplitudes, E_{ro} and E_{to}. Our objective is to find E_{ro} and E_{to} with respect to E_{io}.

We should note that similar equations can be stated for the magnetic field components in the incident, reflected, and transmitted waves but these will be perpendicular to the corresponding electric fields. The electric and magnetic fields anywhere on the wave must be perpendicular to each other as a requirement of electromagnetic wave theory. This means that with $E_{//}$ in the EM wave we have a magnetic field $B_\perp$ associated with it such that $B_\perp = (n/c)E_{//}$. Similarly, $E_\perp$ will have a magnetic field $B_{//}$ associated with it such that $B_{//} = (n/c)E_\perp$.

There are two useful fundamental rules in electromagnetism that govern the behavior of the electric and magnetic fields at a boundary between two dielectric media, which we can arbitrarily label as 1 and 2. These rules are called **boundary conditions**. The first states that the

[17] The definitions of the field components follow those of S. G. Lipson *et al.*, *Optical Physics*, 3rd Edition (Cambridge University Press, 1995) and Grant Fowles, *Introduction to Modern Optics*, 2nd Edition (Dover Publications, Inc., 1975), whose clear treatments of this subject are highly recommended. The majority of authors use a different convention, which leads to different signs later in the equations ("they [Fresnel's equations] must be related to the specific electric field directions from which they are derived." Eugene Hecht, *Optics*, 4th Edition (Addison Wesley, Pearson Education, 2002), p. 115.

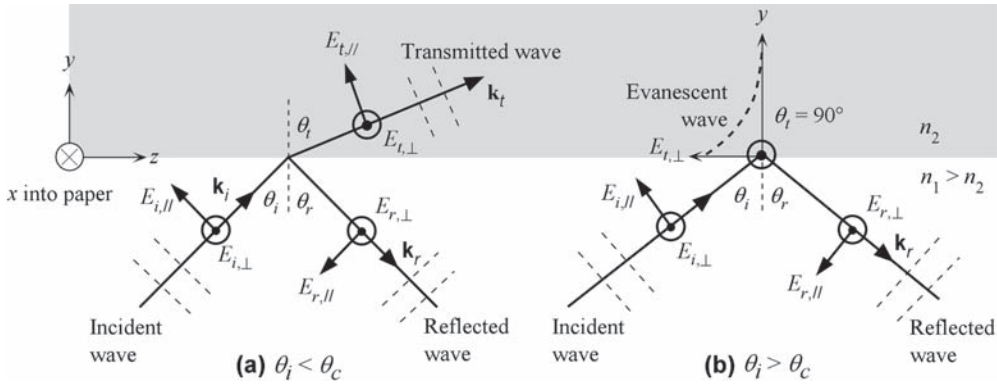


FIGURE 1.15 Light wave traveling in a more dense medium strikes a less dense medium. The plane of incidence is the plane of the paper and is perpendicular to the flat interface between the two media. The electric field is normal to the direction of propagation. It can be resolved into perpendicular ($\perp$) and parallel (//) components. (a) $\theta_i < \theta_c$ then some of the wave is transmitted into the less dense medium. Some of the wave is reflected. (b) $\theta_i > \theta_c$ then the incident wave suffers total internal reflection. There is a decaying evanescent wave into the n_2-medium.

electric field that is tangential to the boundary surface, $E_{\text{tangential}}$, must be continuous across the boundary from medium 1 to 2, that is, at the boundary, $y = 0$ in Figure 1.15,

$$E_{\text{tangential}}(1) = E_{\text{tangential}}(2) \tag{1.6.4}$$

Boundary condition

The second rule is that the tangential component of the magnetic field, $B_{\text{tangential}}$, to the boundary must likewise be continuous from medium 1 to 2, provided that the two media are non-magnetic (relative permeability, $\mu_r = 1$),

$$B_{\text{tangential}}(1) = B_{\text{tangential}}(2) \tag{1.6.5}$$

Boundary condition

Using the boundary conditions above for the fields at $y = 0$ and the relationship between the electric and magnetic fields, we can find the reflected and transmitted waves in terms of the incident wave. The boundary conditions can be satisfied only if the reflection and incidence angles are equal, $\theta_r = \theta_i$, and the angles for the transmitted and incident wave obey Snell's law, $n_1 \sin\theta_1 = n_2 \sin\theta_2$.

Applying the boundary conditions above to the EM wave going from medium 1 to 2, the amplitudes of the reflected and transmitted waves can be readily obtained in terms of n_1, n_2, and the incidence angle θ_i alone.[18] These relationships are called **Fresnel's equations**. If we define $n = n_2/n_1$ as the relative refractive index of medium 2 to that of 1, then the **reflection and transmission coefficients** for $E_\perp$ are

$$r_\perp = \frac{E_{ro,\perp}}{E_{io,\perp}} = \frac{\cos\theta_i - \left[n^2 - \sin^2\theta_i\right]^{1/2}}{\cos\theta_i + \left[n^2 - \sin^2\theta_i\right]^{1/2}} \tag{1.6.6a}$$

Reflection coefficient

[18] These equations are readily available in any electromagnetism textbook. Their derivation from the two boundary conditions involves extensive algebraic manipulation, which we will not carry out here. The electric and magnetic field components on both sides of the boundary are resolved tangentially to the boundary surface and the boundary conditions are then applied. We then use such relations as $\cos\theta_t = \left[1 - \sin\theta_t\right]^{1/2}$ and $\sin\theta_t$ is determined by Snell's law, *etc.*

and

Transmission coefficient

$$t_\perp = \frac{E_{to,\perp}}{E_{io,\perp}} = \frac{2\cos\theta_i}{\cos\theta_i + \left[n^2 - \sin^2\theta_i\right]^{1/2}} \tag{1.6.6b}$$

There are corresponding coefficients for the $E_{//}$ fields with corresponding **reflection and transmission coefficients**, $r_{//}$ and $t_{//}$.

Reflection coefficient

$$r_{//} = \frac{E_{ro,//}}{E_{io,//}} = \frac{\left[n^2 - \sin^2\theta_i\right]^{1/2} - n^2\cos\theta_i}{\left[n^2 - \sin^2\theta_i\right]^{1/2} + n^2\cos\theta_i} \tag{1.6.7a}$$

Transmission coefficient

$$t_{//} = \frac{E_{to,//}}{E_{io,//}} = \frac{2n\cos\theta_i}{n^2\cos\theta_i + \left[n^2 - \sin^2\theta_i\right]^{1/2}} \tag{1.6.7b}$$

Further, the coefficients above are related by

Transmission coefficient

$$r_{//} + nt_{//} = 1 \quad \text{and} \quad r_\perp + 1 = t_\perp \tag{1.6.8}$$

The significance of these equations is that they allow the amplitudes and phases of the reflected and transmitted waves to be determined from the coefficients $r_\perp$, $r_{//}$, $t_{//}$, and $t_\perp$. For convenience we take E_{io} to be a real number so that phase angles of $r_\perp$ and $t_\perp$ correspond to the **phase changes** measured with respect to the incident wave. For example, if $r_\perp$ is a complex quantity then we can write this as $r_\perp = |r_\perp|\exp(-j\phi_\perp)$ in which $|r_\perp|$ and $\phi_\perp$ represent the relative amplitude and phase of the reflected wave with respect to the incident wave for the field perpendicular to the plane of incidence. Of course, when $r_\perp$ is a real quantity, then a positive number represents no phase shift and a negative number is a phase shift of 180° (or π). As with all waves, a negative sign corresponds to a 180° phase shift. Complex coefficients can be obtained only from Fresnel's equations if the terms under the square roots become negative and this can happen only when $n < 1$ (or $n_1 > n_2$), and also when $\theta_i > \theta_c$, the critical angle. Thus, phase changes other than 0 or 180° occur only when there is total internal reflection. Fresnel's equations for normal incidence are greatly simplified. Putting $\theta_i = 0$ into Eqs. (1.6.6) and (1.6.7) we find

Normal incidence

$$r_{//} = r_\perp = \frac{n_1 - n_2}{n_1 + n_2} \quad \text{and} \quad t_{//} = t_\perp = \frac{2n_1}{n_1 + n_2} \tag{1.6.9}$$

Figure 1.16 (a) shows how the magnitudes of the reflection coefficients, $|r_\perp|$ and $|r_{//}|$, vary with the incidence angle θ_i for a light wave traveling from a more dense medium, $n_1 = 1.44$, to a less dense medium, $n_2 = 1.00$, as predicted by Fresnel's equations. Figure 1.16 (b) shows the changes in the phase of the reflected wave, $\phi_\perp$ and $\phi_{//}$, with θ_i. The critical angle θ_c as determined from $\sin\theta_c = n_2/n_1$ in this case is 44°. It is clear that for incidence close to normal (small θ_i), there is no phase change in the reflected wave. The reflection coefficient in Eq. (1.6.9) is a positive quantity for $n_1 > n_2$, which means that the reflected wave suffers no phase change. This is confirmed by $\phi_\perp$ and $\phi_{//}$ in Figure 1.16 (b). As the incidence angle increases, eventually $r_{//}$ becomes zero at an angle of about 35°. We can find this special incidence angle, labeled as θ_p, by solving the Fresnel equation (1.6.7a) for $r_{//} = 0$. The field in the reflected wave is then always perpendicular to the plane of incidence and hence well-defined as illustrated in

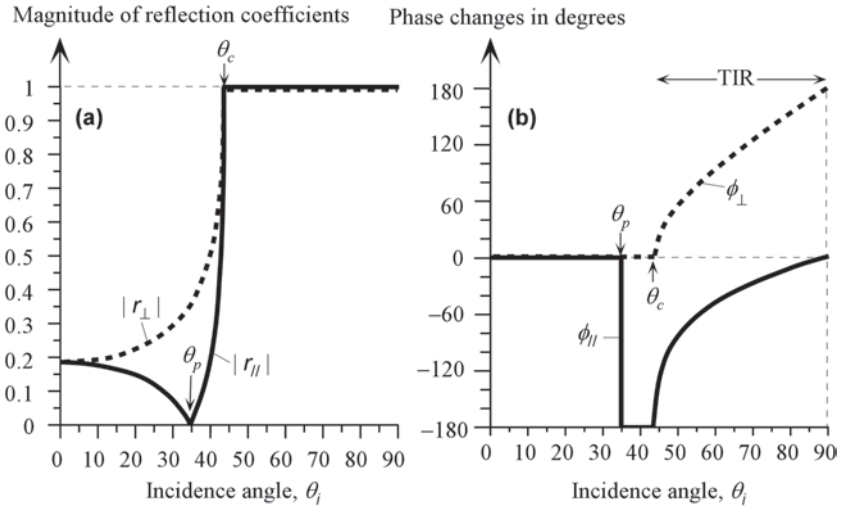


FIGURE 1.16 Internal reflection: (a) Magnitude of the reflection coefficients $r_{//}$ and $r_\perp$ vs. angle of incidence θ_i for $n_1 = 1.44$ and $n_2 = 1.00$. The critical angle is 44°. (b) The corresponding phase changes $\phi_{//}$ and $\phi_\perp$ vs. incidence angle θ_i.

Figure 1.17. This special angle is called the **polarization angle** or **Brewster's**[19] **angle** and from Eq. (1.6.7a) is given by

$$\tan\theta_p = \frac{n_2}{n_1} \tag{1.6.10}$$

Brewster's polarization angle

The reflected wave is then said to be **linearly polarized** because it contains *electric field oscillations that are contained within a well-defined plane*, which is perpendicular to the plane of incidence and also to the direction of propagation. Electric field oscillations in **unpolarized light**, on the other hand, can be in any one of infinite number of directions that are perpendicular

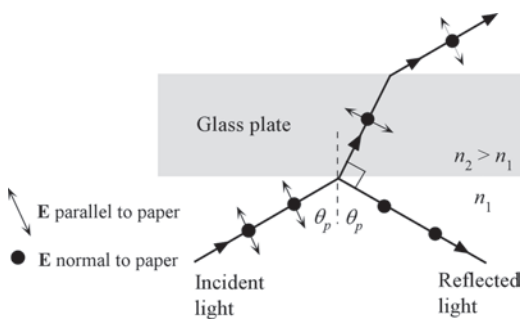


FIGURE 1.17 At the Brewster angle of incidence $\theta_i = \theta_p$, the reflected light contains only field oscillations normal to the plane of incidence (paper).

[19] After Sir David Brewster (1781–1868), a Scottish physicist who was educated in theology at the University of Edinburgh in Scotland. He became interested in the polarization properties of light from 1799 onwards, and reported some of his experiments in scientific journals, including the *Philosophical Transactions of London.*

to the direction of propagation. In linearly polarized light, however, the field oscillations are contained within a well-defined plane. Light emitted from many light sources, such as a tungsten light bulb or an LED diode, is unpolarized.[20] Unpolarized light can be viewed as a stream or collection of EM waves whose fields are randomly oriented in a direction that is perpendicular to the direction of light propagation.

For incidence angles greater than θ_p but smaller than θ_c, Fresnel's equation (1.6.7a) gives a negative number for $r_{//}$, which indicates a phase shift of 180° as shown in $\phi_{//}$ in Figure 1.16 (b). The magnitude of both $r_{//}$ and $r_{\perp}$ increases with θ_i as illustrated in Figure 1.16 (a). At the critical angle and beyond (past 44° in Figure 1.16), that is, when $\theta_i \geq \theta_c$, the magnitudes of both $r_{//}$ and $r_{\perp}$ go to unity so that the reflected wave has the same amplitude as the incident wave. The incident wave has suffered **total internal reflection**, TIR. When $\theta_i > \theta_c$, in the presence of TIR, both Eqs. (1.6.6) and (1.6.7) are complex quantities because then $\sin\theta_i > n$ and the terms under the square roots become negative. The reflection coefficients become complex quantities of the type $r_{\perp} = 1 \cdot \exp(-j\phi_{\perp})$ and $r_{//} = 1 \cdot \exp(-j\phi_{//})$ with the phase angles $\phi_{\perp}$ and $\phi_{//}$ being other than zero or 180°. The reflected wave therefore suffers phase changes, $\phi_{\perp}$ and $\phi_{//}$, in the components $E_{\perp}$ and $E_{//}$. These phase changes depend on the incidence angle, as illustrated in Figure 1.16 (b), and on n_1 and n_2.

Examination of Eq. (1.6.6) for $r_{\perp}$ shows that for $\theta_i > \theta_c$, we have $|r_{\perp}| = 1$, but the phase change $\phi_{\perp}$ is given by

Phase change in TIR

$$\tan\left(\tfrac{1}{2}\phi_{\perp}\right) = \frac{\left[\sin^2\theta_i - n^2\right]^{1/2}}{\cos\theta_i} \tag{1.6.11}$$

For the $E_{//}$ component, the phase change $\phi_{//}$ is given by

Phase change in TIR

$$\tan\left(\tfrac{1}{2}\phi_{//} + \tfrac{1}{2}\pi\right) = \frac{\left[\sin^2\theta_i - n^2\right]^{1/2}}{n^2\cos\theta_i} \tag{1.6.12}$$

We can summarize that in internal reflection ($n_1 > n_2$), the amplitude of the reflected wave from TIR is equal to the amplitude of the incident wave but its phase has shifted by an amount determined by Eqs. (1.6.11) and (1.6.12). The fact that $\phi_{//}$ has an additional π shift that makes $\phi_{//}$ negative for $\theta_i > \theta_c$ is due to the choice for the direction of the reflected optical field $E_{r,//}$ in Figure 1.15. This π shift can be ignored if we by simply invert $E_{r,//}$. (In many books Eq. (1.6.12) is written without the π-shift.)

The reflection coefficients in Figure 1.16 considered the case in which $n_1 > n_2$. When light approaches the boundary from the higher index side, that is $n_1 > n_2$, the reflection is said to be **internal reflection** and *at normal incidence there is no phase change*. On the other hand, if light approaches the boundary from the lower index side, that is $n_1 < n_2$, then it is called **external reflection**. Thus in external reflection light becomes reflected by the surface of an optically denser (higher refractive index) medium. There is an important difference between the two. Figure 1.18 shows how the reflection coefficients $r_{\perp}$ and $r_{//}$ depend on the incidence angle θ_i for external reflection ($n_1 = 1$ and $n_2 = 1.44$). At normal incidence, both coefficients are negative, which means that *in external reflection at normal incidence there is a phase shift of 180°*. Further, $r_{//}$ goes through zero at the **Brewster angle** θ_p given by Eq. (1.6.10). At this

[20] Even light from a tungsten light bulb or an LED has some polarization but this is usually negligibly small.

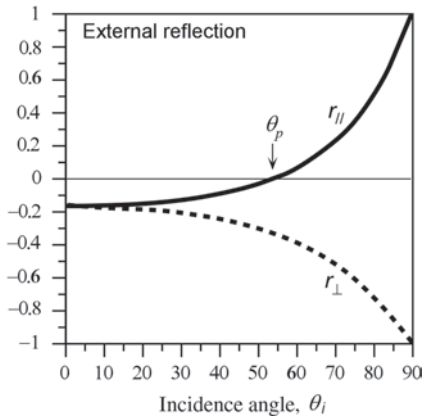


FIGURE 1.18 External reflection coefficients $r_{//}$ and $r_\perp$ vs. angle of incidence θ_i for $n_1 = 1.00$ and $n_2 = 1.44$.

angle of incidence, the reflected wave is polarized in the $E_\perp$ component only. Transmitted light in both internal reflection (when $\theta_i < \theta_c$) and external reflection does not experience a phase shift.

What happens to the transmitted wave when $\theta_i > \theta_c$? According to the boundary conditions, there must still be an electric field in medium 2; otherwise, the boundary conditions cannot be satisfied. When $\theta_i > \theta_c$, the field in medium 2 is a wave that travels near the surface of the boundary along the z direction as shown in Figure 1.19. The wave is called an **evanescent wave** and advances along z with its field decreasing as we move into medium 2; that is,

$$E_{t,\perp}(y, z, t) \propto e^{-\alpha_2 y} \exp j(\omega t - k_{iz} z) \tag{1.6.13}$$

Evanescent wave

in which $k_{iz} = k_i \sin\theta_i$ is the wave vector of the incident wave along the z-axis, and α_2 is an **attenuation coefficient** for the electric field penetrating into medium 2,[21]

$$\alpha_2 = \frac{2\pi n_2}{\lambda_o}\left[\left(\frac{n_1}{n_2}\right)^2 \sin^2\theta_i - 1\right]^{1/2} \tag{1.6.14}$$

Attenuation of evanescent wave

in which λ is the free-space wavelength. According to Eq. (1.6.13), the evanescent wave travels along z and has an amplitude that decays exponentially as we move from the boundary into medium 2 (along y). The field of the evanescent wave is e^{-1} in medium 2 when $y = 1/\alpha_2 = \delta$, which is called the **penetration depth**. It is not difficult to show that the evanescent wave is correctly predicted by Snell's law when $\theta_i > \theta_c$. The evanescent wave propagates along the boundary (along z) with the same speed as the z-component velocity of the incident and reflected waves. In Eqs. (1.6.1) and (1.6.2) we had assumed that the incident and reflected waves were *plane waves*, that is, of *infinite exten*t. If we were to extend the plane wavefronts on the reflected wave, these would cut the boundary as shown in Figure 1.19. The evanescent wave traveling along z can be thought of arising from these plane wavefronts at the boundary as in Figure 1.19. (Evanescent wave is important in light propagation in optical waveguides such as optical fibers.) If the incident wave is a narrow

[21] Normally the term *attenuation coefficient* refers to the attenuation of the irradiance but in this case it refers to the electric field.

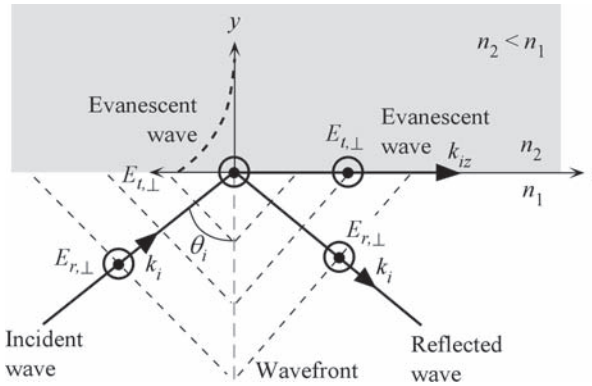


FIGURE 1.19 When $\theta_i > \theta_c$ for a *plane wave* that is reflected, there is an evanescent wave at the boundary whose magnitude decays into the n_2-medium.

beam of light (*e.g.*, from a laser pointer) then the reflected beam would have the same cross-section. There would still be an evanescent wave at the boundary, but it would exist only within the cross-sectional area of the reflected beam at the boundary.

B. Intensity, Reflectance, and Transmittance

It is frequently necessary to calculate the intensity or irradiance[22] of the reflected and transmitted waves when light traveling in a medium of index n_1 is incident at a boundary where the refractive index changes to n_2. In some cases we are simply interested in normal incidence where $\theta_i = 0°$. For example, in laser diodes light is reflected from the ends of an optical cavity where there is a change in the refractive index.

For a light wave traveling with a velocity v in a medium with relative permittivity ε_r, the light intensity I is defined in terms of the electric field amplitude E_o as

Light intensity or irradiance

$$I = \frac{1}{2} v \varepsilon_r \varepsilon_o E_o^2 \tag{1.6.15}$$

Here $\frac{1}{2}\varepsilon_r\varepsilon_o E_o^2$ represents the energy in the field per unit volume. When multiplied by the velocity v it gives the rate at which energy is transferred through a unit area. Since $v = c/n$ and $\varepsilon_r = n^2$ the intensity is proportional to nE_o^2.

Reflectance R measures the intensity of the reflected light with respect to that of the incident light and can be defined separately for electric field components parallel and perpendicular to the plane of incidence. The reflectances $R_\perp$ and $R_{//}$ are defined by

Reflectances

$$R_\perp = \frac{|E_{ro,\perp}|^2}{|E_{io,\perp}|^2} = |r_\perp|^2 \quad \text{and} \quad R_{//} = \frac{|E_{ro,//}|^2}{|E_{io,//}|^2} = |r_{//}|^2 \tag{1.6.16}$$

Although the reflection coefficients can be complex numbers that can represent phase changes, reflectances are necessarily real numbers representing intensity changes. Magnitude of a complex number is defined in terms of its product with its complex conjugate. For example, when $E_{ro,//}$ is a complex number then

$$|E_{ro,//}|^2 = (E_{ro,//})(E_{ro,//})^*$$

in which $(E_{ro,//})^*$ is the complex conjugate of $(E_{ro,//})$.

[22] Strictly the terms intensity and irradiance are not the same as mentioned in footnote 15.

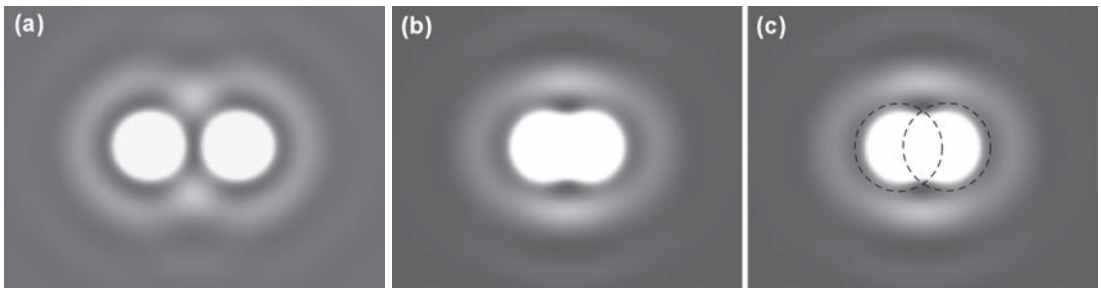


Image of two-point sources captured through a small circular aperture. (a) The two points are fully resolved since the diffraction patterns of the two sources are sufficiently separated. (b) The two images are near the Rayleigh limit of resolution. (c) The first dark ring of one image passes through the center of the bright Airy disk of the other. (Approximate.)

B. Diffraction Grating

A **diffraction grating** in its simplest form is an optical device that has a periodic series of slits in an opaque screen as shown in Figure 1.39 (a). An incident beam of light is diffracted in certain well-defined directions that depend on the wavelength λ and the grating properties. Figure 1.39 (b) shows a typical intensity pattern in the diffracted beam for a finite number of slits. There are "strong beams of diffracted light" along certain directions (θ) and these are labeled according to their occurrence: zero-order (center), first-order, either side of the zero-order, and so on. If there are an infinite number of slits then the diffracted beams have the same intensity. In reality, any periodic variation in the refractive index would serve as a diffraction grating and we will discuss other types later. As in Fraunhofer diffraction we will assume that the observation screen is far away, or that a lens is used to focus the diffracted parallel rays on to the screen (the lens in the observer's eye does it naturally).

We will assume that the incident beam is a plane wave so that the slits become coherent (synchronous) sources. Suppose that the width a of each slit is much smaller than the separation

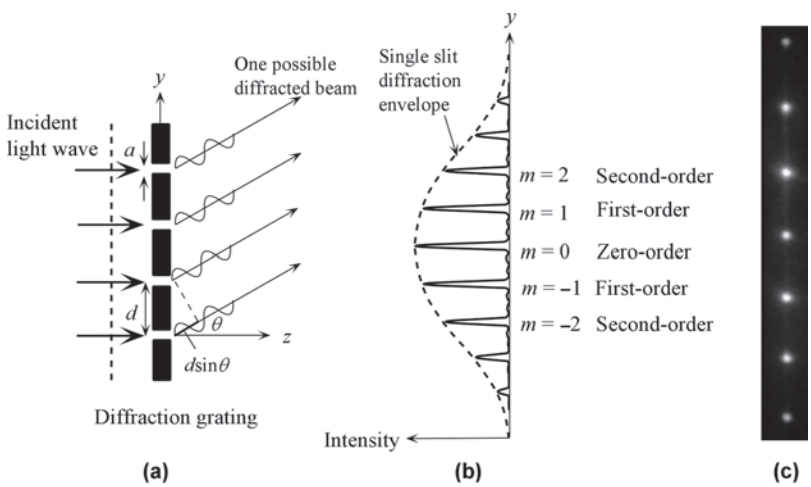


FIGURE 1.39 (a) A diffraction grating with N slits in an opaque screen. Slit periodicity is d and slit width is a; $a \ll d$. (b) The diffracted light pattern. There are distinct, that is diffracted, beams in certain directions (schematic). (c) Diffraction pattern obtained by shining a beam from a red laser pointer onto a diffraction grating. The finite size of the laser beam results in the dot pattern. (The wavelength was 670 nm, red, and the grating has 200 lines per inch.)

d of the slits as shown in Figure 1.39 (a). Waves emanating at an angle θ from two neighboring slits are out of phase by an amount that corresponds to an optical path difference $d\sin\theta$. Obviously, all such waves from pairs of slits will interfere constructively when this is a multiple of the whole wavelength

$$d\sin\theta = m\lambda; \quad m = 0, \pm 1, \pm 2, \ldots \tag{1.12.8}$$

Grating equation

which is the well-know **grating equation**, also known as the **Bragg**[37] **diffraction condition**. The value of m defines the diffraction order; $m = 0$ being zero-order, $m = \pm 1$ being first-order, *etc.* If the grating in Figure 1.39 (a) is in a medium of refractive index n, that is, the incident and diffracted beams are all in the same medium of index n, then we should use λ/n for the wavelength in Eq. (1.12.8), where λ is the free-space wavelength, that is, $d\sin\theta = m\lambda/n$.

The problem of determining the actual intensity of the diffracted beam is more complicated as it involves summing all such waves at the observer and, at the same time, including the diffraction effect of each individual narrow slit. With a smaller than d as in the Figure 1.39 (a), the amplitude of the diffracted beam is modulated by the diffraction amplitude of a single slit since the latter is spread substantially, as illustrated in Figure 1.39 (b). It is apparent that the diffraction grating provides a means of deflecting an incoming light by an amount that depends on its wavelength—the reason for their use in *spectroscopy.*

The diffraction grating in Figure 1.40 (a) is a **transmission grating**. The incident and diffracted beams are on opposite sides of the grating. Typically, parallel thin grooves on a glass plate would serve as a transmission grating as in Figure 1.40 (a). A **reflection grating** has the incident beam and the diffracted beams on the same side of the device as in Figure 1.40 (b). The surface of the device has a periodic reflecting structure, easily formed by etching parallel grooves in a metal film, *etc.* The reflecting unetched surfaces serve as synchronous secondary sources that interfere along certain directions to give diffracted beams of zero-order, first-order, *etc.* Among transmission gratings, it is customary to distinguish between *amplitude gratings* in which the transmission amplitude is modulated, and so-called *phase gratings* where only the refractive index is modulated, without any losses.

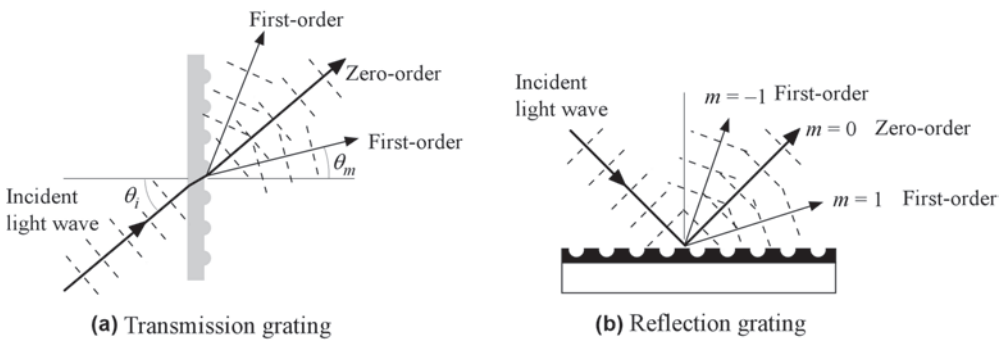


FIGURE 1.40 (a) Ruled periodic parallel scratches on a glass serve as a *transmission grating.* (The glass plate is assumed to be very thin.) (b) A *reflection grating.* An incident light beam results in various "diffracted" beams. The zero-order diffracted beam is the normal reflected beam with an angle of reflection equal to the angle of incidence.

[37] William Lawrence Bragg (1890–1971), Australian-born British physicist, won the Nobel Prize with his father, William Henry Bragg, for his "famous equation" when he was only 25 years old.

When the incident beam is not normal to the diffraction grating, then Eq. (1.12.8) must be modified. If θ_i is the angle of incidence with respect to the normal to the grating, then the diffraction angle θ_m for the m-th mode is given by

Grating equation

$$d(\sin\theta_m - \sin\theta_i) = m\lambda; \quad m = 0, \pm 1, \pm 2, \ldots \tag{1.12.9}$$

The same equation can be used for transmission and reflection gratings provided that we define the angles θ_i and θ_m as positive on either side of the normal as in Figure 1.40 (b).[38] Consider a grating with N slits. The slit width is a (very narrow), and d is the periodicity as before. The detector is at a distance L, far away from the grating. While the periodicity in the slits gives rise to the diffracted beams, the diffraction at each narrow slit defines the envelope of the diffracted intensities as shown in Figure 1.39 (b). If the incident plane wave is normal to the grating, the intensity distribution along y at the screen is given by

Grating diffraction pattern

$$I(y) = I_o\left[\frac{\sin\left(\frac{1}{2}k_y a\right)}{\frac{1}{2}k_y a}\right]^2\left[\frac{\sin\left(\frac{1}{2}Nk_y d\right)}{N\sin\left(\frac{1}{2}k_y d\right)}\right]^2 \tag{1.12.10}$$

where k_y is the scattering wave vector defined by $k_y = (2\pi/\lambda)(y/L) = (2\pi/\lambda)\sin\theta$, and I_o is the maximum intensity along $\theta = 0$. The second term represents the oscillations in the intensity due to interference from different slits. The first term is the envelope of the diffraction pattern, and is the diffraction pattern of a single slit.

The **resolvance** or the **resolving power** R of a diffraction grating is its ability to be able to separate out adjacent wavelengths. If $\lambda_2 - \lambda_1 = \Delta\lambda$ is the minimum wavelength separation that can be measured, as determined by the Rayleigh criterion (the maximum of the intensity distribution at λ_1 is at the first minimum of the intensity distribution at λ_2), and λ is the average wavelength $(1/2)(\lambda_1 + \lambda_2)$ in $\Delta\lambda$, then the **resolving power** is defined by

Resolving power

$$R = \lambda/\Delta\lambda \tag{1.12.11}$$

The separation $\Delta\lambda$ is also called the spectral resolution. If N grooves on a grating are illuminated and the order of diffraction is m, the theoretical resolving power is given simply by $R = mN$. The resolving power is also called the **chromatic resolving power** since it refers to the separation of wavelengths.

Diffraction gratings are widely used in spectroscopic applications because of their ability to provide light deflection that depends on the wavelength. In such applications, the undiffracted light that corresponds to the zero-order beam (Figure 1.40) is clearly not desirable because it wastes a portion of the incoming light intensity. Is it possible to shift this energy to a higher order? Robert William Wood (1910) was able to do so by ruling grooves on glass with a controlled shape as in Figure 1.41 (a) where the surface is angled periodically with a spatial period d. The diffraction condition in Eq. (1.12.9) applies with respect to the normal to the grating plane, whereas the first-order reflection corresponds to reflection from the flat surface, which is at an angle γ. Thus it is possible to "blaze" one of the higher orders (usually $m = 1$) by appropriately choosing γ. Most modern diffraction gratings are of this type. If the angle of incidence is θ_i with respect to the grating normal, then specular reflection occurs at an angle $(\gamma + \theta_i)$ with respect to the face normal and $(\gamma + \theta_i) + \gamma$ with respect to the grating normal. This reflection at $(\gamma + \theta_i) + \gamma$ should occur at diffraction angle θ_m so that

Blazing angle

$$2\gamma = \theta_m - \theta_i \tag{1.12.12}$$

[38] Some books use $d(\sin\theta_m + \sin\theta_i) = m\lambda$ for a transmission grating but the angles become positive on the incidence side and negative on the transmitted side with respect to the normal. It is a matter of sign convention.

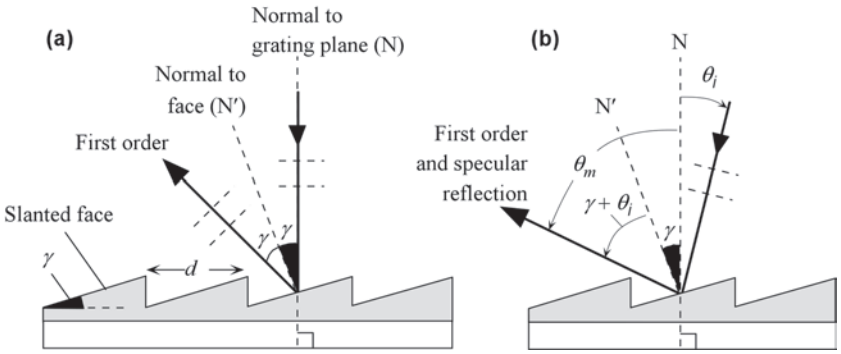
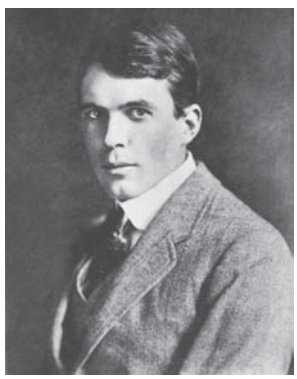


FIGURE 1.41 (a) A blazed grating. Triangular grooves have been cut into the surface with a periodicity *d*. The side of a triangular groove makes an angle γ to the plane of the diffraction angle. For normal incidence, the angle of diffraction must be 2γ to place the specular reflection on the diffracted beam. (b) When the incident beam is not normal, the specular reflection will coincide with the diffracted beam when $(\gamma + \theta_i) + \gamma = \theta_m$.



William Lawrence Bragg (1890–1971), Australian-born British physicist, won the Nobel Prize with his father, William Henry Bragg, for his "famous equation" when he was only 25 years old. *(SSPL via Getty Images.)*

"The important thing in science is not so much to obtain new facts as to discover new ways of thinking about them."

EXAMPLE 1.12.2 A reflection grating

Consider a reflection grating with a period *d* that is 10 μm as in Figure 1.42 (a). Find the diffracted beams if a collimated light wave of wavelength 1550 nm is incident on the grating at an angle of 45° to its normal. What should be the blazing angle γ if we were to use a blazed grating with the same periodicity? What happens to the diffracted beams if the periodicity is reduced to 2 μm?

Solution

If we put $m = 0$ in Eq. (1.12.9) we would find the zero-order diffraction, which is at an angle 45°, as expected, and shown in Figure 1.42 (a). The general Bragg diffraction condition is

$$d(\sin\theta_m - \sin\theta_i) = m\lambda$$

so that

$$(10\ \mu\text{m})(\sin\theta_m - \sin(45°) = (+1)(1.55\ \mu\text{m})$$

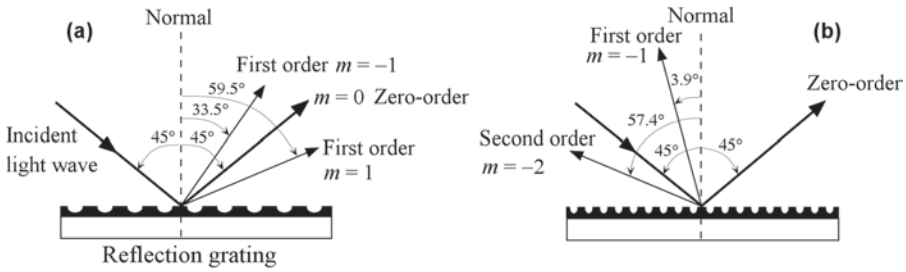


FIGURE 1.42 A light beam is incident at an angle 45° to the normal on a reflection grating. (a) The grating periodicity is 10 μm. (b) The periodicity is 2 μm.

and

$$(10\ \mu\text{m})(\sin\theta_m - \sin(45°) = (-1)(1.55\ \mu\text{m})$$

Solving these two equations, we find $\theta_m = 59.6°$ for $m = 1$, and $\theta_m = 33.5°$ for $m = -1$.

Consider Figure 1.41 (b) in which the secular reflection from the grooved surface coincides with the m_{th} order diffraction when $2\gamma = \theta_m - \theta_i$. Thus

$$\gamma = (1/2)(\theta_m - \theta_i) = (1/2)(59.6° - 45°) = 7.3°$$

Suppose that we reduce d to 2 μm. Recalculating the above we find $\theta_m = -3.9°$ for $m = -1$ and imaginary for $m = +1$. Further, for $m = -2$, there is a second-order diffraction beam at $-57.4°$. Both are shown in Figure 1.42 (b). It is left as an exercise to show that if we increase the angle of incidence, for example, $\theta_i = 85°$ on the first grating, the diffraction angle for $m = -1$ increases from 33.5° to 57.3° and the other diffraction peak ($m = 1$) disappears.

Additional Topics

1.13 INTERFEROMETERS

An **interferometer** is an optical instrument that uses the wave-interference phenomena to produce interference fringes (*e.g.*, dark and bright bands or rings) which can be used to measure the wavelength of light, surface flatness, or small distances. A nearly monochromatic light wave is split into two coherent waves traveling two different paths, and then the two waves are brought together and made to interfere on a screen (or a detector array); the result is an interference pattern as in the Young's fringes in Figure 1.31. In some interferometers, the intensity of the resultant interference is measured at one location, and this intensity is monitored due to changes in one of the optical path lengths. Even a small change in the optical path, distance × refractive index, nL, can cause a measurable shift in the diffraction pattern or a displacement in the fringes, which can be used to infer on nL. There are many types of interferometers.

The **Fabry–Perot interferometer** is a Fabry–Perot cavity–based interferometer that produces an interference ring pattern (bright and dark rings) when illuminated from a broad monochromatic light source as illustrated in Figure 1.43. The resonator consists of two parallel flat glass plates facing each other and separated by an adjustable spacer. A piezoelectric transducer can provide small changes in the spacing between the plates. The inside surface of the glass plates are coated to enhance reflections within the cavity. (Dielectric mirrors can also be used on the inner surfaces.)

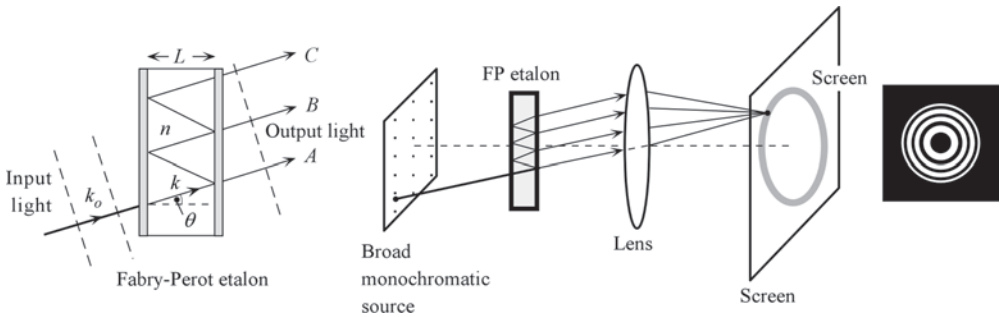


FIGURE 1.43 Fabry–Perot interferometer.

The light entering the cavity will experience multiple reflections inside the cavity. The transmitted waves *A*, *B*, *C*, *etc.*, are focused by a lens onto a point *P* on a screen. If *A*, *B*, *C*, *etc.*, are in phase then the point *P* will be a bright spot. Suppose that *L* is the separation of the plates, θ is the angle at which refracted rays inside the cavity are propagating with respect to the normal, $k = 2\pi n/\lambda$ is the propagation constant in the cavity (λ is the free-space wavelength), and *n* is the refractive index of the medium inside the cavity. We can then show that the phase difference between A and B is $2kL\cos\theta$, which must be $2m\pi$, where *m* is an integer, for constructive interference. Recall that for normal incidence this was $2kL$. Thus, *P* is bright if

$$2nL\cos\theta = m\lambda; \quad m = 0, \pm 1, \ldots \tag{1.13.1}$$

All such points with the same θ lie on a circle about the etalon axis. It is apparent that the interference pattern consists of dark and bright rings. The interference ring diameter depends on the wavelength and the optical separation *nL* (refractive index × distance) of the plates of the etalon. The interferometer can be used in spectroscopic applications such as the measurement of the source wavelength.

In Figure 1.44, showing a **Mach–Zehnder interferometer**, a coherent light beam is split by a beam splitter into two paths at *O*. These beams are then reflected by two mirrors M_1 and M_2 onto a second beam splitter where they are combined at *C* to give rise to an output light beam onto a detector *D*. The resultant field at *C* depends on the phase difference between the waves traveling the paths *OAC* and *OBC*. We can easily monitor the changes in *n* or the length *d* of a sample by placing it into one of the paths, such as in *OAC*. If $k_o = 2\pi/\lambda$ is the propagation constant in vacuum and $k = 2\pi n/\lambda$ is that in the sample, then the phase difference Φ between the weaves arriving at *C* is[39]

$$\Phi = k_o(\overline{OAC} - d) + kd - k_o\overline{OBC}$$

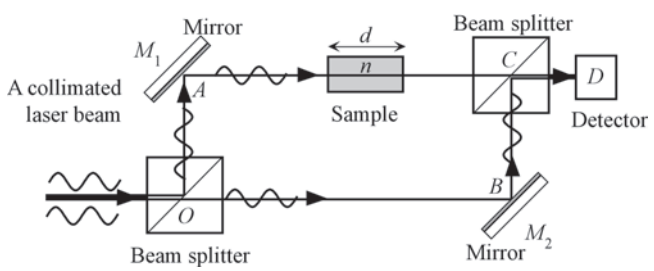


FIGURE 1.44 Mach–Zehnder interferometer.

[39] $\overline{OAC}$ stands for the distance from *O* to *A* to *C*.

The change $\Delta\Phi$ in the phase angle due changes in the sample would be

$$\Delta\Phi = \Delta(kd) = \frac{2\pi}{\lambda}\Delta(nd)$$

Thus, the change in the detected signal represents the change in *nd.* On the other hand, if we can somehow modulate n or d, we have the means of modulating the intensity at the detector. The *Mach–Zehnder modulators* used in communications rely on changing n in a suitable crystal (such as lithium niobate) by the application of an electric field through the electro-optic effect as discussed later in the book.

1.14 THIN FILM OPTICS: MULTIPLE REFLECTIONS IN THIN FILMS

There is much interest in the reflection of light from, and transmission through, a thin film coating of a transparent material (called an optical coating) on a substrate such as a semiconductor crystal; or even on glass as in antireflection coated lenses. The simplest case is shown in Figure 1.45 where light traveling in a medium of refractive index n_1 is incident on a thin film coating of index n_2 on a substrate of index n_3. There are multiple reflections in the thin film, and the problem is almost identical to the Fabry–Perot optical cavity in which there are similar multiple interferences.

Suppose the thickness of the coating is *d.* For simplicity, we will assume normal incidence. The phase change in traversing the coating thickness *d twice* is $\phi = 2 \times (2\pi/\lambda)n_2 d$ where λ is the free-space wavelength. The wave has to be multiplied by $\exp(-j\phi)$ to account for this phase difference in crossing the thickness twice. The reflection and transmission coefficients for the present n_1-n_2-n_3 system are given by,

$$r_1 = r_{12} = \frac{n_1 - n_2}{n_1 + n_2} = -r_{21}, \quad r_2 = r_{23} = \frac{n_2 - n_3}{n_2 + n_3}, \tag{1.14.1a}$$

and

$$t_1 = t_{12} = \frac{2n_1}{n_1 + n_2}, \quad t_1' = t_{21} = \frac{2n_2}{n_1 + n_2}, \quad t_2 = t_{23} = \frac{2n_2}{n_2 + n_3} \tag{1.14.1b}$$

where

$$1 - t_1 t_1' = r_1^2 \tag{1.14.1c}$$

The first reflected beam, $A_1 = A_0 \times r$, the second is $A_2 = A_o \times t_1 \times t_1' \times r_2 \times e^{-j\phi}$, and so on.

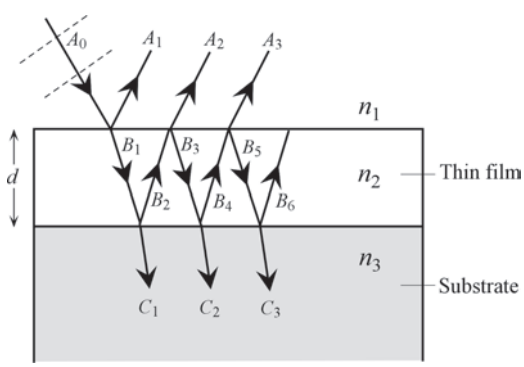


FIGURE 1.45 Light traveling in a medium of refractive index n_1 is incident on a thin film coating of index n_2 on a substrate of index n_3.

The amplitude of the reflected beam is

$$A_{\text{reflected}} = A_1 + A_2 + A_3 + A_4 + \ldots$$

that is,

$$A_{\text{reflected}}/A_0 = r_1 + t_1 t_1' r_2 e^{-j\phi} - t_1 t_1' r_1 r_2^2 e^{-j2\phi} + t_1 t_1' r_1^2 r_2^3 e^{-j3\phi} + \ldots \qquad \textbf{(1.14.2)}$$

which is a geometric series. Using Eq. (1.14.1c), Eq. (1.14.2) can be conveniently summed to obtain the overall reflection coefficient r

$$r = \frac{r_1 + r_2 e^{-j\phi}}{1 + r_1 r_2 e^{-j\phi}} \qquad \textbf{(1.14.3)}$$

Thin film reflection coefficient

Similarly, we can sum for the amplitude of the transmitted beam as

$$C_{\text{transmitted}} = C_1 + C_2 + C_3 + \ldots$$

that is,

$$C_{\text{transmitted}}/A_0 = t_1 t_2 e^{-j\phi/2} - t_1 t_2 r_1 r_2 e^{-j3\phi/2} + t_1 t_2 r_1^2 r_2^2 e^{-j5\phi/2} + \ldots \qquad \textbf{(1.14.4)}$$

which is a geometric series that sums to

$$t = \frac{t_1 t_2 e^{-j\phi/2}}{1 + r_1 r_2 e^{-j\phi}} \qquad \textbf{(1.14.5)}$$

Thin film transmission coefficient

Equations (1.14.3) and (1.14.5) describe the reflected and transmitted waves. The reflectance and transmittance are then

$$R = |r|^2 \quad T = (n_3/n_1)|t|^2 \qquad \textbf{(1.14.6)}$$

Thin film reflectance and transmittance

Figure 1.46 (a) shows R and T as a function of ϕ for $n_1 < n_2 < n_3$. Clearly, R is minimum, and T is maximum, whenever $\phi = \pi \times$ (odd number) or $\phi = 2(2\pi/\lambda)n_2 d = \pi(2m + 1)$, where $m = 0, 1, 2, \ldots$ The latter leads to

$$d = \frac{\lambda}{4n_2}(2m + 1); \quad m = 0, 1, 2, \ldots \qquad \textbf{(1.14.7)}$$

Thickness for minimum reflection

which is the thickness required to minimize the reflection, and maximize the transmission of light when n_2 is intermediate between n_1 and n_3. The oscillations in R and T with ϕ (*e.g.*, as the wavelength is scanned) are sometime referred to as an **interference fringes** in wavelength.

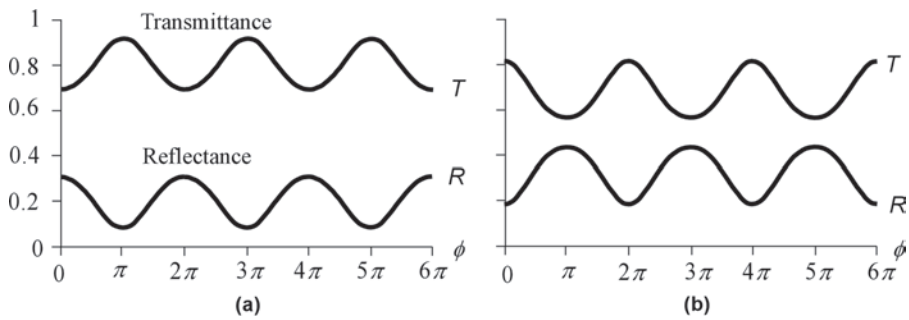


FIGURE 1.46 (a) Reflectance R and transmittance T vs. $\phi = 2n_2 d/\lambda$, for a thin film on a substrate where $n_1 = 1$ (air), $n_2 = 2.5$, $n_3 = 3.5$, and $n_1 < n_2 < n_3$. (b) R and T vs. ϕ for a thin film on a substrate where $n_1 = 1$ (air), $n_2 = 3.5$, $n_3 = 2.5$, and $n_2 > n_3 > n_1$.

Figure 1.46 (b) represents the reflectance and transmittance vs. ϕ of light through a thin layer of high index material on a low index substrate where $n_1 < n_3 < n_2$; for example, a semiconductor film on a glass substrate. Notice the difference between the two cases, especially the locations of the maxima and minima. (Why is there a difference?)

The phase change ϕ, of course, depends on three factors, d, n_2, and λ. The minimum and maximum reflectances in Figure 1.46 at those particular ϕ values can be found by using Eqs. (1.14.3) and (1.14.7). For $n_1 < n_2 < n_3$ as in Figure 1.46 (a)

Minimum and maximum reflectance

$$R_{\min} = \left(\frac{n_2^2 - n_1 n_3}{n_2^2 + n_1 n_3}\right)^2; \quad R_{\max} = \left(\frac{n_3 - n_1}{n_3 + n_1}\right)^2 \tag{1.14.8}$$

and the transmittance can be found from $R + T = 1$. When $n_1 < n_3 < n_2$ then $R_{\min}$ and $R_{\max}$ equations are interchanged. While $R_{\max}$ appears to be independent from n_2, the index n_2 is nonetheless still involved in determining maximum reflection inasmuch as R reaches $R_{\max}$ when $\phi = 2(2\pi/\lambda)n_2 d = \pi(2m)$; when $\phi = \pi \times$ (even number).

In transmission spectra measurements, a spectrophotometer is used to record the transmittance of a light beam as a function of wavelength through a sample. If the sample is a thin film on a substrate, there will be multiple reflections and interferences in the thin film, and the measured transmittance will exhibit maxima and minima as in Figure 1.46 as the wavelength (or ϕ) is scanned. The locations of the maxima and minima, with the knowledge of the substrate index (n_3), can be used to find d and n.

EXAMPLE 1.14.1 Thin film optics

Consider a semiconductor device with $n_3 = 3.5$ that has been coated with a transparent optical film (a dielectric film) with $n_2 = 2.5$, $n_1 = 1$ (air). If the film thickness is 160 nm, find the minimum and maximum reflectances and transmittances and their corresponding wavelengths in the visible range. (Assume normal incidence.)

Solution

This case corresponds to Figure 1.46 (a). Minimum reflectance $R_{\min}$ occurs at $\phi = \pi$ or odd multiple of π, and maximum reflectance $R_{\max}$ at $\phi = 2\pi$ or an integer multiple of 2π. From Eq. (1.14.8) we have

$$R_{\min} = \left(\frac{n_2^2 - n_1 n_3}{n_2^2 + n_1 n_3}\right)^2 = \left(\frac{2.5^2 - (1)(3.5)}{2.5^2 + (1)(3.5)}\right)^2 = 0.080 \quad \text{or} \quad 8.0\%$$

and

$$R_{\max} = \left(\frac{n_3 - n_1}{n_3 + n_1}\right)^2 = \left(\frac{3.5 - 1}{3.5 + 1}\right)^2 = 0.31 \quad \text{or} \quad 31\%$$

Corresponding transmittances are,

$$T_{\max} = 1 - R_{\min} = 0.92 \quad \text{or} \quad 92\%$$

and

$$T_{\min} = 1 - R_{\max} = 0.69 \quad \text{or} \quad 69\%$$

Without the thin film coating, the reflectance would be 31%, the maximum reflectance.

Given $\phi = 2dn_2(2\pi/\lambda)$, and $d = 160$ nm, and the fact that the minimum reflectance corresponds to $\phi =$ odd integer $\times\ \pi$, we choose $\phi = 3\pi$ (by trial and error). The wavelength λ_{min} is then given by

$$\lambda_{min} = 4\pi dn_2/\phi = 4\pi(160\text{ nm})(2.5)/(3\pi) = 533\text{ nm (green)}$$

which is in the visible. The maximum reflectance in the visible occurs when $\phi = 4\pi$, giving

$$\lambda_{max} = 4\pi dn_2/\phi = 4\pi(160\text{ nm})(2.5)/(4\pi) = 400\text{ nm (violet)}$$

1.15 MULTIPLE REFLECTIONS IN PLATES AND INCOHERENT WAVES

The interference of light waves in a thin film takes place because the waves have much longer coherence lengths than the thickness of the film so that the waves exhibit mutual coherence. We can add the electric field and interference leads to bright and dark fringes. A film would be considered too thick if it does not exhibit any interference phenomena due to the coherence length of the waves being shorter than the thickness. Such cases easily arise when we pass light through a transparent (or partially transparent) plate or when the light source is incoherent. In such cases, we cannot add the electric field to find the reflected and transmitted light irradiances. The overall reflectance and transmittance would be independent of the round-trip phase change ϕ inside the plate and would not exhibit the behavior in Figure 1.46. We have to use Eq. (1.10.5).

Consider a light beam of unit intensity that is passed through a thick plate of transparent material of index n_2 in a medium of index n_1 as in Figure 1.47. The first transmitted light intensity into the plate is $(1 - R)$, and the first transmitted light out is $(1 - R) \times (1 - R)$ or $(1 - R)^2$. However, there are internal reflections as shown, so that the second transmitted light is $(1 - R) \times R \times R \times (1 - R) = R^2(1 - R)^2$, so that the transmitted intensity through the plate is

$$T_{plate} = (1 - R)^2 + R^2(1 - R)^2 + R^4(1 - R)^2 + \ldots = (1 - R)^2\left[1 + R^2 + R^4 + \ldots\right]$$

or

$$T_{plate} = \frac{(1 - R)^2}{1 - R^2} \qquad \textbf{(1.15.1)}$$

Transmittance for a thick plate or incoherent light

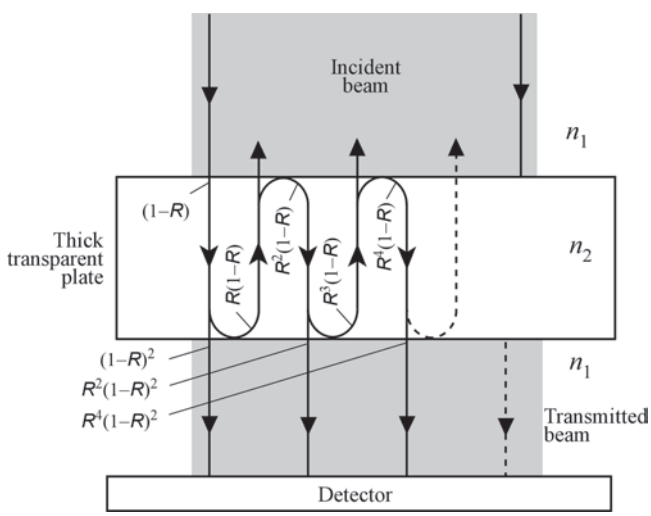


FIGURE 1.47 Transmitted and reflected light through a slab of material in which there is no interference.

Transmittance for a thick plate for incoherent light

By substituting for R in terms of the indices, we can write this as

$$T_{\text{plate}} = \frac{2n_1n_2}{n_1^2 + n_2^2} \tag{1.15.2}$$

Transmittance for a thick plate or incoherent beam of light

For example, for a glass plate of $n_2 = 1.60$ in air ($n_1 = 1$), $T_{\text{plate}} = 89.9\%$ while the simple transmittance through an n_1–n_2 interface would give 94.7%. The overall reflectance is $1 - T_{\text{plate}}$ so that

$$R_{\text{plate}} = \frac{(n_1 - n_2)^2}{n_1^2 + n_2^2} \tag{1.15.3}$$

One of the simplest ways to determine the refractive index of a plate is to measure the transmittance T_{plate} in Eq. (1.15.1), from which we can calculate n_2.

1.16 SCATTERING OF LIGHT

When a light beam propagates in a medium in which there are small particles or inhomogeneities, such as local changes in the refractive index of the medium, some of the power in the beam is radiated away from the direction of propagation, that is some of the power becomes scattered. **Scattering** is a process by which some of the power in a propagating electromagnetic wave is redirected as secondary EM waves in various directions away from the original direction of propagation as illustrated in Figure 1.48 (a). There are a number of scattering processes, which are usually classified in terms of the size of the scattering particles in relation to the wavelength of light that is scattered. In **Rayleigh scattering**, the scattering particle size, or the scale of inhomogeneities in a medium, is much smaller than the wavelength of light. The intensity of the scattered light at an angle θ to the original beam depends on the scattering process; the Rayleigh scattering case is shown in Figure 1.48 (b) in which the scattering is not spherically symmetric.

Consider what happens when a propagating wave encounters a molecule, an impurity in a crystal or a small dielectric particle (or region), which is smaller than the wavelength of light. The electric field in the wave polarizes the particle by displacing the lighter electrons with respect to the heavier positive nuclei. The electrons in the molecule couple and oscillate with the electric field in the wave (ac electronic polarization). The oscillation of charge "up" and "down," or the oscillation of the induced dipole, radiates EM waves all around the molecule as illustrated in Figure 1.48 (a). We should remember that an oscillating charge is like an alternating current which always radiates EM waves (like an antenna). The net effect is that the incident

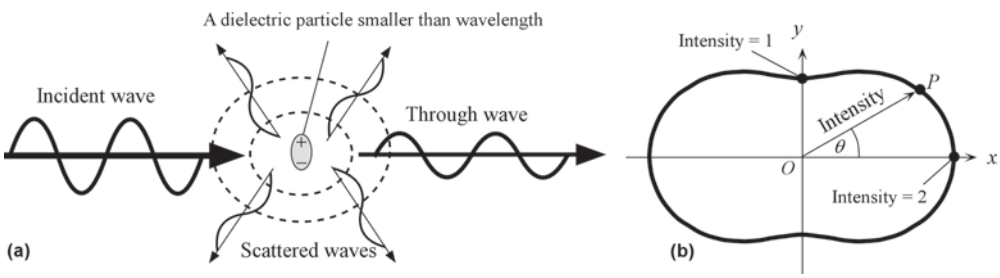


FIGURE 1.48 (a) Rayleigh scattering involves the polarization of a small dielectric particle or a region that is much smaller than the light wavelength. The field forces dipole oscillations in the particle (by polarizing it) which leads to the emission of EM waves in "many" directions so that a portion of the light energy is directed away from the incident beam. (b) A polar plot of the dependence of the intensity of the scattered light on the angular direction θ with respect to the direction of propagation x in Rayleigh scattering (in a polar plot, the radial distance OP is the intensity).



Lord Rayleigh (John William Strutt) was an English physicist (1877–1919) and a Nobel Laureate (1904) who made a number of contributions to wave physics of sound and optics. He formulated the theory of scattering of light by small particles and the dependence of scattering on $1/\lambda^4$ circa 1871. Then, in a paper in 1899 he provided a clear explanation on why the sky is blue. Ludvig Lorentz, around the same time, and independently, also formulated the scattering of waves from a small dielectric particle, though it was published in Danish (1890).[40] *(© Mary Evans Picture Library/Alamy.)*

wave becomes partially reradiated in different directions and hence loses intensity in its original direction of propagation. (We may think of the process as the particle absorbing some of the energy via electronic polarization and reradiating it in different directions.) It may be thought that the scattered waves constitute a spherical wave emanating from the scattering molecule, but this is not generally the case as the re-emitted radiation depends on the shape and polarizability of the molecule in different directions. We assumed a small particle so that at any time the field has no spatial variation through the particle, whose polarization then oscillates with the electric field oscillation. Whenever the size of the scattering region, whether an inhomogeneity, a small particle, a molecule, or a defect in a crystal, is much smaller than the wavelength λ of the incident wave, the scattering process is generally termed **Rayleigh scattering.** Typically, the particle size is smaller than one-tenth of the light wavelength.

Rayleigh scattering of light propagating in a glass medium is of particular interest in photonics because it results in the attenuation of the transmitted light pulses in optical fibers. The glass structure is such that there are small random spatial variations in the refractive index about some average value. There are therefore local fluctuations in the relative permittivity and polarizability, which effectively act if there are small inhomogeneities in the medium. These dielectric inhomogeneities arise from fluctuations in the relative permittivity that is part of the intrinsic glass structure. As the fiber is drawn by freezing a liquid-like flow, random thermodynamic fluctuations in the composition and structure that occur in the liquid state become frozen into the solid structure. Consequently, the glass fiber has small fluctuations in the relative permittivity which leads to Rayleigh scattering. A small inhomogeneous region acts like a small dielectric particle and scatters the propagating wave in different directions. Nothing can be done to eliminate Rayleigh scattering in glasses as it is part of their intrinsic structure.

It is apparent that the scattering process involves electronic polarization of the molecule or the dielectric particle. We know that this process couples most of the energy at ultraviolet frequencies where the dielectric loss due to electronic polarization is maximum and the loss is due to EM wave radiation. Therefore, as the frequency of light increases, the scattering becomes more severe. In other words, *scattering decreases with increasing wavelength.* The intensity of the scattered radiation is proportional to $1/\lambda^4$. For example, blue light, which has a shorter wavelength than red light, is scattered more strongly by air molecules. When we look at the sun directly, it appears yellow because the blue light has been scattered in the direct light more than

[40] Pedro Lilienfeld's "A Blue Sky History", in *Optics and Photonics News*, *15*(6), 32, 2004, is highly recommended; it also provides the original references.

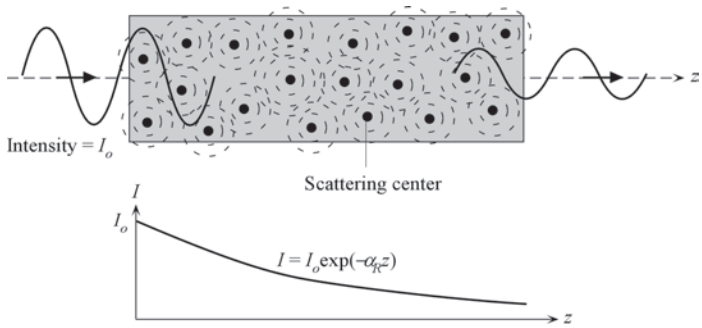


FIGURE 1.49 When a light beam propagates through a medium in which there are small particles, it becomes scattered as it propagates and loses power in the direction of propagation. The light becomes attenuated.

the red light. When we look at the sky in any direction but the sun, our eyes receive scattered light which appears blue; hence the sky is blue. At sunrise and sunset, the rays from the sun have to traverse the longest distance through the atmosphere and have the most blue light scattered, which gives the sun its red color at these times.

The intensity of a light beam in a medium with small particles (or inhomogeneities) decreases as the beam propagates due to Rayleigh scattering from these small particles as illustrated in Figure 1.49. The intensity at a position z inside the medium from the radiation receiving face is given by

Attenuation by scattering

$$I = I_o \exp(-\alpha_R z) \tag{1.16.1}$$

where α_R is the attenuation coefficient due to Rayleigh scattering. α_R depends on the concentration of scattering particles N, their radius a, wavelength λ and the mismatch between the refractive index n of the scattering spheres and the index n_o of the medium

Rayleigh attenuation coefficient

$$\alpha_R \propto N \cdot a^6 \cdot \frac{1}{\lambda^4} \cdot \left(\frac{n^2 - n_o^2}{n^2 + n_o^2} \right)^2 \tag{1.16.2}$$

Equation (1.16.2) has been written in terms of products to identify those factors that affect α_R.

Mie scattering refers to the scattering of light from scatterers that have dimensions comparable with, or greater than, the wavelength of light. For example, Mie scattering would occur for light scattering from long organic molecules in a solution, or scattering from various particulate pollutants (as in smog) in the atmosphere, including dust particles. The scatterers are assumed to have a refractive index significantly different from that of the surrounding medium. The scattering depends on the ratio of the scattering particle diameter to the wavelength of light, and favors scattering in the forward direction. The dependence on the wavelength is weaker than Rayleigh scattering.

The scattering of light from various particles suspended in a liquid is usually referred to as **turbidity**. For example, small solid particles suspended in water will scatter light and would make the water appear cloudy. The quality of water is often monitored for small particles by measuring the turbidity of water.

1.17 PHOTONIC CRYSTALS

The dielectric mirror, or the Bragg reflector, in Figure 1.24 is a stack of alternating layers of high and low refractive index material in a periodic manner. The one-dimensional periodic variation in the refractive index n in Figure 1.24 represents one of the simplest periodic structures within

a general class of optical materials called **photonic crystals**. A **photonic crystal** (PC) is a material that has been structured to possess a periodic modulation of the refractive index n, just like in Figure 1.24, so that the structure influences the propagation and confinement of light within it. The periodicity can be in one- (1D), two- (2D), or three-dimensional (3D); Figures 1.50 (a)–(c) illustrates 1D, 2D, and 3D simple photonic crystals as examples. In fact, quite complicated structures can be constructed that have very interesting optical properties. The dielectric mirror or the Bragg reflector can be viewed as one of the simplest 1D photonic crystals.[41] For the structure to influence the propagation of the EM wave, it has to diffract the wave, which means that the scale of periodic variations must be on the wavelength scale. As apparent from Figure 1.24, the 1D PC has a band of frequencies over which it reflects the light and, conversely, there is a **stop band** over which no transmission is possible through the dielectric stack. There is a band of frequencies that represent waves that are not allowed to go through this periodic structure in the direction of refractive index variation, along z in Figure 1.50; this band is called an **optical** or **photonic bandgap**.

The periodic variation in n in Figure 1.50 is normally assumed to extend indefinitely, whereas in practice, the PCs have a finite size, for example, a certain number of layers in the dielectric mirror, not infinite. As in normal crystals, the periodic structures in Figure 1.50 have a **unit cell**, which repeats itself to generate the whole lattice—that is, the whole crystal structure. For the 1D PC in Figure 1.50 (a), for example, two adjacent layers, n_1n_2, form the unit cell. We can move this unit cell along z by a distance Λ, the **period** (or **periodicity**), many times to generate the whole 1D photonic crystal.

The periodicity of a photonic crystal implies that any property at a location z will be the same at $z \pm \Lambda$, $z \pm 2\Lambda$ and so on; that is, there is *translational symmetry* along z (in 1D).

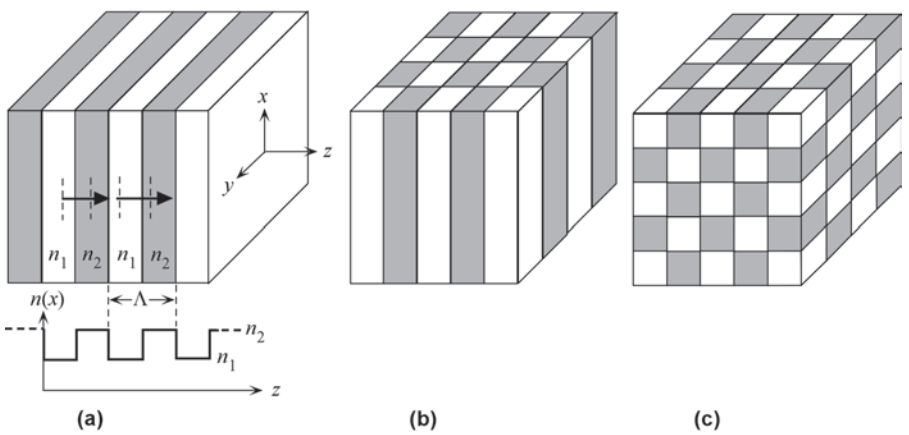


FIGURE 1.50 Photonic crystals in (a) 1D, (b) 2D, and (c) 3D, D being the dimension. Gray and white regions have different refractive indices and may not necessarily be the same size. Λ is the periodicity. The 1D photonic crystal in (a) is the well-known Bragg reflector, a dielectric stack.

[41] The propagation of light through such a one-dimensional (1D) periodic variation of n has been well-known, dating back to the early work of Lord Rayleigh in 1887. Eli Yablonovitch has suggested that the name "photonic crystal" should really only apply to 2D and 3D periodic structures with a large dielectric (refractive index) difference. (E. Yablonovitch, "Photonic crystals: What's in a name?," *Opt. Photon. News*, *18*, 12, 2007.) Nonetheless, the dielectric mirror in Figure 1.50 (a) is often considered as the simplest 1D photonic crystal to derive the concept of a "photonic bandgap," essentially a stop band.

The EM waves that are allowed to propagate along z through the periodic structure are called the **modes** of the photonic crystal. They have a special waveform that must bear the periodicity of the structure, and are called **Bloch waves**. Such a wave for the field E_x, for example, has the form $E_x(z, t) = A(z)\exp(-jkz)$, which represents a traveling wave along z and $A(z)$ is an amplitude function that has the periodicity of the structure, that is, it is periodic along z with a period Λ. $A(z)$ depends on the periodic refractive index function $n(z)$. As we will see in Chapter 3, the electron motion in a semiconductor crystal is also described by Bloch waves (electron wavefunctions).

As in the case of the dielectric mirror, the 1D PC has a band of frequencies over which there can be no propagation along z. In a homogeneous medium of refractive index n, the relationship between the frequency ω and the propagation constant k is simple, that is, $c/n = \omega/k$, where k is the propagation constant inside the medium The dispersion behavior of the medium, that is, ω vs. k, is a straight line with a slope c/n. The dispersion characteristic of a 1D PC for waves along z in Figure 1.50 (a) is shown in Figure 1.51 (a). We notice several important characteristics. At low frequencies (long wavelengths), the waves propagate as if they are in a homogenous medium with a constant phase velocity (dashed straight line). As expected, there is a band of frequencies $\Delta\omega = \omega_2 - \omega_1$ (between S_1 and S_2) over which no propagation along z is allowed, which is a **photonic bandgap** along z. The whole ω vs. k curves are periodic in k with a period $2\pi/\Lambda$. The point P is equivalent to P' because $k = k' + (2\pi/\Lambda)$. We only need to consider k-vales from $-\pi/L$ to π/L. This region is called the **first Brillouin zone**.

At low frequencies or long wavelengths (small k values) in Figure 1.51 (a), the wavelength is so much longer than the variations in n that the propagating wave experiences essentially some average refractive index, n_{av}, that is $(n_1 + n_2)/2$ if the n_1 and n_2 layers have the same thickness, and propagates through the structure as if the structure was a homogenous medium with some effective refractive index, n_{av}. Its phase velocity is ω/k, which is c/n_{av}, and its group velocity, the slope of the ω vs. k characteristic, is the same as c/n_{av}, in this region (ignoring the wavelength dependence of n_1 and n_2). As the wavelength decreases (ω increases), partial reflections of the waves from the boundaries become important and interfere with propagation. At sufficiently small wavelengths, diffraction becomes important as all these partially reflected waves interfere with each other and give rise to significant reflection. Eventually, a critical wavelength (corresponding to ω_1) is reached where the waves become fully diffracted or reflected backwards. A backward traveling (in $-z$ direction) wave experiences the same reflection. These forward and backward diffracted waves give rise to a standing wave in the structure; indeed, only the latter can exist and waves cannot propagate, that is, travel freely.



Eli Yablonovitch at the University of California at Berkeley, and Sajeev John (shown later in this chapter) at the University of Toronto, carried out the initial pioneering work on photonic crystals. Eli Yablonovitch has suggested that the name "photonic crystal" should apply to 2D and 3D periodic structures with a large dielectric (refractive index) difference. (E. Yablonovitch, "Photonic crystals: What's in a name?," *Opt. Photon. News*, *18*, 12, 2007.) Their original papers were published in the same volume of *Physical Review Letters* in 1987. According to Eli Yablonovitch, "Photonic Crystals are semiconductors for light." *(Courtesy of Eli Yablonovitch.)*

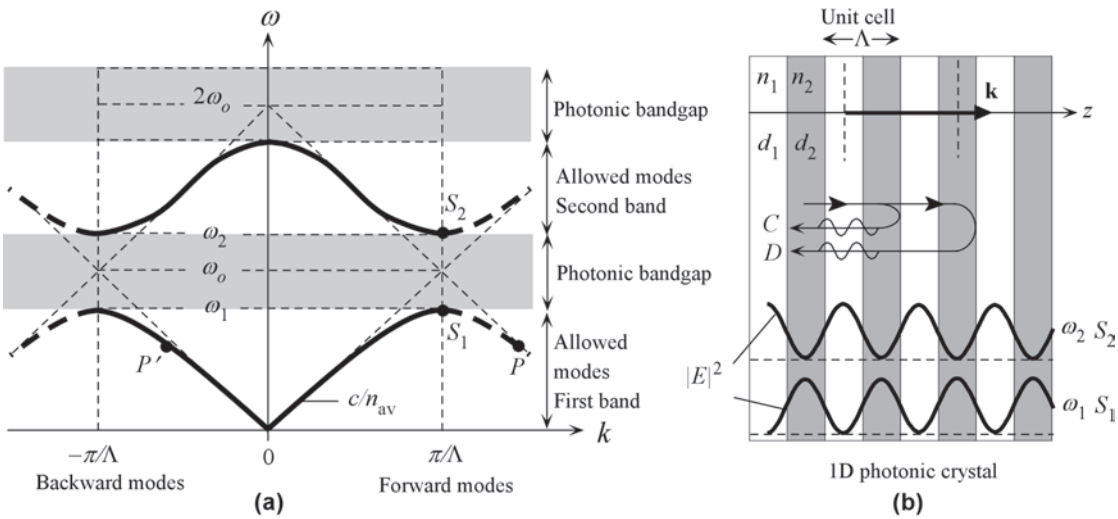


FIGURE 1.51 (a) Dispersion relation, ω vs. k, for waves in a 1D PC along the z-axis. There are allowed modes and forbidden modes. Forbidden modes occur in a band of frequencies called a photonic bandgap. (b) The 1D photonic crystal corresponding to (a), and the corresponding points S_1 and S_2 with their stationary wave profiles at ω_1 and ω_2.

Consider what happens when reflections such as C and D from two successive unit cells interfere constructively and give rise to a backward diffracted (reflected) wave as illustrated in Figure 1.51 (b). Reflections from two successive unit cells must be in phase for full reflection. This means that the phase difference between C and D $(2\pi/\lambda)(2n_1d_1 + 2n_2d_2)$ must be $2m\pi$, where $m = 1, 2, \ldots$ is an integer. If we define $n_{\text{av}} = (n_1d_1 + n_2d_2)/\Lambda$, and $k = n_{\text{av}}(2\pi/\lambda)$ then reflection occurs when $2k\Lambda = 2m\pi$, that is, $k = m\pi/\Lambda$. These are the wave vectors or propagation constants of the waves that cannot be propagated. The waves suffer Bragg reflection in 1D. What happens to these diffracted waves?

The diffracted waves in $+z$ and $-z$ directions set up a standing wave in the structure. If we write the two reflected waves in opposite directions as $A\exp(jkz)$ and $A\exp(-jkz)$ they would add as $A\exp(jkz) \pm A\exp(-jkz)$, which shows that there two possibilities S_1 and S_2. One of them, S_1, has most of its energy inside the n_2, high refractive index layers, and hence has a lower frequency ω_1. S_2 has most of its energy in the n_1 layers and has a higher frequency ω_2; these are shown in Figure 1.51 (b). There are no waves in the $\omega_2 - \omega_1$ interval, which we know as the **photonic bandgap** (PBG). The bandgap $\Delta\omega = \omega_2 - \omega_1$ increases linearly with the index difference $\Delta n = n_2 - n_1$.

It is clear that the 1D crystal exhibits a photonic bandgap for light propagation along z. It should also be apparent that there would be no PBGs for propagation along the x and y direction along which there are no periodic n-variations. Since we can resolve any **k**-vector along x, y, and z directions, overall there is no *net* PBG for light propagation in a 1D PC. The PBG for a 1D crystal is called a **pseudo PBG**.

The above ideas can be readily extended to 2D and 3D periodic structures. Again, in principle, there is no full PBG in the 2D PC. In the case of 3D periodic structures, there would be photonic bandgaps along x, y, and z directions, and for difference polarizations of the electric field. If the refractive index contrast and the periodicity in the 3D structure are such that these photonic bandgaps overlap in *all directions* and for *all polarizations* of light, as schematically depicted in Figure 1.52, the overlap frequency range $\Delta\omega$ becomes a **full photonic bandgap** in all directions for all polarizations of light—no light can propagate through this structure over this

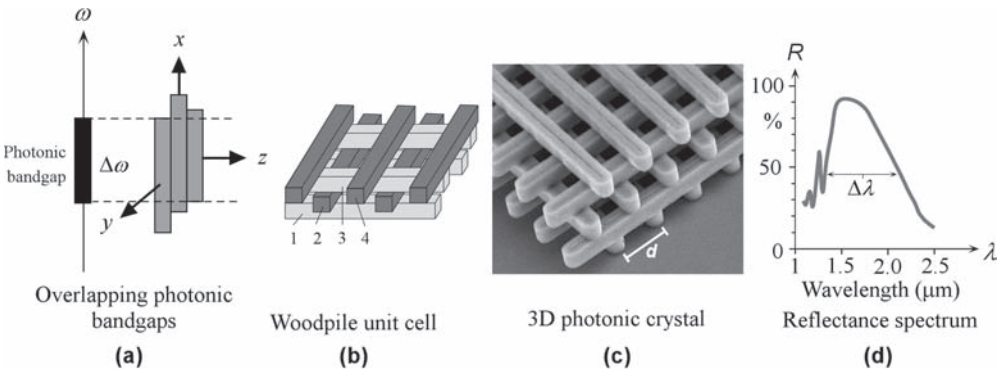
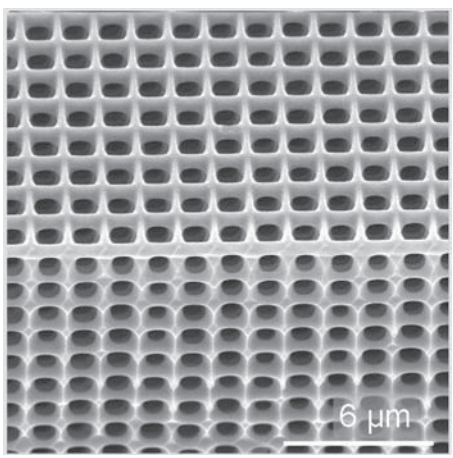
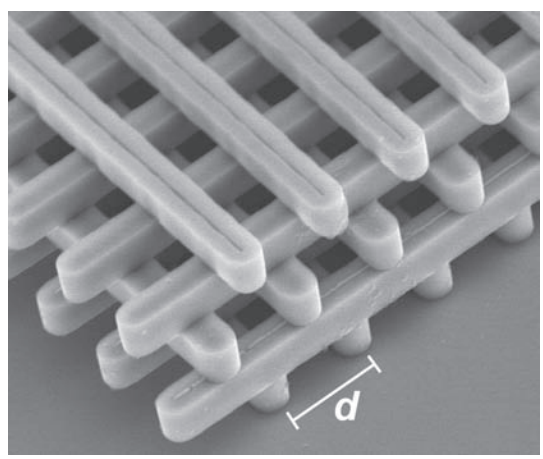


FIGURE 1.52 (a) The photonic bandgaps along *x, y,* and *z* overlap for all polarizations of the field, which results in a full photonic bandgap $\Delta\omega$ (an intuitive illustration). (b) The unit cell of a *woodpile* photonic crystal. There are 4 layers, labeled 1–4 in the figure, with each later having parallel "rods." The layers are at right angles to each other. Notice that layer 3 is shifted with respect to 1, and 4 with respect to 2. (c) An SEM image of a woodpile photonic crystal based on polycrystalline Si; the rod-to-rod pitch *d* is on the micron scale. (*Courtesy of Sandia National Laboratories.*) (d) The optical reflectance of a woodpile photonic crystal showing a photonic bandgap between 1.5 μm and 2 μm. The photonic crystal is similar to that in (c) with five layers and $d \approx 0.65$ μm. (*Source:* The reflectance spectrum was plotted using the data appearing in Fig. 3 in S-Y. Lin and J.G. Fleming, *J. Light Wave Technol.*, *17*, 1944, 1999.)

frequency range, $\Delta\omega$. The structure then has a **full photonic bandgap**. As it turns out not *any* 3D periodic structure results in a full photonic bandgap. Only certain 3D periodic structures allow full photonic bandgaps to develop. One such structure is the *wood pile* periodic structure shown in Figure 1.52 (b). The unit cell has four layers of rods. The rods are parallel in each layer and the layers are



An SEM image of a 3D photonic crystal made from porous silicon in which the lattice structure is close to being simple cubic. The silicon squares, the unit cells, are connected at the edges to produce a cubic lattice. This 3D PC has a photonic bandgap centered at 5 μm and about 1.9 μm wide. *(Courtesy of Max-Planck Institute for Microstructure Physics.)*



An SEM image of a 3D photonic crystal that is based on the *wood pile* structure. The rods are polycrystalline silicon. Although five layers are shown, the unit cell has four layers, *e.g.*, the four layers starting from the bottom layer. Typical dimensions are in microns. In one similar structure with rod-to-rod pitch $d = 0.65$ μm with only a few layers, the Sandia researchers were able to produce a photonic bandgap $\Delta\lambda$ of 0.8 μm centered around 1.6 μm within the telecommunications band. *(Courtesy of Sandia National Laboratories.)*

at 90° to each other. An SEM image of a woodpile photonic crystal is shown in Figure 1.52 (c). The optical reflectance, that is, 1 – transmittance, of this woodpile PC is shown in Figure 1.52 (d). It is apparent that there is a photonic bandgap, a stop band, over a range of frequencies, or wavelengths, $\Delta\lambda$, around 3 μm. The width and the location of the reflectance or the transmittance band depends on the structure of the photonic crystal, that is, the periodicity, unit cell structure and refractive index contrast. The colors of certain butterflies and insects arise not from pigments or colorants but from a photonic crystal effect with the right periodicity.

An important practical aspect of PCs, analogous to semiconductor crystals, is the importance of defects. Point and line defects that occur in normal crystals also occur in PCs, as illustrated in Figure 1.53. They are intentionally introduced to endow the PC structure with certain optical properties. A defect is a discontinuity in the periodicity of the PC. For example, if we upset the periodicity by removing a unit cell, we create a so-called **point defect**. This void can act as an *optical cavity*, trapping the EM radiation within the cavity as illustrated in Figure 1.53. We can, of course, remove a group of unit cells, or modify the refractive index over a few unit cells, which would create an optical microcavity. Defects introduce localized electromagnetic modes with frequencies within the photonic bandgap. Defects can tightly confine a mode of light in a very small cavity volume. We can also enhance the refractive index locally, which would also classify as a point defect.

Line defects are formed when a long row of unit cells are missing, or the refractive index stays constant over a long line in the crystal. Such a line in which the index is constant allows propagating EM modes within the photonic bandgap. Since EM waves can propagate within and along the line defect, the line defect acts as an **optical wave guide**, guiding the radiation along its length as shown in Figure 1.53. The EM wave cannot spread into the perfect photonic crystal region since, in this region, the frequency falls into the stop band.

One very important property of photonic crystals is their ability to *suppress* or *inhibit spontaneous emission*. We can understand this effect intuitively by considering a 3D PC made from a semiconductor. In spontaneous emission, an electron falls from the conduction band

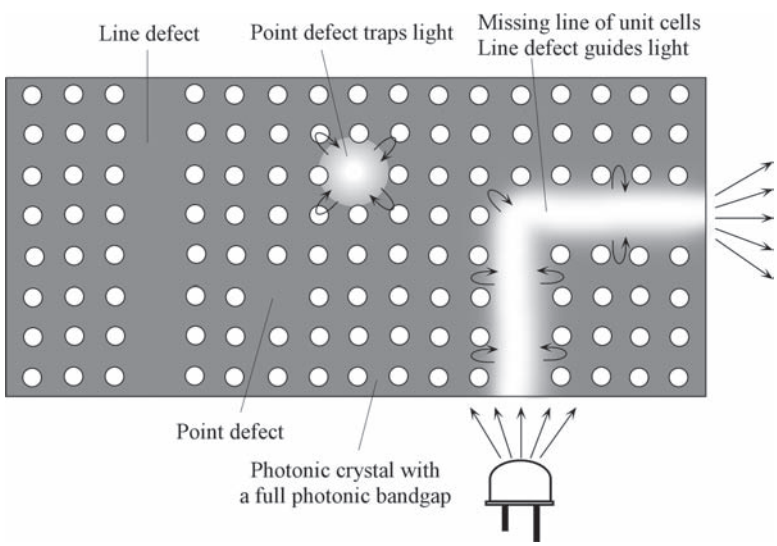


FIGURE 1.53 Schematic illustration of point and line defects in a photonic crystal. A point defect acts as an optical cavity, trapping the radiation. Line defects allow the light to propagate along the defect line. The light is prevented from dispersing into the bulk of the crystal since the structure has a full photonic bandgap. The frequency of the propagating light is in the bandgap, that is, in the stop band.



Sajeev John, at the University of Toronto, along with Eli Yablonovitch (shown earlier in the chapter) carried out the initial pioneering work in the development of the field of photonics crystals. Sajeev John was able to show that it is possible to trap light in a similar way the electron is captured, that is localized, by a trap in a semiconductor. Defects in photonic crystals can confine or localize electromagnetic waves; such effects have important applications in quantum computing and integrated photonics. *(Courtesy of Sajeev John.)*

to the valence band spontaneously and emits a photon of energy $\hbar\omega$ that corresponds to the bandgap energy E_g However, if the photon frequency $\hbar\omega$ falls into the bandgap of the PC, then this photon is not allowed to propagate or "exist" in the structure. It is prevented from being emitted—the photons have no place to go.

Photonic crystals have also been shown to exhibit a so-called *superprism effect*. Under appropriate conditions, the dispersion of light by a prism-shaped photonic crystal is considerably enhanced over that corresponding to a homogeneous prism having the same average refractive index as the photonic crystal. This originates from the strong curvature of the ω–k curve near the edge of the Brillouin Zone as apparent in Figure 1.51 (a). The latter may also be viewed as a high refractivity variation with wavelength.

Questions and Problems

1.1 Maxwell's wave equation and plane waves

(a) Consider a traveling sinusoidal wave of the form $E_x = E_o\cos(\omega t - kz + \phi_o)$. The latter can also be written as $E_x = E_o\cos\left[k(vt - z) + \phi_o\right]$, where $v = \omega/k$ is the velocity. Show that this wave satisfies Maxwell's wave equation, and show that $v = 1/(\mu_o\varepsilon_o\varepsilon_r)^{1/2}$.

(b) Consider a traveling function of any shape, even a very short delta pulse, of the form $E_x = f\left[k(vt - z)\right]$, where f is any function, which can be written is $E_x = f(\phi)$, $\phi = k(vt - z)$. Show that this traveling function satisfies Maxwell's wave equation. What is its velocity? What determines the form of the function f?

1.2 Propagation in a medium of finite small conductivity An electromagnetic wave in an isotropic medium with a dielectric constant ε_r and a finite conductivity σ and traveling along z obeys the following equation for the variation of the electric field E perpendicular to z

$$\frac{d^2E}{dz^2} - \varepsilon_o\varepsilon_r\mu_o\frac{\partial^2E}{\partial t^2} = \mu_o\sigma\frac{\partial E}{\partial t} \qquad \textbf{(P1.1)}$$

Show that one possible solution is a plane wave whose amplitude decays exponentially with propagation along z, that is, $E = E_o\exp(-\alpha' z)\exp\left[j(\omega t - kz)\right]$. Here $\exp(-\alpha' z)$ causes the envelope of the amplitude to decay with z (attenuation) and $\exp\left[j(\omega t - kz)\right]$ is the traveling wave portion. Show that in a medium in which α is small, the wave velocity and the attenuation coefficient of the field are given by

$$v = \frac{\omega}{k} = \frac{1}{\sqrt{\mu_o\varepsilon_o\varepsilon_r}} \quad \text{and} \quad \alpha' = \frac{\sigma}{2\varepsilon_o cn}$$

where n is the refractive index ($n = \varepsilon_r^{1/2}$). What is the attenuation coefficient α that describes the decay of the light intensity? (Metals with high conductivities are excluded.)

1.3 Point light source What is the irradiance measured at a distance of 1 m and 2 m from a 1 W light point source?

1.4 Gaussian beam Estimate the divergence and Rayleigh range of a Gaussian beam from a He-Ne Laser with $\lambda = 633$ nm and a beam width of 1.00 nm at $z = 0$. After traversing 10 m through vacuum, what will the beam width be?

1.5 Gaussian beam in a cavity with spherical mirrors Consider an optical cavity formed by two aligned spherical mirrors facing each other as shown in Figure 1.54. Such an optical cavity is called a *spherical mirror*

resonator, and is most commonly used in gas lasers. Sometimes, one of the reflectors is a plane mirror. The two spherical mirrors and the space between them form an optical resonator because only certain light waves with certain frequencies can exist in this optical cavity. The radiation inside a spherical mirror cavity is a *Gaussian beam*. The actual or particular Gaussian beam that fits into the cavity is that beam whose wavefronts at the mirrors match the curvature of the mirrors. Consider the symmetric resonator shown in Figure 1.54 in which the mirrors have the same radius of curvature R. When a wave starts at A, its wavefront is the same as the curvature of A. In the middle of the cavity it has the minimum width and at B the wave again has the same curvature as B. Such a wave in the cavity can replicate itself (and hence exist in the cavity) as it travels between the mirrors provided that it has right beam characteristics, that is the right curvature at the mirrors. The radius of curvature R of a Gaussian beam wavefront at a distance z along its axis is given by

$$R(z) = z\left[1 + (z_o/z)^2\right]; \quad z_o = \pi w_o^2/\lambda \text{ is the Rayleigh range}$$

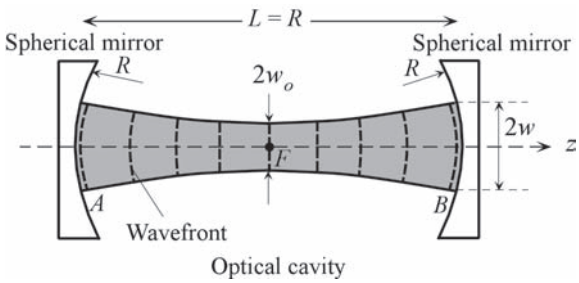


FIGURE 1.54 Two spherical mirrors reflect waves to and from each other. The optical cavity contains a Gaussian beam. This particular optical cavity is symmetric and confocal; the two focal points coincide at F.

Consider a confocal symmetric optical cavity in which the mirrors are separated by $L = R$.

(a) Show that the cavity length L is $2z_o$, that is, it is the same as twice the Rayleigh range, which is the reason the latter is called the **confocal length**.

(b) Show that the waist of the beam $2w_o$ is fully determined only by the radius of curvature R of the mirrors, and given by

$$2w_o = (2\lambda R/\pi)^{1/2}$$

(c) If the cavity length $L = R = 50$ cm, and $\lambda = 633$ nm, what is the waist of the beam at the center and also at the mirrors?

1.6 Cauchy dispersion equation Using the Cauchy coefficients and the general Cauchy equation, calculate refractive index of a silicon crystal at wavelengths of 200 μm and at 2 μm, over two orders of magnitude wavelength change. What is your conclusion?

1.7 Sellmeier dispersion equation Using the Sellmeier equation and the coefficients, obtain a graph of the refractive index of fused silica (SiO_2) versus its wavelength in the range of 500 nm to 1550 nm.

1.8 Sellmeier dispersion equation The Sellmeier dispersion coefficient for pure silica (SiO_2) and 86.5% SiO_2-13.5% GeO_2 are given in Table 1.2. Write a program on your computer or calculator, or use a math software package or even a spreadsheet program (*e.g.*, Excel) to obtain the refractive index n as a function of λ from 0.5 μm to 1.8 μm for both pure silica and 86.5% SiO_2-13.5% GeO_2. Obtain the group index, N_g, vs. wavelength for both materials and plot it on the same graph. Find the wavelength at which the material dispersion, defined as the derivative of the group velocity with respect to the wavelength, becomes zero in each material.

1.9 The Cauchy dispersion relation for zinc selenide ZnSe is a II–VI semiconductor and a very useful optical material used in various applications such as optical windows (especially high power laser windows), lenses, and prisms. It transmits over 0.50–19 μm. n in the 1–11 μm range is described by a Cauchy expression of the form

$$n = 2.4365 + \frac{0.0485}{\lambda^2} + \frac{0.0061}{\lambda^4} - 0.0003\lambda^2$$

in which λ is in μm. What are the n_{-2}, n_0, n_2 and n_4 coefficients? What is ZnSe's refractive index n and group index N_g at 5 μm?

1.10 Refractive index, reflection, and the Brewster's angle

(a) Consider light of free-space wavelength 1300 nm traveling in pure silica medium. Calculate the phase velocity and group velocity of light in this medium. Is the group velocity ever greater than the phase velocity?

(b) What is the Brewster angle (the polarization angle θ_p) and the critical angle (θ_c) for total internal reflection when the light wave traveling in this silica medium is incident on a silica–air interface. What happens at the polarization angle?

(c) What is the reflection coefficient and reflectance at normal incidence when the light beam traveling in the silica medium is incident on a silica–air interface?

(d) What is the reflection coefficient and reflectance at normal incidence when a light beam traveling in air is incident on an air–silica interface? How do these compare with part (c) and what is your conclusion?

1.11 Snell's law and lateral beam displacement What is the lateral displacement when a laser beam passes through two glass plates, each of thickness 1 mm, with refractive indices of $n_1 = 1.570$ and $n_2 = 1.450$ respectively, if the angle of incidence is 45°?

1.12 Snell's law and lateral beam displacement An engineer wants to design a refractometer (an instrument for measuring the refractive index) using the lateral displacement of light through a glass plate. His initial experiments involve using a plate of thickness L, and measuring the displacement of a laser beam when the angle of incidence θ_i is changed, for example, by rotating (tilting) the sample. For $\theta_i = 40°$ he measures a displacement of 0.60 mm, and when $\theta_i = 80°$ he measures 1.69 mm. Find the refractive index of the plate and its thickness. (*Note:* You need to solve a nonlinear equation for n numerically.)

1.13 Snell's law and prisms Consider the prism shown in Figure 1.55 that has an apex angle $\alpha = 60°$. The prism has a refractive index of n and it is in air.

(a) What are Snell's law at interfaces at A (incidence and transmittance angles of θ_i and θ_t) and B (incidence and transmittance angles of θ_i' and θ_t')?

(b) Total deflection $\delta = \delta_1 + \delta_2$ where $\delta_1 = \theta_i - \theta_t$ and $\delta_2 = \theta_t' - \theta_i'$. Now, $\beta + \theta_i' + \theta_t = 180°$ and $\alpha + \beta = 180°$. Find the deflection of the beam for an incidence angle of 45° for the following three colors at which n is known: Blue, $n = 1.4634$ at $\lambda = 486.1$ nm; yellow, $n = 1.4587$ at $\lambda = 589.2$ nm; red, $n = 1.4567$ at $\lambda = 656.3$ nm. What is the separation in distance between the rays if the rays are projected on a screen 1 m away.

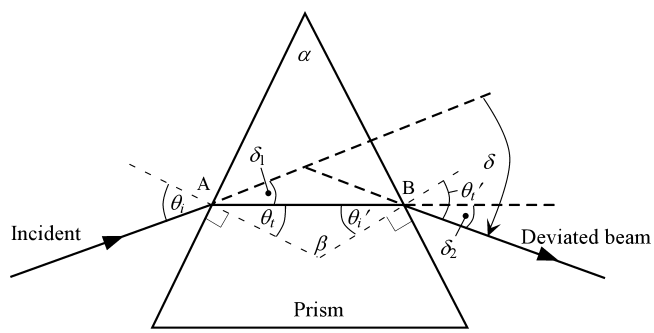


FIGURE 1.55 A light beam is deflected by a prism through an angle δ. The angle of incidence is θ_i. The apex angle of the prism is α.

1.14 Fermat's principle of least time Fermat's principle of least time in simple terms states that *when light travels from one point to another it takes a path that has the shortest time*. In going from a point A in some medium with a refractive index n_1 to a point B in a neighboring medium with refractive index n_2 as in Figure 1.56, the light path AOB involves refraction at O that satisfies Snell's law. The time it takes to travel from A to B is minimum only for the path AOB such that the incidence and refraction angles θ_i and θ_t satisfy Snell's law. Let's draw a straight line from A to B cutting the x-axes at O'. The line $AO'B$ will be our reference line and we will place the origin of x and y coordinates at O'. Without invoking Snell's law, we will vary point O along the x-axis (hence OO' is a variable labeled x), until the time it takes to travel AOB is minimum, and thereby derive Snell's law. The time t it takes for light to travel from A to B through O is

$$t = \frac{AO}{c/n_1} + \frac{OB}{c/n_2} = \frac{\left[(x_1 - x)^2 + y_1^2\right]^{1/2}}{c/n_1} + \frac{\left[(x_2 + x)^2 + y_2^2\right]^{1/2}}{c/n_2} \qquad \textbf{(P1.2)}$$

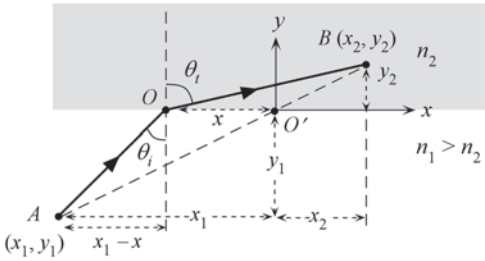


FIGURE 1.56 Consider a light wave traveling from point $A(x_1, y_2)$ to $B(x_1, y_2)$ through an arbitrary point O at a distance x from O'. The principle of least time from A to B requires that O is such that the incidence and refraction angles obey Snell's law.

The incidence and transmittance angles are given by

$$\sin\theta_i = \frac{x_1 - x}{\left[(x_1 - x)^2 + y_1^2\right]^{1/2}} \quad \text{and} \quad \sin\theta_i = \frac{(x_2 + x)}{\left[(x_2 + x)^2 + y_2^2\right]^{1/2}} \tag{P1.3}$$

Differentiate Eq. (P1.2) with respect to x to find the condition for the "least time" and then use Eq. (P1.3) in this condition to derive Snell's law.



Pierre de Fermat (1601–1665) was a French mathematician who made many significant contributions to modern calculus, number theory, analytical geometry, and probability. *(Courtesy of Mary Evans Picture Library/Alamy.)*

1.15 Antireflection (AR) coating

(a) A laser beam of wavelength 1550 nm from air is launched to a single mode optical fiber with a core refractive index $n_1 = 1.45$. Estimate the refractive index and thickness of film required for an anti-reflecting coating on this fiber.

(b) A Ge photodiode is designed to operate at 1550 nm, and it is required to have AR coatings to minimize reflected light. Two possible materials are available for AR coating: SiO_2 with a refractive index of 1.46, and TiO_2 with a refractive index of 2.2. Which would be better suited? What would be the thickness for the AR coating on this photodiode? The refractive index of Ge is about 4.

(c) Consider a Ge photodiode that is designed for operation around 1200 nm. What are the best AR coating refractive index and thickness if the refractive index of Ge is about 4.0?

1.16 Single- and double-layer antireflection V-coating For a single-layer AR coating of index n_2 on a material with index $n_3 (> n_2 > n_1)$, as shown in Figure 1.57 (a), the minimum reflectance at normal incidence is given by

$$R_{\min} = \left[\frac{n_2^2 - n_1 n_3}{n_2^2 + n_1 n_3}\right]^2$$

when the reflections $A, B, \ldots$ all interfere as destructively as possible. $R_{\min} = 0$ when $n_2 = (n_1 n_3)^{1/2}$. The choice of materials may not always be the best for a single-layer AR coating. *Double-layer AR coatings*, as shown in

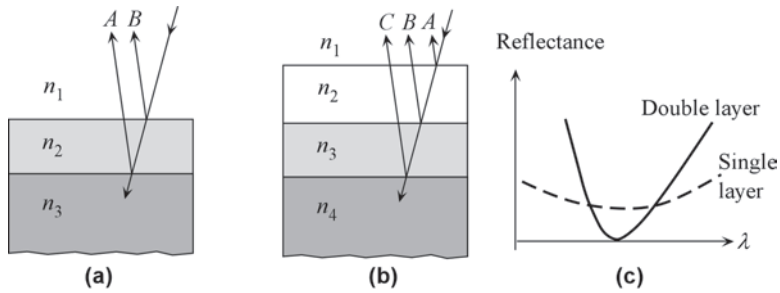


FIGURE 1.57 (a) A single-layer AR coating. (b) A double-layer AR coating and (c) its V-shaped reflectance spectrum over a wavelength range.

Figure 1.57 (b), can achieve lower and sharper reflectance at a specified wavelength as in Figure 1.57 (c). To reduce the reflection of light at the n_1–n_4 interface, two layers n_2 and n_3, each quarter wavelength in the layer (λ/n_2 and λ/n_3) are interfaced between n_1 and n_4. The reflections A, B, and C for normal incidence result in a minimum reflectance given by

$$R_{\min} = \left[\frac{n_3^2 n_1 - n_4 n_2^2}{n_3^2 n_1 + n_4 n_2^2}\right]^2$$

The double-layer reflectance vs. wavelength behavior usually has a V-shape, and such coatings are called *V-coatings.*

(a) Show that double-layer reflectance vanishes when

$$(n_2/n_3)^2 = n_1/n_4$$

(b) Consider an InGaAs, a semiconductor crystal with an index 3.8, for use in a photodetector. What is the reflectance without any AR coating?

(c) What is the reflectance when InGaAs is coated with a thin AR layer of Si_3N_4? Which material in Table 1.3 would be ideal as an AR coating?

TABLE 1.3 Typical AR materials and their approximate refractive indices over the visible wavelengths

	MgF_2	SiO_2	Al_2O_3	CeF_3	Sb_2O_3	Si_3N_4	SiO	ZrO_2	ZnS	TiO_2	CdS
n	1.38	1.46	1.65	1.65	1.9–2.1	1.95	2.0	2.05	2.35	2.35	2.60

(d) What two materials would you choose to obtain a V-coating? Note: The choice of an AR coating also depends on the technology involved in depositing the AR coating and its effects on the interface states between the AR layer and the semiconductor. $Si_{1-x}N_x$ is a common AR coating on devices inasmuch as it is a good passive dielectric layer, its deposition technology is well established and changing its composition (x) changes its index.

1.17 Single-, double-, and triple-layer antireflection coatings Figure 1.58 shows the reflectance of an uncoated glass, and glass that has a single- (1), double- (2) and triple- (3) layer AR coatings? The coating details are in the figure caption. Each layer in single- and double-layer AR coatings has a thickness of $\lambda/4$, where λ is the wavelength in the layer. The triple-layer AR layer has three coatings with thicknesses $\lambda/4$, $\lambda/2$, and $\lambda/4$. Can you qualitatively explain the results by using interference? What applications would need single-, double-, and triple-layer coatings?

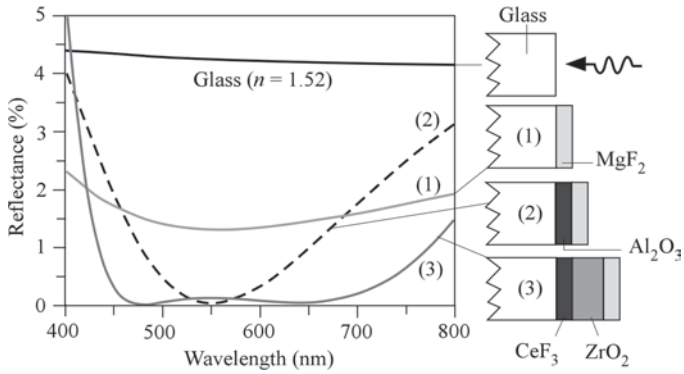


FIGURE 1.58 Reflectance vs. wavelength for a glass plate, $n = 1.52$, with and without AR coatings. (1) Single-layer AR coating is a quarter wavelength ($\lambda/4$) thick MgF_2, $n = 1.38$. (2) Double-layer coating is $\lambda/4$ thick MgF_2 and $\lambda/4$ thick Al_2O_3, $n = 1.69$. (3) Triple-layer coating is $\lambda/4$ thick MgF_2, $\lambda/2$ thick ZrO_2, $n = 2.05$, and a $\lambda/4$ thick CeF_3, $n = 1.64$. (*Source:* Plotted from data appearing in Figure 2.2 in S. Chattopadhyay *et al*., *Mater. Sci. Engin. R*, *69*, 1, 2010.)

1.18 Reflection at glass–glass and air–glass interfaces A ray of light that is traveling in a glass medium of refractive index $n_1 = 1.460$ becomes incident on a less dense glass medium of refractive index $n_2 = 1.430$. Suppose that the free-space wavelength of the light ray is 850 nm.

(a) What should the minimum incidence angle for TIR be?

(b) What is the phase change in the reflected wave when the angle of incidence $\theta_i = 85°$ and when $\theta_i = 90°$?

(c) What is the penetration depth of the evanescent wave into medium 2 when $\theta_i = 85°$ and when $\theta_i = 90°$?

(d) What is the reflection coefficient and reflectance at normal incidence ($\theta_i = 0°$) when the light beam traveling in the glass medium ($n = 1.460$) is incident on a glass–air interface?

(e) What is the reflection coefficient and reflectance at normal incidence when a light beam traveling in air is incident on an air–glass ($n = 1.460$) interface? How do these compare with part (d) and what is your conclusion?

1.19 Dielectric mirror A dielectric mirror is made up of a quarter wave layer of GaAs with $n_H = 3.38$ and AlAs with $n_L = 3.00$ at around 1550 nm. The light is incident on the mirror from another semiconductor of refractive index $n_0 = 3.40$. Find out the number of pairs of layers N needed to get 90% reflectance. Find out the bandwidth of the reflected light.

1.20 TIR and polarization at water–air interface

(a) Given that the refractive index of water is about 1.33, what is the polarization angle for light traveling in air and reflected from the surface of the water?

(b) Consider a diver in sea pointing a flashlight towards the surface of the water. What is the critical angle for the light beam to be reflected from the water surface?

1.21 Reflection and transmission at a semiconductor–semiconductor interface A light wave with a wavelength of 890 nm (free-space wavelength) that is propagating in GaAs becomes incident on AlGaAs. The refractive index of GaAs is 3.60, that of AlGaAs is 3.30.

(a) Consider normal incidence. What are the reflection and transmission coefficients and the reflectance and transmittance? (From GaAs into AlGaAs.)

(b) What is the Brewster angle (the polarization angle θ_p) and the critical angle (θ_c) for total internal reflection for the wave in (a); the wave that is traveling in GaAs and incident on the GaAs–AlGaAs interface?

(c) What is the reflection coefficient and the phase change in the reflected wave when the angle of incidence $\theta_i = 79°$?

(d) What is the penetration depth of the evanescent wave into medium 2 when $\theta_i = 79°$ and when $\theta_i = 89°$? What is your conclusion?

1.22 Phase changes on TIR Consider a light wave of wavelength 870 nm traveling in a semiconductor medium (GaAs) of refractive index 3.60. It is incident on a different semiconductor medium (AlGaAs) of refractive index

3.40, and the angle of incidence is 80°. Will this result in total internal reflection? Calculate the phase change in the parallel and perpendicular components of the reflected electric field.

1.23 Fresnel's equations Fresnel's equations are sometimes given as follows:

$$r_{\perp} = \frac{E_{ro,\perp}}{E_{io,\perp}} = \frac{n_1\cos\theta_i - n_2\cos\theta_t}{n_1\cos\theta_i + n_2\cos\theta_t}$$

$$r_{//} = \frac{E_{ro,//}}{E_{io,//}} = \frac{n_1\cos\theta_t - n_2\cos\theta_i}{n_1\cos\theta_t + n_2\cos\theta_i}$$

$$t_{\perp} = \frac{E_{to,\perp}}{E_{io,\perp}} = \frac{2n_1\cos\theta_i}{n_1\cos\theta_i + n_2\cos\theta_t}$$

and

$$t_{//} = \frac{E_{to,//}}{E_{io,//}} = \frac{2n_1\cos\theta_i}{n_1\cos\theta_t + n_2\cos\theta_i}$$

Show that these reduce to Fresnel's equation given in Eqs. (1.6.6) and (1.6.7).

Using Fresnel's equations, find the reflection and transmission coefficients for normal incidence and show that

$$r_{\perp} + 1 = t_{\perp} \quad \text{and} \quad r_{//} + nt_{//} = 1$$

where $n = n_2/n_1$.

1.24 Fresnel's equations Consider a light wave traveling in a glass medium with an index $n_1 = 1.440$ and it is incident on the glass–air interface. Using Fresnel's equations only, that is, Eqs. (1.6.6a) and (1.6.6b), calculate the reflection coefficients $r_{\perp}$ and $r_{//}$ and hence reflectances $R_{\perp}$ and $R_{//}$ for (a) $\theta_i = 25°$ and (b) $\theta_i = 50°$. In the case of $\theta_i = 50°$, find the phase change $\phi_{\perp}$ and $\phi_{//}$ from the reflection coefficients by writing $r = |r|\exp(-j\phi)$. Compare $\phi_{\perp}$ and $\phi_{//}$ from $r_{\perp}$ and $r_{//}$ calculations with those calculated from Eqs. (1.6.11) and (1.6.12).

1.25 Goos-Haenchen phase shift A ray of light which is traveling in a glass medium (1) of refractive index $n_1 = 1.460$ becomes incident on a less dense glass medium (2) of refractive index $n_2 = 1.430$. Suppose that the free-space wavelength of the light ray is 850 nm. The angle of incidence $\theta_i = 85°$. Estimate the lateral Goos-Haenchen shift in the reflected wave for the perpendicular field component. Recalculate the Goos-Haenchen shift if the second medium has $n_2 = 1$ (air). What is your conclusion? Assume that the virtual reflection occurs from a virtual plane in medium B at a distance d that is roughly the same as the penetration depth. Note that d actually depends on the polarization, the direction of the field, but we will ignore this dependence.

1.26 Evanescent wave Total internal reflection of a plane wave from a boundary between a more dense medium (1) n_1 and a less dense medium (2) n_2 is accompanied by an evanescent wave propagating in medium 2 near the boundary. Find the functional form of this wave and discuss how its magnitude varies with the distance into medium 2.

1.27 TIR and FTIR

(a) By considering the electric field component in medium B in Figure 1.21, explain how you can adjust the amount of transmitted light through a thin layer between two higher refractive index media.

(b) What is the critical angle at the hypotenuse face of a beam splitter cube made of glass with $n_1 = 1.6$ and having a thin film of liquid with $n_2 = 1.3$. Can you use 45° prisms with normal incidence?

(c) Explain how a light beam can propagate along a layer of material between two different media as shown in Figure 1.59 (a). Explain what the requirements are for the indices n_1, n_2, n_3. Will there be any losses at the reflections?

(d) Consider the prism coupler arrangement in Figure 1.59 (b). Explain how this arrangement works for coupling an external light beam from a laser into a thin layer on the surface of a glass substrate. Light is then propagated inside the thin layer along the surface of the substrate. What is the purpose of the adjustable coupling gap?

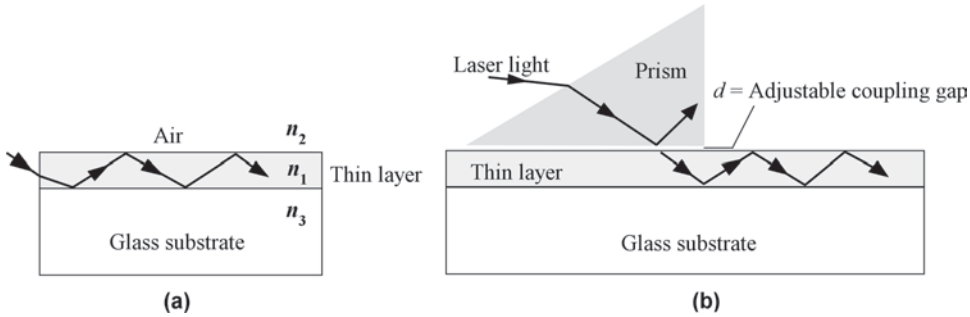


FIGURE 1.59 (a) Light propagation along an optical guide. (b) Coupling of laser light into a thin layer—optical guide—using a prism. The light propagates along the thin layer.

1.28 Complex refractive index and dielectric constant The complex refractive index $N = n - jK$ can be defined in terms of the *complex relative permittivity* $\varepsilon_r = \varepsilon_{r1} - j\varepsilon_{r2}$ as

$$\mathrm{N} = n - jK = \varepsilon_r^{1/2} = (\varepsilon_{r1} - j\varepsilon_{r2})^{1/2}$$

Show that

$$n = \left[\frac{(\varepsilon_{r1}^2 + \varepsilon_{r2}^2)^{1/2} + \varepsilon_{r1}}{2}\right]^{1/2} \quad \text{and} \quad K = \left[\frac{(\varepsilon_{r1}^2 + \varepsilon_{r2}^2)^{1/2} - \varepsilon_{r1}}{2}\right]^{1/2}$$

1.29 Complex refractive index Spectroscopic ellipsometry measurements on a germanium crystal at a photon energy of 1.5 eV show that the real and imaginary parts of the complex relative permittivity are 21.56 and 2.772, respectively. Find the complex refractive index. What is the reflectance and absorption coefficient at this wavelength? How do your calculations match with the experimental values of $n = 4.653$ and $K = 0.298$, $R = 0.419$ and $\alpha = 4.53 \times 10^6\,\mathrm{m}^{-1}$?

1.30 Complex refractive index Figure 1.26 shows the infrared extinction coefficient K of CdTe. Calculate the absorption coefficient α and the reflectance R of CdTe at 60 μm and 80 μm.

1.31 Refractive index and attenuation in the infrared region—Reststrahlen absorption Figure 1.26 shows the refractive index n and the extinction coefficient K as a function of wavelength λ in the infrared for a CdTe crystal due to lattice absorption, called Reststrahlen absorption. It results from the ionic polarization of the crystal induced by the optical field in the light wave. The relative permittivity ε_r due to positive (Cd^{2+}) and negative (Te^{2-}) ions being made to oscillate by the optical field about their equilibrium positions is given in its simplest form by

$$\varepsilon_r = \varepsilon_r' - j\varepsilon_r'' = \varepsilon_{rH} + \frac{\varepsilon_{rH} - \varepsilon_{rL}}{\left(\dfrac{\omega}{\omega_T}\right)^2 - 1 + j\dfrac{\gamma}{\omega_T}\left(\dfrac{\omega}{\omega_T}\right)} \qquad \textbf{(P1.4)}$$

where ε_{rL} and ε_{rH} are the relative permittivity at low (L) and high (H) frequencies, well below and above the infrared peak, γ is a loss coefficient characterizing the rate of energy transfer from the EM wave to lattice vibrations (phonons), and ω_T is a transverse optical lattice vibration frequency that is related to the nature of bonding between the ions in the crystal. Table 1.4 provides some typical values for CdTe and GaAs. Equation (P1.4) can be used to obtain a reasonable approximation to the infrared refractive index n and extinction coefficient K due to Reststrahlen absorption. (a) Consider CdTe, and plot n and K vs. λ from 40 μm to 90 μm and compare with the experimental results in Figure 1.26 in terms of the peak positions and the width of the extinction coefficient peak. (b) Consider

TABLE 1.4 Ionic polarization resonance parameters for CdTe and GaAs

	ε_{rL}	ε_{rH}	$\omega_T(\mathrm{rad\ s^{-1}})$	$\gamma(\mathrm{rad\ s^{-1}})$
CdTe	10.20	7.10	2.68×10^{13}	0.124×10^{13}
GaAs	13.0	11.0	5.07×10^{13}	0.045×10^{13}

GaAs, and plot n and K vs. λ from 30 μm to 50 μm. (c) Calculate n and K for GaAs at $\lambda = 38.02$ μm and compare with the experimental values $n = 7.55$ and $K = 0.629$. (You might want to use a logarithmic scale for K.)

1.32 Coherence length A narrow band pass filter transmits wavelengths in the range 5000 ± 0.5 A°. If this filter is placed in front of a source of white light, what is the coherence length of the transmitted light?

1.33 Spectral widths and coherence

(a) Suppose that frequency spectrum of a radiation emitted from a source has a central frequency υ_o and a spectral width $\Delta\upsilon$. The spectrum of this radiation in terms of wavelength will have a central wavelength λ_o and a spectral width $\Delta\lambda$. Clearly, $\lambda_o = c/\upsilon_o$. Since $\Delta\lambda \ll \lambda_o$ and $\Delta\upsilon \ll \upsilon_o$, using $\lambda = c/\upsilon$, show that the line width $\Delta\lambda$ and hence the coherence length l_c are

$$\Delta\lambda = \Delta\upsilon\frac{\lambda_o}{\upsilon_o} = \Delta\upsilon\frac{\lambda_o^2}{c} \quad \text{and} \quad l_c = c\Delta t = \frac{\lambda_o^2}{\Delta\lambda}$$

(b) Calculate $\Delta\lambda$ for a lasing emission from a He-Ne laser that has $\lambda_o = 632.8$ nm and $\Delta\upsilon \approx 1.5$ GHz. Find its coherence time and length.

1.34 Coherence lengths Find the coherence length of the following light sources:

(a) An LED emitting at 1550 nm with a spectral width 150 nm;
(b) A semiconductor laser diode emitting at 1550 nm with a spectral width 3 nm;
(c) A quantum well semiconductor laser diode emitting at 1550 nm with a spectral width of 0.1 nm;
(d) A multimode He-Ne laser with a spectral frequency width of 1.5 GHz;
(e) A specially designed single mode and stabilized He-Ne laser with a spectral width of 100 MHz.

1.35 Fabry–Perot optical cavity Consider an optical cavity formed between two identical mirrors, each with reflectance = 0.97. The refractive index of the medium enclosed between the mirrors is 1. Find out the minimum length of the optical cavity which can resolve spectral lines of a sodium lamp with line width $\Delta\lambda = 0.6$ nm and $\Delta\lambda = 589.3$ nm. Further, estimate the mode separation in frequency and wavelength. What are the finesse F and Q factors for this cavity?

1.36 Fabry–Perot optical cavity from a ruby crystal Consider a ruby crystal of diameter 1 cm and length 10 cm. The refractive index is 1.78. The ends have been silvered and the reflectances are 0.99 and 0.95 each. What is the nearest mode number that corresponds to a radiation of wavelength 694.3 nm? What is the actual wavelength of the mode closest to 694.3 nm? What is the mode separation in frequency and wavelength? What are the finesse F and Q factor for the cavity?

1.37 Fabry–Perot optical cavity spectral width Consider an optical cavity of length 40 cm. Assume the refractive index is 1, and use Eq. (1.11.3) to plot the peak closest to 632.8 nm for 4 values of $R = 0.99, 0.90, 0.75$ and 0.6. For each case find the spectral width $\delta\lambda_m$, the finesse F and Q. How accurate is Eq. (1.11.5) in predicting $\delta\lambda_m$. (You may want to use a graphing software for this problem.)

1.38 Diffraction A collimated beam of light of wavelength 632.8 nm is incident on a circular aperture of 250 μm. Find out the divergence of the transmitted beam. Obtain the diameter of the transmitted beam at a distance of 10 m. What would be the divergence if the aperture is a single slit of width 250 μm?

1.39 Diffraction intensity Consider diffraction from a uniformly illuminated circular aperture of diameter D. The far field diffraction pattern is given by a Bessel function of the first kind and first order, J_1, and the intensity at a point P on the angle θ_i with respect to the central axis through the aperture is

$$I(\gamma) = I_o\left(\frac{2J_1(\gamma)}{\gamma}\right)^2$$

where I_o is the maximum intensity, $\gamma = (1/2)kD\sin\theta$ is a variable quantity that represents the angular position θ on the screen as well as the wavelength ($k = 2\pi/\lambda$) and the aperture diameter D. $J_1(\gamma)$ can be calculated from

$$J_1(\gamma) = \frac{1}{\pi}\int_0^{\pi}\cos(\alpha - \gamma\sin\alpha)d\alpha$$

where α is an integration variable. Using numerical integration, or a suitable mathematics software program, plot $\left[J_1(\gamma)/\gamma\right]$ vs. γ for $\gamma = 0 - 8$ and confirm that zero-crossings occur at $\gamma = 3.83, 7.02$ and the maxima at $\gamma = 0, 5.14$. What is the intensity ratio of the first bright ring (at $\gamma = 5.14$) to that at the center of the Airy disk ($\gamma = 0$)? (You can use a very small γ instead of zero for the center intensity calculation.) Using the first zero at $\gamma = 3.83$, verify Eq. (1.12.5), $\sin\theta_o = 1.22\lambda/D$, where θ_o is the angular position of the first dark ring, as defined in Figure 1.36 (b).

2.32 Cut-back method of attenuation measurement Cut-back method is a destructive measurement technique for determining the attenuation α of a fiber. The first part of the experiment involves measuring the optical power P_{far} coming out from the fiber at the far end as shown in Figure 2.54. Then, in the second part, keeping everything the same, the fiber is cut close to the launch or the source end. The output power P_{near} is measured at the near end from the shortcut fiber. The attenuation is then given by

$$\alpha = (-10/L) \log (P_{\text{far}}/P_{\text{near}})$$

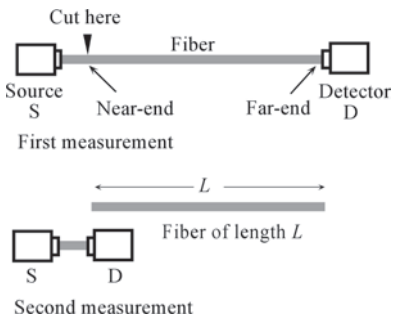


FIGURE 2.54 Illustration of the cut-back method for measuring the fiber attenuation. S is an optical source and D is a photodetector.

in which L is the separation of the measurement points, the length of the cut fiber, and α is in dB per unit length. The output P_{near} from the shortcut fiber in the second measurement is actually the input into the fiber under test in the first experiment. Usually a mode scrambler (mode stripper) is used for multimode fibers before the input. The power output from a particular fiber is measured to be 13 nW. Then, 10 km of fiber is cut out and the power output is measured again and found to be 43 nW. What is the attenuation of the fiber?

2.33 Intrinsic losses

(a) Consider a silica-based standard single-mode fiber with a NA of 0.15. What is its attenuation at 1550 nm? How does this compare with the attenuation quotes for the Corning SMF@28e$^+$, 0.16–0.19 dB km^{-1}at 1550 nm?

(b) Consider a step index fiber $n_1 = 1.47$ and $n_2 = 1.46$, core diameter $= 50$ μm. Corresponding to the maximum value of the incident angle for which light rays will be guided through this fiber, calculate the number of reflections and the loss that would take place in one kilometre length of the fiber assuming 0.01% loss of power at each reflection.

2.34 Scattering losses and fictive temperature Rayleigh scattering process decreases with wavelength, and as mentioned in Chapter 1, it is inversely proportional to λ^4. The expression for the attenuation α_R in a single component glass such as silica due to Rayleigh scattering is approximately given by two sets of different equations in the literature[32]

$$\alpha_R \approx \frac{8\pi^3}{3\lambda^4} n^8 p^2 \beta_T k_B T_f \quad \text{and} \quad \alpha_R \approx \frac{8\pi^3}{3\lambda^4} (n^2 - 1)^2 \beta_T k_B T_f$$

in which λ is the free-space wavelength, n is the refractive index at the wavelength of interest, β_T is the isothermal compressibility (at T_f) of the glass, k_B is the Boltzmann constant, and T_f is a quantity called the *fictive temperature* (or the glass transition temperature) at which the liquid structure during the cooling of the fiber is frozen to become the glass structure. Fiber is drawn at high temperatures. As the fiber cools, eventually the temperature drops sufficiently for the atomic motions to be so sluggish that the structure becomes essentially "frozen-in" and remains like this even at room temperature. Thus, T_f marks the temperature below which the liquid structure is frozen and hence the density fluctuations are also frozen into the glass structure. Use these two equations and calculate the attenuation in dB km^{-1} due to Rayleigh scattering at $\lambda = 1.55\,\mu$m given that pure silica (SiO_2) has the following properties: $T_f \approx 1180°C$; $\beta_T \approx 7 \times 10^{-11}\,\text{m}^2\,\text{N}^{-1}$ (at high temperatures); $n = 1.5$ at 1.5 μm; $p = 0.28$. The lowest reported attenuation around this wavelength is about 0.14 dB km^{-1}. What is your conclusion? (Note that α_B above is in Nepers per meter.)

2.35 Bending loss Bending losses always increase with the mode field diameter. Since the MFD increases for decreasing V, $2w \approx 2 \times 2.6a/V$, smaller V fibers have higher bending losses. How does the bending loss

[32] For example, R. Olshansky, *Rev. Mod. Phys.*, *51*, 341, 1979.

α vs. radius of curvature R behavior look like on a semilogarithmic plot [as in Figure 2.39 (a)] for two values of the V-number V_1 and V_2 if $V_2 > V_1$? It is found that for a single-mode fiber with a cutoff wavelength $\lambda_c = 1180\,\text{nm}$, operating at 1300 nm, the microbending loss reaches 1 dB m^{-1} when the radius of curvature of the bend is roughly 6 mm for $\Delta = 0.00825$, 12 mm for $\Delta = 0.00550$, and 35 mm for $\Delta = 0.00275$. Explain these findings.

2.36 Bend loss reduced fibers Consider the bend loss measurements listed in Table 2.8 for four different types of fiber. The trench fibers have a trench placed in the cladding where the refractive index is lowered as shown in Figure 2.39. The nanoengineered fiber is shown in Figure 2.55. There is a ring of region in the cladding in which there are gas-filled nanoscale voids. (They are introduced during fabrication.) A void in the ring has a circular cross-section but has a length along the fiber that can be a few meters. These voids occupy a volume in the ring that is only 1–10%. Plot the bending loss semilogarithmically (α on a log scale and R on a linear scale) and fit the data to $\alpha_{\text{micobend}} = A\exp(-R/R_c)$ and find A and R_c. What is your conclusion? Suppose that we set our maximum acceptable bending loss to 0.1 dB turn^{-1} in installation (the present goal is to bring the bending loss to below 0.1 dB turn^{-1}). What are the allowed radii of curvature for each turn?

TABLE 2.8 Bend radius R in mm, α in dB turn^{-1}. Data over 1.55–1.65 μm range

Standard SMF[a] 1550 nm		Trench Fiber 1[b] 1650 nm		Trench Fiber 2[c] 1625 nm		Nanoengineered Fiber[a] 1550 nm	
R (mm)	α (dB turn^{-1})	R (mm)	α (dB turn^{-1})	R (mm)	α (dB turn^{-1})	R (mm)	α (dB turn^{-1})
5.0	15.0	7.50	0.354	5.0	0.178	5.0	0.031
7.0	4.00	10.0	0.135	7.5	0.0619	7.5	0.0081
10.0	0.611	15.0	0.020	10.0	0.0162	10.0	0.0030
12.5	0.124			15.0	0.00092	15.0	0.00018
16.0	0.0105						
17.5	0.0040						

(*Source:* Data used from a number of sources: [a]M. -J. Li *et al.*, *J. Light Wave Technol.*, *27*, 376, 2009; [b]K. Himeno *et al.*, *J. Light Wave Technol.*, *23*, 3494, 2005; [c]L. -A. de Montmorillon, *et al.* "Bend-Optimized G.652D Compatible Trench-Assisted Single-Mode Fibers," *Proceedings of the 55th IWCS/Focus*, pp. 342–347, November, 2006.)

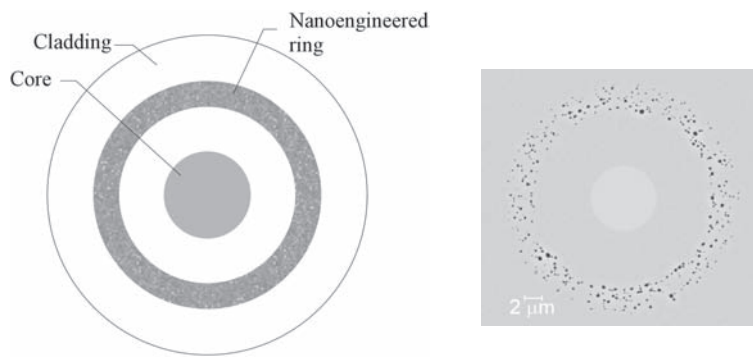


FIGURE 2.55 Left: The basic structure of bend-insensitive fiber with a nanoengineered ring in the cladding. Right: An SEM picture of the cross-section of a nanoengineered fiber with reduced bending losses. *(Courtesy of Ming-Jun Li, Corning Inc. For more information see U.S. Patent 8,055.110, 2011.)*

2.37 Microbending loss Microbending loss α_B depends on the fiber characteristics and wavelength. We will calculate α_B approximately given various fiber parameters using the single-mode fiber microbending loss equation (D. Marcuse, *J. Op. Soc. Am.*, *66*, 216, 1976)

$$\alpha_B \approx \frac{\pi^{1/2}\kappa^2}{2\gamma^{3/2}V^2\left[K_1(\gamma a)\right]^2} R^{-1/2} \exp\left(-\frac{2\gamma^3}{3\beta^2} R\right)$$

where R is the bend radius of curvature; a is the fiber radius; β is the propagation constant, determined by b, normalized propagation constant, which is related to V, $\beta = n_2k\left[1 + b\Delta\right]$; $k = 2\pi/\lambda$ is the free-space wave vector; $\gamma = [\beta^2 - n_2^2k^2]^{1/2}$; $\kappa = [n_1^2k^2 - \beta^2]^{1/2}$; and $K_1(x)$ is a first-order modified Bessel function, available in math software packages. The normalized propagation constant b can be found from $b = (1.1428 - 0.996V^{-1})^2$. Consider a single-mode fiber with $n_1 = 1.450, n_2 = 1.446, 2a$ (diameter) $= 3.9\,\mu$m. Plot α_B vs. R for $\lambda = 633$ nm and 790 nm from $R = 2$ mm to 15 mm. Figure 2.56 shows the experimental results on an SMF that has the same properties as the fiber above. While the data in Figure 2.56 are for a 10 cm fiber, the equation above gives the attenuation per meter. We need to divide the calculated values by 10 to compare the two sets of data. What is your conclusion? (Note that while the data in Figure 2.56 are for a 10 cm fiber, the equation above gives the attenuation per meter. We need to divide the calculated attenuation values by 10 to compare the two sets of data. You might wish to compare your calculations with the experiments of A. J. Harris and P. F. Castle, *IEEE J. Light Wave Technol., LT4*, 34, 1986).

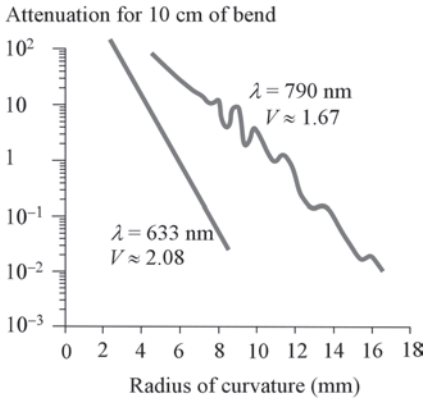


FIGURE 2.56 Measured microbending loss (attenuation) for a 10-cm fiber bent by different amounts of radius of curvature R. Single-mode fiber with a core diameter of 3.9 μm, cladding radius of 48 μm, $\Delta = 0.00275$, NA ≈ 0.10, $V \approx 1.67$ and 2.08. (*Source:* Data extracted from A. J. Harris and P. F. Castle, "Bend Loss Measurements on High Aperture Single-Mode Fibers as a Function of Wavelength and Bend Radius," *IEEE J. Light Wave Technol.*, *LT14*, 34, 1986, and replotted with a smoothed curve; [see original article for the discussion of peaks in α_B vs. R at 790 nm]).

2.38 Fiber Bragg Grating (FBG) A silica fiber-based FBG is required to operate at 850 nm. What should be the periodicity of the grating Λ? If the amplitude of the index variation Δn is 2×10^{-5} and total length of the FBG is 5 mm, what are the maximum reflectance at the Bragg wavelength and the bandwidth of the FBG? Assume that the effective refractive index $\overline{n}$ is 1.460. What are the reflectance and the bandwidth if Δn is 2×10^{-4}?



Fiber Bragg Grating (FBG)–based optical strain sensor for monitoring strains in civil structures, for example, bridges, dams, buildings, and tunnels. The fiber is mounted on a steel carrier and maintained stretched. *(Courtesy of Micron Optics.)*

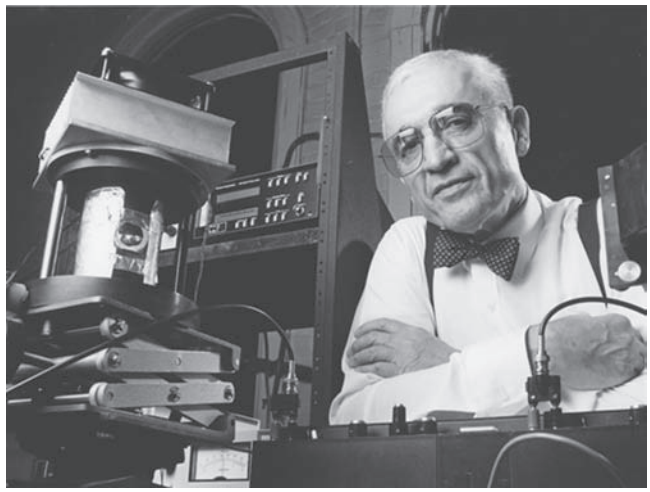
2.39 Fiber Bragg Grating sensor array Consider a FBG strain sensor array embedded in a silica fiber that is used to measure strain at various locations on an object. Two neighboring sensors have grating periodicities of $\Lambda_1 = 534.5$ nm and $\Lambda_2 = 539.7$ nm. The effective refractive index is 1.450 and the photoelastic coefficient is 0.22. What is the maximum strain that can be measured? What would be the main problem with this sensor array? What is the strain at fracture if the fiber fractures roughly at an applied stress of 700 MPa and the elastic modulus is 70 GPa? What is your conclusion?



Fiber Bragg Grating (FBG)–based optical temperature sensors for use from −200°C to 250°C. FBGs are mounted in different packages: top left, all dielectric; top right, stainless steel; and bottom, copper. The sensors operate over 1510–1590 nm. *(Courtesy of Lake Shore Cryotronics Inc.)*

I wanted visible light, because I knew that if I could get red light, other colors would be possible.

—Nick Holonyak Jr.



Nick Holonyak Jr. carried out the early work in the development of practical light-emitting diodes (LEDs) in the visible spectrum during the 1960s while working as a consulting research scientist for General Electric Co. in Syracuse. He made his first visible laser-LED in 1962, which emitted red light.[1] It served as the basis for the concept of the "ultimate lamp." Since 1963, he has been at the University of Illinois at Urbana-Champaign, where he currently holds the John Bardeen Endowed Chair. In the February 1963 issue of *Readers Digest*, Nick Holonyak Jr. suggested that the incandescent light bulb will eventually be replaced by the LED. The above photo was taken around 1970–1975. *(Courtesy of University of Illinois at Urbana-Champaign.)*

[1] N. Holonyak Jr. and S. F. Bevacqua, "Coherent (Visible) Light Emission from a Ga(As_xP_{1-x}) Junction," *Appl. Phys. Letts., 1*, 82, 1962.

3 SEMICONDUCTOR SCIENCE AND LIGHT-EMITTING DIODES

3.1 REVIEW OF SEMICONDUCTOR CONCEPTS AND ENERGY BANDS

A. Energy Band Diagrams, Density of States, Fermi–Dirac Function and Metals

We know from modern physics that the energy of the electron in an atom is quantized and can only have certain discrete values as illustrated for the Li atom in Figure 3.1. The isolated Li atom has two electrons in the 1*s* shell and one electron in the 2*s* subshell. The same concept also applies to the electron energy in a molecule with several atoms. Again the electron energy is quantized. However, when we bring together something like 10^{23} Li atoms to form the metal crystal, the interatomic interactions result in the formation of electron **energy bands**. The 2*s* energy level splits into some 10^{23} closely spaced energy levels that effectively form an **energy band**, which is called the 2*s* band. Similarly other higher energy levels also form bands as illustrated in Figure 3.1. These energy bands overlap to form one continuous energy band which represents the energy band structure of a metal. The 2*s* energy level in the Li atom is half full (2*s* subshell needs 2 electrons but has 1) which means that the 2*s* band in the crystal will also be half full. Metals characteristically have partially filled energy bands.

In the case of metals, the energy band diagram is straightforward and essentially one band of energies in which the energy levels from the bottom of the band to a certain level are all filled at zero Kelvin (0 K), as indicated in Figure 3.1. Each energy level in this band corresponds to a wavefunction for the electron inside the crystal, and represents a **quantum state** for the electron. Thus, the energy levels are often referred to as energy states. In the solid, the bottom of the band is normally taken as the zero of energy so that electrons fill all the states until a certain energy called the **Fermi energy**, denoted as E_F—the highest occupied energy level at 0 K. The states above E_F are empty up to the vacuum level. The energy required to remove an electron from the metal is the energy involved in taking an electron from E_F to the vacuum level, which is called the **work function** Φ of the metal. For example, a photon with an energy $h\upsilon$ greater than Φ that is incident on a metal can cause an electron at E_F to become ejected out from the metal to become free, a phenomenon called the **photoelectric effect**. The electron is said to be **photoemitted.** Many photomultiplier tubes in optoelectronics have a cathode material with the right Φ that allows electrons to be photoemitted when light within the desired range of wavelengths is incident on the cathode.

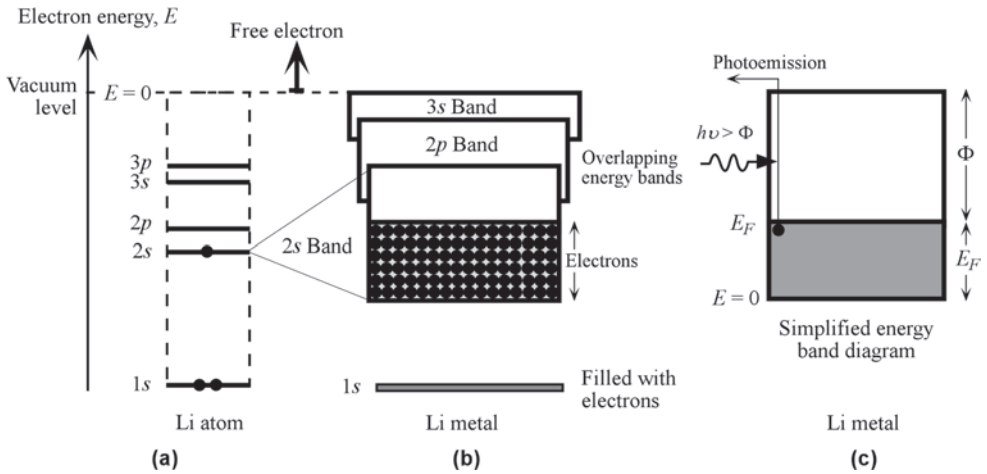


FIGURE 3.1 (a) Energy levels in a Li atom are discrete. (b) The energy levels corresponding to outer shells of isolated Li atoms form an energy band inside the crystal—for example, the 2*s* level forms a 2*s* band. Energy levels form a quasi-continuum of energy within the energy band. Various energy bands overlap to give a single band of energies that is only partially full of electrons. There are states with energies up to the vacuum level where the electron is free. (c) A simplified energy band diagram for a metal, and the photoelectric effect.

The photoemitted electrons are then accelerated by the applied field inside the multiplier tube and are made incident on other electrodes to cause secondary electron emission and hence multiplication. (One incident photon can give rise to 10^6 collected electrons.)

The electrons around E_F control many of the properties of metals such as electrical and thermal conductivity. When an electric field E is applied, an electron at E_F can gain energy from the external force eE and be able to move to a higher energy state since these are empty and available. Electrical conduction occurs by the drift of electrons at around E_F.

Many important properties of both metals (and semiconductors below) are described by considering the behavior of electrons within an energy band. In the case of metals, this is called the **free electron model**; that is, the electrons are free inside the metal and hence electron potential energy is constant. These electrons are what are normally called conduction electrons.

There are two important concepts in describing free electrons in an energy band. The **density of states** (DOS) $g(E)$ represents the number of electronic states (electron wavefunctions) in a band per unit energy per unit volume of the crystal. We can use quantum mechanics to calculate the DOS by considering how many electron wavefunctions there are within a given energy range per unit volume of the crystal. Figure 3.2 (a) shows the energy band diagram of a metal, and (b) shows, in a simplified way, how $g(E)$ depends on the electron energy in the band. According to quantum mechanics, for an electron confined within a three-dimensional potential energy well, as one might expect for a conduction electron in the crystal of a metal, DOS $g(E)$ increases with energy from the bottom of the band and is given by

Density of states in a band

$$g(E) = 4\pi (2m_e)^{3/2} h^{-3} E^{1/2} = AE^{1/2} \tag{3.1.1}$$

where E is the electron energy from the bottom of the band, m_e is the mass of the electron, h is Planck's constant, and A is a constant that represents the quantities multiplying $E^{1/2}$ in the second term. The DOS gives information on available states, that is, wavefunctions, for the electron and not on their actual occupation.

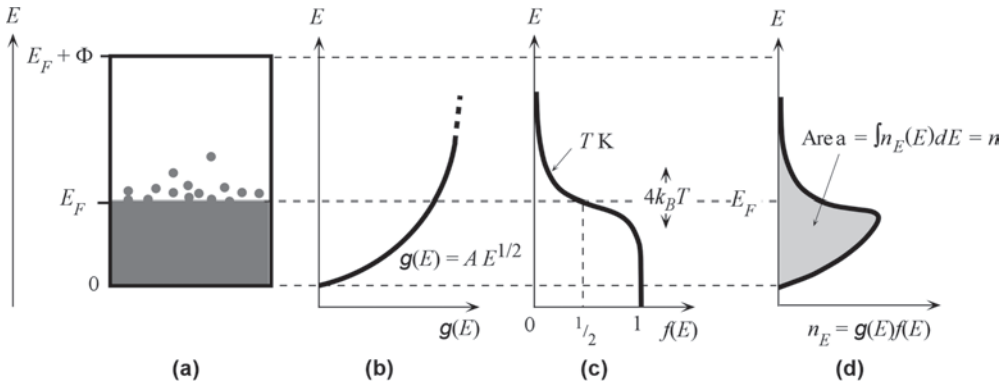


FIGURE 3.2 (a) Above 0 K, due to thermal excitation, some of the electrons are at energies above E_F. (b) The density of states, $g(E)$ vs. E in the band. (c) The probability of occupancy of a state at an energy E is $f(E)$. The product $g(E)f(E)$ is the number of electrons per unit energy per unit volume or electron concentration per unit energy. The area under the curve with the energy axis is the concentration of electrons in the band, n.

The **Fermi–Dirac function** $f(E)$ is the probability of finding an electron in a quantum state with energy E (state implies a wavefunction). It is a fundamental property of a collection of interacting electrons in *thermal equilibrium.* It is given by

$$f(E) = \frac{1}{1 + \exp\left(\dfrac{E - E_F}{k_B T}\right)} \tag{3.1.2}$$

Fermi–Dirac function

where k_B is the Boltzmann constant, T is the temperature (K), and E_F is the Fermi energy. Figure 3.2 (c) shows the dependence of $f(E)$ on the energy E. At low energies well below E_F, all states are occupied and at high energies, well above E_F, the states are empty. At E_F, $f(E_F) = 1/2$, half the states are occupied. Changes in f(E) occur around E_F over several k_BTs of energy, as indicated in Figure 3.2 (c). The higher the temperature, the greater is the reach of $f(E)$ to higher energies, and therefore more electrons at higher energies.

Consider the product $g(E)f(E)$, density of states at $E \times$ probability of finding an electron in a state at E. The product $g(E)f(E)$ represents the number electrons per unit energy per unit volume, $n_E(E)$, in the band which is shown in Figure 3.2 (d). It is essentially the energy distribution of electrons in the band. Thus, $n_E\,dE = g(E)f(E)dE$ is the number of electrons in the energy range E to $E + dE$. Integrating this from the bottom to the top of the band gives the conduction electron concentration n in the band,

$$n = \int_0^{E_F+\Phi} g(E)f(E)dE \tag{3.1.3}$$

Equations (3.1.1) and (3.1.2) can be substituted in Eq. (3.1.3) to evaluate the integration. Once integrated, the resulting equation will relate n to the Fermi energy E_F at a temperature T. Put differently, E_F depends on n, the total number of electrons per unit volume in the band. At 0 K, the Fermi energy E_{FO} is

$$E_{FO} = \left(\frac{h^2}{8m_e}\right)\left(\frac{3n}{\pi}\right)^{2/3} \tag{3.1.4}$$

Fermi energy at 0 K for metals

E_{FO} is typically a few electron volts, that is, orders of magnitude greater than k_BT. Above 0 K, E_F decreases slightly with T, though we ignore this tiny change in using metal band diagrams in optoelectronics. Inasmuch as the electrons in the metal are free, they have kinetic energies only. An electron at the Fermi energy E_{FO} has several eVs of kinetic energy, which is orders of magnitude greater than the classical kinetic theory (roughly $3k_BT/2$ for a free particle). The reason is that the electrons in the metal must obey the *Pauli exclusion principle*, which prevents more than two electrons (with spins up and down) from occupying a given energy state. Thus, the electrons must distribute themselves among available states and need to reach higher energies to satisfy the Pauli exclusion principle.

B. Energy Band Diagrams of Semiconductors

The electron energies in a semiconductor crystal, however, are distinctly different than that for metals. Figure 3.3 (a) shows a simplified two-dimensional view of the silicon crystal which has each Si atom bonding to four neighbors. All the four valence electrons per atom are used in these bonds. The interactions between the Si atoms and their valence electrons result in the electron energy in the crystal falling into two distinct energy bands called the **valence band** (VB) and **conduction band** (CB) that are separated by an energy gap, **bandgap** (E_g), as shown in Figure 3.3 (b). There are no allowed electron energies in the **bandgap**; it represents the forbidden electron energies in the crystal. The valence band represents electron wavefunctions in the crystal that correspond to bonds between the atoms. Electrons that occupy these wavefunctions are the valence electrons. Since at absolute zero of temperature all the bonds are occupied by valence electrons (there are no broken bonds), all the energy levels in the VB are normally filled with these electrons. The CB represents electron wavefunctions in the crystal that have higher energies than those in the VB and are normally empty at zero Kelvin. The top of the VB is labeled E_v, bottom of conduction band E_c, so that $E_g = E_c - E_v$ is the bandgap. The width of the CB is called the **electron affinity** χ.

An electron placed in the CB is free to move around the crystal and also to respond to an electric field because there are plenty of neighboring empty energy levels. This electron can easily gain energy from the field and move to higher energy levels because these states are

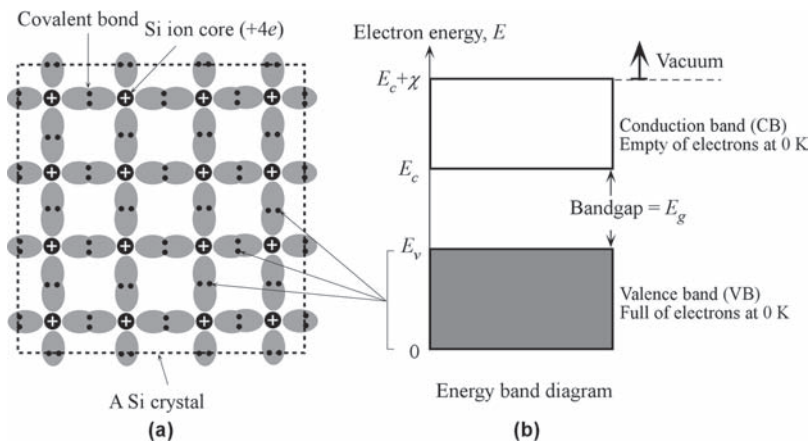


FIGURE 3.3 (a) A simplified two-dimensional view of a region of the Si crystal showing covalent bonds. (b) The energy band diagram of electrons in the Si crystal at absolute zero of temperature. The bottom of the VB has been assigned a zero of energy.

empty in the CB. Generally we can treat an electron in the CB as if it were free within the crystal by simply assigning an **effective mass** m_e^* to it. This effective mass is a quantum mechanical quantity, which takes into account that the electron in the CB interacts with a periodic potential energy as it moves through the crystal so that its inertial resistance to acceleration (definition of mass) is not the same as if it were free in vacuum.

Since the only empty states are in the CB, the excitation of an electron from the VB requires a minimum energy of E_g. Figure 3.4 illustrates what happens when an incident photon of energy $h\upsilon > E_g$ interacts with an electron in the VB. This electron absorbs the incident photon and gains sufficient energy to surmount the energy gap E_g and reach the CB. Consequently, a free electron in the CB and a "hole," corresponding to a missing electron in the VB, are created. In some semiconductors, such as Si and Ge, the photon absorption process also involves lattice vibrations (vibrations of the Si atoms) which we have not shown in Figure 3.4.

The empty electronic state, or the missing electron in the bond, is what we call a **hole** in the valence band. The free electron, which is in the CB, can wander around the crystal and contribute to the electrical conduction when an electric field is applied. The region remaining around the hole in the VB is positively charged because a charge of $-e$ has been removed from an otherwise neutral region of the crystal. This hole, denoted as h^+, can also wander around the crystal as if it were "free." This is because an electron in a neighboring bond can "jump," that is, tunnel into the hole, to fill the vacant electronic state at this site and thereby create a hole at its original position as illustrated in Figure 3.5. This is effectively equivalent to the hole being displaced in the opposite direction. Thus, conduction in semiconductors occurs by both electrons and holes with charges $-e$ and $+e$ and their own effective masses m_e^* and m_h^*.

When a wandering electron in the CB meets a hole in the VB, it has found an empty electronic state of lower energy and it therefore occupies it. The electron falls from the CB to the VB to fill the hole. This is called **recombination**, which results in the annihilation of an electron in the CB and a hole in the VB as in Figure 3.5. The excess energy of the electron falling from CB to VB in certain semiconductors such as GaAs and InP is emitted as a photon. In Si and Ge the excess energy is lost as lattice vibrations (heat), that is, as phonons. In the steady state, the thermal generation rate is balanced by the recombination rate so that the electron concentration n in the CB and hole concentration p in the VB remain constant; both n and p depend on the temperature.

Although in the specific example in Figure 3.4, a photon of energy $h\upsilon > E_g$ creates an electron–hole pair, other sources of energy can also lead to an electron–hole pair creation. In fact, in the absence of radiation, there is still an electron and hole pair generation process going on in the sample as a result of **thermal generation**. Due to thermal energy, the atoms in the crystal

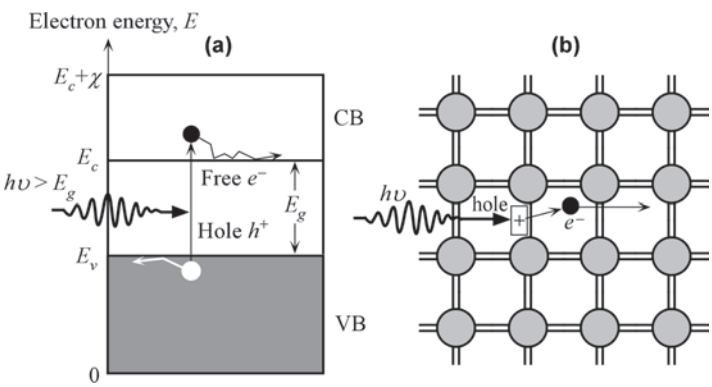


FIGURE 3.4 (a) A photon with an energy $h\upsilon$ greater than E_g can excite an electron from the VB to the CB. (b) Each line with Si-Si atoms is a valence electron in a bond. When a photon breaks a Si-Si bond, a free electron and a hole in the Si-Si bond are created. The result is the photogeneration of an electron–hole pair (EHP).